GENERAL COLLEGE CHEMISTRY LAB MANUAL (SEMESTER 2)

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Los Medanos College

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Laboratory Manual Chemistry 26

Traditional Format

May 2023

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A detailed breakdown of this resource's licensing can be found in ***Back Matter/Detailed Licensing****.*

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Disclaimer: These experiments are intended to be used with experienced supervision in a laboratory classroom setting. Los Medanos College and the Contra Costa Community College

District assumes no responsibility or liability from the use or misuse of these experimental procedures either in or out of the laboratory classroom.

In other words: if you think it is a good idea to go in your back yard (or anywhere else) and mix some random chemicals, we inform you firmly that it is a bad idea. Do not sue us if or when you get hurt. We warned you.

Several sources were used for the experiments in this laboratory manual:

Permission was granted from the Laney College Department of Chemistry to adapt some of their experiments. Experiments 1, 12, 15, 16, 17, 21, 34 and Volume Measurements from the 30A manual; Experiments 3 and 4 from the

30B manual; and Experiment 17 from the 1A manual were adapted for use.

Permission was granted from the Santa Monica College physical science department to adapt their experiments from their Chemistry 11 laboratory manuals. Experiments 1, 2, 5, 7, 12, 15, 17, and 18 were adapted.

Permission was granted from Long Island University to use images in Experiment 2510-T.

The Techniques section is modified from the LMC Chemistry 06/07

Laboratory Manual. Additional descriptions of techniques are available from https://chem.libretexts.org/Bookshelves/Ancillary_Materials/Demos_Techniques_and_Experiments/General_Lab_Techniques.

The 2020-2021 LMC staff of student stockroom workers were of great assistance in testing out and improving these experiments.

Los Medanos faculty and staff who assisted this effort include Melinda Capes, Dennis Gravert, Edward Haven, Nancy Peters, Girlie Sison and Paul West.

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Organization and Version Control

Laboratory Manual Organization

This laboratory manual is organized in two parts:

Part 1 is laboratory experiments. These are available to instructors for laboratory class assignments. There are more experiments than are necessary for most classes, so students may not do all experiments in this manual.

Part 2 is for supplemental documents. This section is for more detailed explanations, additional problems sets, and abridged reporting sheets for specific experiments. This section is intended for faculty to add customized supplemental documents.

Pagination is done within each experiment.

Version Control

Because this laboratory manual is intended to be improved over time, all documents use version control. Users of this manual should be aware of how version control has been implemented. In the upper right hand area of all pages, there is a version number. Version numbers are changed when a revision is done, and are also incorporated into the filenames of the individual experiments. The magnitude of the difference done during an update will determine how much the version number is changed.

A minor change such as a misspelled word would cause the version to change in the third place. For example, if an experiment went from version 1.3.2 to 1.3.3, it indicates a minor change between the two versions.

A moderate change, such as adding a section or changing a diagram or photograph would cause the version to change in the second place, and resetting the third place. For example, a technique changed from version 1.2.5 to version 1.3 indicates that a significant change occurred, but that the underlying procedure is similar.

A major change, such as a full rewrite of the procedure, would be a large and significant change. For example, a change from version 2.3.4 to version 3 would indicate that the two versions are fundamentally different.

Organization and Version Control

The most current version of experiments and techniques will be offered as the individual file (available for uploading to Canvas). An annual update is planned for a single production file for easy printing.

CHAPTER OVERVIEW

Experiments

2601 Freezing Point Depression

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- describe colligative properties.
- find the freezing point depression of a solution.
- determine the molar mass of an unknown solute using freezing point depression data.

1.2 Background

Colligative properties of solutions can be determined by comparing the properties of the pure solvent with the properties of the solvent upon addition of various solute particles. Properties such as boiling point and freezing point depend on the intermolecular attractive forces between solvent molecules. Changes in these physical properties occur when the presence of a solute disrupts these interactions.

When a small amount of nonvolatile solute is dissolved in a solvent, the vapor pressure of the solution will be less than the vapor pressure of the pure solvent at the same temperature. As a result, the boiling point of the solution, T_b, is higher than the boiling point of the pure solvent, $T_b°$. The amount by which the boiling point of the solution exceeds the boiling point of the pure solvent, $\Delta T_b = T_b – T_b°$, is called the boiling point elevation. Similarly, because of the reduction in vapor pressure over the solution, the freezing point of the solution, T_f, is lower than the freezing point of the pure solvent, $T_f°$. The amount by which the freezing point of the solution is decreased from that of the pure solvent, $\Delta T_f = T_f° – T_f$, is called the freezing point depression. A sample vapor pressure diagram for a pure solvent and its solution is shown below. (Figure 1)

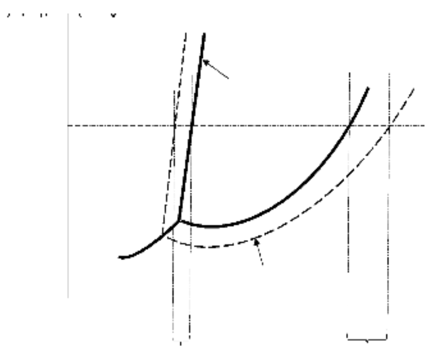


Figure 1. Vapor pressure curve

Focusing on freezing point depression, the magnitude of the freezing point depression attributed to the added solute is proportional to its molality.

$\Delta T_f = K_f m$ (Equation 1)

Molality, *m*, has units of moles of solute per kilogram of solvent, and is more useful for the calculation of this colligative property because, unlike molarity, it is independent of temperature. K_f is known as the freezing point depression constant and depends on the solvent used.

In this experiment you will determine the molar mass of an unknown solute by adding it into a solvent and measuring the resulting freezing point depression of the solvent. From the measured ΔT_f and the known K_f value of the solvent, you can then determine the value of *m* using the above equation.

Solutes, however, dissolve in solvent differently. Most molecular substances such as ethanol, C_2H_5OH, dissolve in water and remain as discrete solvated molecules, $C_2H_5OH(aq)$. Other solutes, like ionic substances, dissociate or break apart in solution to form ions. An example is table salt, NaCl, which dissolves in water to form $Na^+(aq)$ and $Cl^-(aq)$ ions.

Since the colligative properties of solution depend on the concentration of dissolved solute particles, an additional factor, *i*, (also called the van't Hoff factor) is included in the colligative property equations to account for the formation of multiple particles as electrolytes dissociate:

$\Delta Tf = iK_f m$ (Equation 2)

The solvent that will be used in this experiment is laboratory water that has a K_f value of 4.5 °C·kg·mol^{-1}. Since only non-dissociating solutes will be used in this experiment, the value of *i* for your unknown solute will be considered equal to 1.

To determine the freezing point of this pure solvent, you will measure the temperature as a function of time as the liquid cools in an insulating jacket. At first, the temperature will fall quite rapidly. When the freezing point is reached, solid will begin to form, and the temperature will tend to hold steady until the sample is all solid. This behavior is shown graphically in Figure 2. The freezing point of the pure liquid is the constant temperature observed while the liquid is freezing to a solid.

The cooling behavior of a solution is somewhat different from that of a pure liquid, also shown in Figure 2. As discussed earlier, the temperature at which a solution freezes is lower than that for the pure solvent. In addition, there is a slow gradual fall in temperature as freezing proceeds. The best value for the freezing point of the solution is obtained by drawing two straight lines connecting the points on the temperature-time graph. The first line connects points where the solution is all liquid. The second line connects points where the solid and liquid coexist. The point where the two lines intersect is the freezing point of the solution.

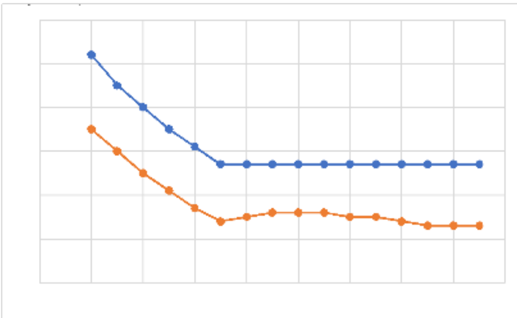


Figure 2. Cooling curve for a pure solvent (top) and for a solution (bottom)

Note that when the solid first appears the temperature may fall below the freezing point, but then it comes back up as more of the solid forms. This effect is called supercooling, a phenomenon that may occur with both the pure liquid and the solution. When drawing the straight line in the solid-liquid region of the graph, ignore points where supercooling is observed. To establish the proper straight line in the solid-liquid region it is necessary to record the temperature until the trend with time is smooth and clearly established.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

3.0 CHEMICALS AND SolutionS

3.0 CHEMICALS AND SolutionS

Chemical	**Concentration**	**Approximate Amount**	**Notes**
Stearic acid	N/A	15 g	
Lauric acid (unknown)	N/A	4 g	
Myristic acid (unknown)	N/A	4 g	
Palmitic acid (unknown)	N/A	4 g	

4.0 GLASSWARE AND APPARATUS

50 mL beaker	Hot plate
100 mL beaker	Balance
150 mL beaker or small Pyrex bowl (70x50)	Crucible tong
Scoopula	Paper towel (cut-to-fit)
Digital thermometer	

5.0 PROCEDURE

5.1 Determining the Freezing Point of Pure Stearic Acid (a Fatty Acid)

1. Fill a 150 mL beaker with about 30 mL tap water and heat on a hot plate. This will be your hot water bath.
2. Prepare an insulating jacket by wrapping a piece of paper towel around a clean dry 50 mL beaker and fitting it in a 100 mL beaker. Remove the 50 mL beaker and reserve the 100 mL beaker and the paper towel as the insulating jacket (see Figure 3). The insulating jacket prevents premature cooling due to contact with the skin or other surface.

Figure 3. Set-up for an Insulating Jacket

3. Weigh the 50 mL beaker using an electronic balance and record its mass. Transfer about 15 grams (half-full) of fatty acid to the beaker. Reweigh and record the mass of the beaker with the fatty acid. Calculate the mass of stearic acid by difference.
4. Place the beaker containing fatty acid into the hot water bath to melt it. After the fatty acid has completely melted, place the thermometer into the ‘melt’ and heat until the temperature reaches 85^{o}C. From this point on, the thermometer is not removed from the sample to prevent loss of material and contamination of bench tops with fatty acids. Remove the beaker from the water bath and dry the outside.
5. Place the beaker containing fatty acid in the previously prepared insulating jacket. Stir the fatty acid slowly but continuously to help minimize supercooling. Record the initial temperature of the sample and then every 30 seconds for 8-10 minutes. Temperatures are recorded until the

temperature of the sample remains constant, changing by 0.1ºC per reading, for 3 minutes (6 readings).

6. Perform a second trial by putting back this beaker containing the solidified fatty acid into the hot water bath and repeating steps 4 and 5.

2. Determining the Freezing Point of Solutions (Stearic Acid + Unknown)

7. Remove the 50 mL beaker containing fatty acid from the insulating jacket and cool to room temperature. Carefully place and rest the thermometer on a clean paper towel on your bench top; making sure that no solids sticking on the tip of the thermometer are dislodged.
8. (Solution I) To the fatty acid sample used above, add approximately 1.8 g of an unknown sample (a different fatty acid). Exactly 1.8 g is not needed, but make sure to record its mass to the nearest 0.1 mg. You may use the 'tare' function of the balance in this step.
9. Repeat steps 4 to 5 above. If time permits, a second trial (step 6) can be performed.
10. (Solution II) To the same fatty acid mixture, an additional 1.8 g of the same unknown sample is transferred. Make sure to cool the beaker to room temperature before placing on the balance. Similarly, exactly 1.8 g is not needed but make sure to record its mass to the nearest 0.1 mg.
11. Repeat steps 4 to 5 above. If time permits, a second trial (step 6) can be performed.

5.3 CLEAN UP

12. The solidified solutions can be carefully scraped off the beaker using a spatula. All solids can go down the trash bin. Discard any remaining residue from the beaker and thermometer directly into the trash bin.
13. Return the digital thermometer to the cart/stockroom.

6.0 DATA RECORDING SHEET *Show the equations you used and the calculations you performed below.*

Last Name	**First Name**		**Partner Name(s)**	**Date**

Table 1. Mass Measurements

Unknown ID	
5.1 Freezing Point of Pure Stearic Acid	
Mass of 50 mL beaker and stearic acid	
Mass of empty 50 mL beaker	
Mass of stearic acid (by difference)	
5.2 Freezing Point of Solution	
Identification code for unknown	
Mass of unknown sample (step 8; for solution I)	
Mass of additional unknown sample (step 10)	
Total mass of unknown sample (for solution II)	

Table 2. Temperature Measurements. Record the temperature every 30 seconds as the pure solvent and solutions are cooled. Note the temperature at which any crystal first appears. (*Use additional sheet(s), if necessary.*)

Time Elapsed (minutes)	Temperature (^oC)		
	Pure Solvent	Solution I	Solution II
0			
0.5			
1			
1.5			
2			
2.5			
3			
3.5			
4			
4.5			
5			
5.5			
6			
6.5			
7			
7.5			
8			
8.5			
9			
9.5			
10			

7.0 CALCULATIONS

7.1 Graphical Analysis of Data

1. Use Excel to create three (or four) separate graphs of "Temperature versus Time" for the pure solvent and the two solutions studied. Each graph should have an appropriate title and labeled axes with an appropriate scale. Add two trendlines to the data points of each graph. You can do this by hand with a ruler or by using Excel. The first line is applied to data points that correspond

to the cooling of the liquid state – these are the points on the steep part of the graph. The second line is applied to data points that correspond to the co-existence of both the solid and liquid (freezing) – these are the points on the part of the graph where the temperature levels out. Extrapolate the two trendlines towards each other until they intersect. The temperature at the point of intersection is the solvent freezing point and should be clearly shown on each graph.

2. **Attach your three graphs to this report.**
3. Record the freezing point temperatures obtained from the graphs below:

Pure stearic acid: _________ °C

Solution I: _________ °C

Solution II: ______________ °C

7.2 Calculation of Molar Mass

Complete the table below with the results of your calculations. Be sure to include all units. Note that K_f stearic acid = 4.5 °C·kg·mol^{-1}.

	Solution I	Solution II
1. Mass of stearic acid		
2. Total mass of unknown added		
3. Freezing point of pure stearic acid		
4. Freezing point of solution		
5. Freezing point depression, ΔT_f		
6. Molality of solution		
7. Moles of unknown in solution		
8. Molar Mass of unknown		

Unknown ID: ______________________ has an average molar mass of ____________.

8.0 POST-LAB QUESTIONS

1. Using the known freezing point of stearic acid (69^{o}C), determine the percentage error in your experimentally measured freezing point. Show your calculation below and be sure to report your answer to the correct number of significant figures.

2. Obtain the actual molar mass of your unknown solute (from your instructor) and determine the percentage error in your experimental average value. What factors could have affected such deviation from the actual molar mass of your unknown solute?

3. The commercial antifreeze used in cars is typically 50% v/v water and ethylene glycol. Discuss why this solution is useful for protecting car engines from both extreme summer and winter temperatures.

4. Hydrogen chloride (HCl) is soluble in both water and in benzene (C_6H_6). The freezing points and K_f values are given below for each solvent.

Solvent	FP (°C)	K_f (°C/·kg·mol^{-1})
Water	0.00	1.86
C_6H_6	5.50	5.12

1. For a 0.01 *m* HCl(aq) solution, the freezing point depression is about 0.04°C. Calculate the van 't Hoff factor for HCl in water. Is this van 't Hoff factor what you would expect for HCl dissolved in water? Explain why or why not.

2. For a 0.01 *m* HCl dissolved in benzene solution, the freezing point depression is about 0.05°C. Calculate the van 't Hoff factor for HCl in benzene.

3. Compare the van 't Hoff factor for HCl when water is the solvent (a) to the van 't Hoff factor for HCl when benzene is the solvent (b). Is there any difference between these two values? Account for the difference in terms of intermolecular forces between HCl and each of the solvents.

2602 Factors Affecting Reaction Rates

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to study the effects of concentration, temperature, and catalysts on a reaction rate.

1.2 Background

This experiment will allow you to study the effects of concentration, temperature,

and catalysts on a reaction rate. The reaction whose rate you will study is the reduction of permanganate ion by oxalic acid. The overall balanced equation is:

$2\ MnO_4^-(aq) + 5\ H_2C_2O_4\,(aq) + 6\ H^+(aq) \rightarrow 2\ Mn^{2+}(aq) + 10\ CO_2(aq) + 8\ H_2O$

Measuring reaction rate entails measuring some unique property of one of the

products or reactants. In this experiment, the unique property you will look for is the

decolorization of the permanganate solution. The change in color from pink to colorless can be used as a signal to measure the rate of the reaction. The time from mixing the reactants to the disappearance of the pink color can easily be measured. This

length of time is the time required by the reaction to reduce MnO_4^- to Mn^{2+}.

As the experiment will show, the time required varies with the concentrations of the

reactants, with temperature, and with the presence of a catalyst.

To determine the effect of concentration, the reaction will be carried out with the reactants at several different initial concentrations. This will be done by varying the amount of either sulfuric acid (source of H^+) or oxalic acid added to the reaction mixture. The total volume will be kept at 45 mL in each case by adding appropriate amount of laboratory water.

To determine the effect of temperature, the reaction will be performed at a specific concentration of reactants (held constant) and the temperature will be varied. A small change in temperature can cause a large change in reaction rate. For a typical reaction near room temperature, a 10°C increase in temperature doubles the rate

of reaction. This is because, in a typical reaction, very few of the collisions between

reactant ions or molecules take place with sufficient energy to cause reaction. The

energy necessary for a collision to result in reaction is called the energy of activation, Ea,

of that reaction. A small increase in temperature can have a considerable effect on the

number of molecules that have this required amount of energy to undergo reaction when

collision occurs.

A catalyst is a substance that can increase the rate of reaction by changing the mechanism of the reaction. Usually, a catalyzed mechanism will have a lower activation energy than an uncatalyzed mechanism. At any given temperature, there will be more molecules having sufficient energy to react via the catalyzed mechanism than via the uncatalyzed route. Therefore, the rate of the catalyzed reaction will be faster.

In this experiment, the catalyst will be copper metal. Both the catalyzed and the uncatalyzed reactions will be performed at several lengths of a copper wire while keeping the initial concentrations of the reactants and the reaction temperature constant. In this way, the effect of the presence and amount of catalyst on the reaction rate can be determined.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

All solutions from this experiment should be poured into the INORGANIC WASTE container in the fume hood.

Copper wires must be separated from the solution, dried, and disposed of in the SOLID WASTE container in the fume hood.

3.0 CHEMICALS AND SolutionS

3.0 CHEMICALS AND SolutionS

Chemical	**Concentration**	**Approximate Amount**	**Notes**
Laboratory water	Pure	150 mL	Be sure to use laboratory water
$KMnO_4$(aq)	0.0014 M	50 mL	
$H_2C_2O_4$(aq)	0.20 M	50 mL	
H_2SO_4(aq)	2.0 M	50 mL	
Copper wire	Various lengths	10 in	

4.0 GLASSWARE AND APPARATUS

50 mL beakers (4) (or any size ≤ 250 mL)	Droppers (plastic or glass with rubber bulb)
400 mL beaker (for water bath)	Thermometer
Test tubes (medium)	Hot plate

5.0 PROCEDURE

This experiment should be performed by two people working together. During each run, one person can watch the reaction mixture and the other can watch the clock or timer.

5.1 Effect of H^+ Concentration

1. Using the four clean and dry 50-mL beakers (or any size ≤ 250 mL available), obtain about 50 mL of each of the four reagents needed to prepare the mixtures listed in the table below. Label each beaker appropriately. Assign one dropper for each reagent as well.

2. Prepare each reaction mixture by first mixing $KMnO_4$, H_2SO_4, and laboratory water in a clean and dry test tube. Then, quickly add $H_2C_2O_4$ to the test tube, mix by flicking, and start the timer. Record the time when the pink color has completely disappeared. Monitor the reaction for about 10 minutes. It will be helpful to use a white paper as background to see the decolorization process. Enter the time on your data recording sheet.

Reaction #	$KMnO_4$	H_2O	H_2SO_4	$H_2C_2O_4$
1A	15 drops	25 drops	5 drops	0
2A	15 drops	20 drops	5 drops	5 drops
3A	15 drops	10 drops	15 drops	5 drops
4A	15 drops	0	25 drops	5 drops

5.2 Effect of $H_2C_2O_4$ Concentration

Repeat step 2 in Part 5.1 above using the reaction mixtures below. Be sure to use clean and dry test tubes.

Reaction #	$KMnO_4$	H_2O	H_2SO_4	$H_2C_2O_4$
1B	15 drops	25 drops	0	5 drops
2B	15 drops	10 drops	15 drops	5 drops
3B	15 drops	5 drops	15 drops	10 drops
4B	15 drops	0	15 drops	15 drops

5.3 Effect of Temperature

1. Set up a hot-water bath in a half-filled (with tap water) 400 mL beaker placed on a hot plate. Heat the water bath until the temperature of the water is between 40°C and 50°C. Try to maintain this temperature interval by heating or letting the water bath to cool. The exact temperature at which the reaction is run is not crucial.
2. In this part of the experiment, the concentrations of reactants will be the same for both runs but the temperature will be higher than room temperature.
3. Prepare each reaction mixture by first mixing $KMnO_4$, H_2SO_4, and laboratory water in a clean and dry test tube. Place this test tube into the water bath for several minutes to allow the solution to reach the bath temperature. Then, remove the test tube from the bath and quickly add $H_2C_2O_4$, mix by flicking, and return the test tube to the bath. Start the timer. Record the time when the pink color has completely disappeared.

Run #	$KMnO_4$	H_2O	H_2SO_4	$H_2C_2O_4$	Temp
1C	15 drops	10 drops	15 drops	5 drops	40°C – 50°C
2C	15 drops	10 drops	15 drops	5 drops	40°C – 50°C

5.4 Effect of Catalyst

1. In this part of the experiment, the concentrations of reactants will be the same for all runs and the reaction will be kept at room temperature.
2. Repeat step 2 in Part 5.1 above. Be sure to use clean and dry test tubes. Copper wire will be added as indicated in the table below. The exact length of the copper wire is not crucial. You may coil the wire to allow for maximum contact with the solution.

Run #	$KMnO_4$	H_2O	H_2SO_4	$H_2C_2O_4$	Cu Length
1D	15 drops	10 drops	15 drops	5 drops	0
2D	15 drops	10 drops	15 drops	5 drops	1 – 3 in.
3D	15 drops	10 drops	15 drops	5 drops	4 – 6 in.

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

6.1 Effect of H^+ Concentration

Reaction #	Notes/Observations	Reaction Time
1A		
2A		
3A		
4A		

6.2 Effect of $H_2C_2O_4$ Concentration

Reaction #	Notes/Observations	Reaction Time
1B		
2B		
3B		
4B		

6.3 Effect of Temperature

Reaction #	Notes/Observations	Reaction Time
1C		
2C		

6.4 Effect of Catalyst

Reaction #	Notes/Observations	Reaction Time
1D		
2D		
3D		

7.0 POST-LAB QUESTIONS

1. What is the effect of the concentration of H^+ (from sulfuric acid) on the rate of the reaction? What is the effect of the concentration of $H_2C_2O_4$ on the rate of the reaction?

2. Did you observe any similarities or differences between the effect of H^+ and $H_2C_2O_4$ concentrations? Explain your answer fully.

3. What are the effects of the temperature and the presence of Cu metal on the reaction rate? Provide a physical explanation to each of these observations. What is occurring on the

molecular level?

4. Which of the factors you determined experimentally has the most profound effect on the reaction rate? Explain your choice.

2603 Rate Law and the Initial Rate Method

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- determine the rate law of a chemical reaction using the initial rate method.
- determine the energy of activation (E_a) of the reaction by finding the value of the rate constant, k, at different temperatures.

1.2 Background

1.2.1 Finding the Rate Law using the Initial Rate Method

The rate law of a chemical reaction is a mathematical equation that describes how the reaction rate depends upon the concentration of each reactant. Two methods are commonly used in the experimental determination of the rate law: the initial rate method and the graphical method. In this experiment, we shall use the initial rate method to determine the rate law of a reaction. It will be helpful to review this concept in the lecture.

The reaction to be studied in this experiment is represented by the following balanced ionic equation:

$6\ I^-(aq) + BrO_3^-(aq) + 6\ H^+(aq) \rightarrow 3\ I_2(aq) + Br^-(aq) + 3\ H_2O(l)$ (Equation 1)

The rate law for this reaction is of the form:

rate = $k\ [I^-]^m[BrO_3^-]^n[H^+]^p$ (Equation 2)

where the value of the rate constant, *k*, is dependent upon the temperature at which the

reaction is run. The values of the reaction orders, *m*, *n*, and *p* are usually, though not always, small integers. The values of *m*, *n*, *p* and k must be determined experimentally to specify the rate law for this reaction completely.

The initial rate method allows the values of the reaction orders to be found by running the reaction multiple times under controlled conditions and measuring the rate of the reaction in each case. All variables are held constant from one run to the next, except for the concentration of one reactant. The order of that reactant concentration in the rate law can be determined by observing how the reaction rate varies as the concentration of that one reactant is varied. This method is repeated for each reactant until all the orders are determined. At that point, the rate law can be used to find the value of *k* for each trial. If the temperatures are the same for each trial, then the values of *k* should be the same too.

The rate of the reaction can be expressed in terms of the rate of decrease of the concentration of BrO_3^-.

rate of reaction=-Δ[BrO3-]/Δt (Equation 3)

Note that this is the average rate of the reaction over the time interval Δt since the concentration of BrO_3^- and therefore the rate of the reaction is continuously decreasing during Δt. In this experiment, we can assume that $\Delta[BrO_3^-]$ is negligible compared to $[BrO_3^-]$ thus allowing us to approximate our measured average rate as being equal to the initial, instantaneous rate.

In addition, the rate of the reaction will be measured indirectly by running a second reaction, known as a 'clock' reaction, simultaneously along with the reaction of interest. The 'clock' reaction must be inherently fast relative to the reaction of interest, and it must consume at least one of the products (I_2) of the reaction of interest. In this case, the parallel reactions are the following:

$6\ I^-(aq) + BrO_3^-(aq) + 6\ H^+(aq) \rightarrow 3\ I_2(aq) + Br^-(aq) + 3\ H_2O(l)$ *slow*

('clock' reaction) $3\ I_2(aq) + 6\ S_2O_3^{2-}(aq) \rightarrow 6\ I^-(aq) + 3\ S_4O_6^{2-}(aq)$ *fast*

Overall reaction:

$BrO_3^-(aq) + 6\ S_2O_3^{2-}(aq) + 6\ H^+(aq) \rightarrow Br^-(aq) + 3\ H_2O(l) + 3\ S_4O_6^{2-}(aq)$ (Equation 4)

Note that since the 'clock' reaction is relatively fast, the $I_2(aq)$ will be consumed by the clock reaction as quickly as it can be produced by the reaction of interest thus holding the I_2 concentration at a very low value close to zero. Only after the $S_2O_3^{2-}$ has been entirely consumed will the concentration of I_2 start to increase. After the $S_2O_3^{2-}$ has been consumed, the concentration of I_2 will increase rapidly allowing the I_2 to react with a starch indicator resulting in the solution to acquire a deep blue color.

In this experiment, you will measure the time required for the solution to turn blue.

This is essentially the amount of time required for all the $S_2O_3^{2-}$ to be reacted.

A stoichiometric calculation can be used to determine the number of moles of BrO_3^- reacted at the point in time when all the $S_2O_3^{2-}$ has been consumed. By dividing the number of moles of BrO_3^- reacted by the total volume of the reaction mixture we can obtain the change in the BrO_3^- concentration. We can obtain the reaction rate by dividing this change in concentration by the amount of time required for the blue color to appear (Δt).

1. Measure the time required for the solution to turn blue. (all $S_2O_3^{2-}$ reacted)
2. Calculate moles BrO_3^- reacted
3. Divide moles BrO_3^- by total volume of reaction mixture = $[BrO_3^-]$
4. Reaction rate = $[BrO_3^-]$ divided by a. = $[BrO_3^-]/(\Delta t)$

1.2.2 Determining the Activation Energy of the Reaction

The value of the rate constant, *k*, measured previously is dependent upon the

temperature at which the reaction occurs according to the Arrhenius Equation:

$k=Ae^{-Ea/RT}$ (Equation 5)

where A is the frequency factor and is related to the number of properly aligned collisions

that occur per second between reactant molecules; E_a is the activation energy of the

reaction and is the minimum energy that must be present in a collision for a reaction to

occur; and R is the universal gas constant with a value of $8.3145\ x\ 10^{-3}$ kJ·mol^{-1}.

Taking the natural logarithm of both sides of Equation 5 gives:

$\ln \ln k=-E_a/R\ (1/T)+\ln \ln A$ (Equation 6)

Note that this equation is that of a straight line of the form, $y=mx+b$, where:

$y = \ln k$, $x = 1/T$, $m = -E_a/R$, and $b = \ln \ln A$.

Thus, if the rate constant, *k*, is measured at several temperatures and ln *k* is plotted as a function of 1/T, the slope of the resulting line will allow the value of E_a for the reaction to be determined.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Laboratory water	Pure	150 mL	Be sure to use laboratory water
KI(aq)	0.010 M	100 mL	
$Na_2S_2O_3$(aq)	0.0010 M	100 mL	
$KBrO_3$(aq)	0.040 M	100 mL	
HCl(aq)	0.10 M	100 mL	
Ice	Pure		For preparing ice/water bath
Starch solution	0.5 – 1 %	3-4 drops for each reaction mixture	For use as indicator

4.0 GLASSWARE AND APPARATUS

Squirt bottle	10 mL graduated cylinder
250 mL beakers (4) (or any size ≥ 150 mL)	Large test tubes (2)
250 mL Erlenmeyer flask	600 or 1000 mL beaker (for the hot water bath)
125 mL Erlenmeyer flask	Thermometer
Medium test tubes (4)	Hot plate

5.0 PROCEDURE

5.1 Preparation of Glassware

- Because soap residue and other chemicals can interfere with the reaction we are observing, it is critical that all glassware used in this experiment be rinsed several times using laboratory water (and not soap!) prior to performing the experiment. Use a squirt bottle to minimize waste; never rinse glassware directly under the laboratory water tap. In general, there is no need to dry glassware after rinsing because most solutions that we use are aqueous. But shake them gently several times to remove almost all the water.

- Prior to use, make sure to rinse the graduated cylinder with about 2-3 mL of the reagent, pouring this rinse into the sink.

5.2 Determining the Rate Law Using the Initial Rate Method

As shown in the table below, four (4) reaction mixtures will be prepared. The table summarizes the volume of each reagent that should be used in each reaction mixture.

Reaction Mixture	**Flask I (250 mL)**			**Flask II (125 mL)**		
	0.010 M KI	0.0010 M $Na_2S_2O_3$	Laboratory H_2O	0.040 M $KBrO_3$	0.10 M HCl	1% Starch
1	10.0 mL	10.0 mL	10.0 mL	10.0 mL	10.0 mL	3-4 drops
2	20.0 mL	10.0 mL	---	10.0 mL	10.0 mL	3-4 drops
3	10.0 mL	10.0 mL	---	20.0 mL	10.0 mL	3-4 drops
4	10.0 mL	10.0 mL	---	10.0 mL	20.0 mL	4. drops

1. Using the four 250-mL beakers (or any size ≥ 150 mL available), obtain about 100 mL of each of the four reagents needed to prepare the mixtures listed in the table (except laboratory water). Rinse each beaker with about 5 mL of the particular reagent solution you will store in it first, pouring this rinse into the sink, then fill the beaker with the 100 mL you will need. Label each beaker appropriately. You are now ready to prepare Reaction Mixture 1.
2. Using a clean 10-mL graduated cylinder, measure 10.0 mL of laboratory water (from your squirt bottle) and transfer it into the 250-mL Erlenmeyer flask (labeled Flask I).
3. Into the 250-mL Erlenmeyer flask containing the 10.0 mL of laboratory water (Flask I), separately measure and add 10.0 mL of the 0.010 M KI and 10.0 mL of the 0.0010 M $Na_2S_2O_3$.
4. Into the 125-mL Erlenmeyer flask (labeled Flask II), in a similar manner, combine 10.0 mL of the 0.040 M $KBrO_3$ reagent solution, 10.0 mL of the 0.10 M HCl solution, and 3 to 4 drops of starch indicator. Shake the starch solution thoroughly before use.
5. The next step requires two experimenters: one to operate the stopwatch or timer and another to mix and swirl the contents of the two flasks. Be certain that the person using the stopwatch or timer knows how to operate it by testing it once or twice before proceeding.
6. Pour the contents of Flask II into Flask I rapidly and then swirl the solution to mix thoroughly. Start your stopwatch or timer the moment the two solutions are combined. Set the flask containing the reaction mixture down on a sheet of white paper and watch carefully for the blue color of the starch-iodine complex to appear. It should take about one to three minutes. Stop the timer the instant that the blue color appears. Record the elapsed time on your data recording sheet.
7. Using your thermometer, measure the temperature of the reaction mixture immediately following the reaction to the nearest tenth of a degree and record this value on your data recording sheet.

8. Dispose of the contents of the flask in the sink. Rinse both Erlenmeyer flasks and your thermometer as described in the preparation of glassware section (Part 5.1).
9. Repeat steps 2 to 7, performing a second trial for Reaction Mixture 1. If the time you measure for this second trial differs by more than 10% from that of your first trial, repeat the procedure. If after three trials you still unable to obtain two trials with reaction times that differ by less than 10%, see your instructor.
10. Now repeat this procedure for the other three reaction mixtures listed in the table above. You need to perform only one trial for each of these mixtures. Compare the times you obtain for each of these mixtures with those obtained by other teams. Repeat any trials where the reaction time differs significantly from those obtained by other teams. Record these data on your data recording sheet. Do not forget to measure the temperature immediately following each trial.
11. Dispose of the contents of the flasks in the sink. Rinse the flasks, and thermometer before returning into your locker or to the stockroom.

5.3 Determining the Activation Energy (E_a) of the Reaction (Optional)

1. Prepare the solutions given for Reaction Mixture 1 in the table but instead of a 250 mL Erlenmeyer flask, use large test tubes (labeled TT 1 and TT 2).

2. Prepare a water bath at room temperature using tap water and a large (600 mL or 1000 mL) beaker. Use utility clamps to secure the large test tubes, and a thermometer into the water bath. Make sure that the solution levels are at or below the water bath level. Let them sit for about five minutes to equilibrate to the bath temperature. After equilibration, record the water bath temperature on your data recording sheet. This will be the reaction temperature.

3. Remove TT 1 from the clamp and quickly transfer the solution into TT 2. Swirl the reaction mixture to mix thoroughly. Start your timer the moment you combine the two solutions. Keep the flask containing the mixed solution in the water bath and observe carefully for the blue color to appear. Stop the timer the instant that this faint blue color first appears. Record the elapsed time and the final temperature of this mixture on your data recording sheet.

4. Dispose of the contents of the flask in the sink. Rinse the 125 mL Erlenmeyer flask and your thermometer as described in the preparation of glassware section (Part 5.1). The test tube does not need to be cleaned since it will always contain the same solution mixture.

5. Perform two more trials at elevated temperatures using a hot-water bath in place of the room temperature water bath. Each trial should be performed using the quantities given for Reaction Mixture 1 in the table. Heat both test tube and 125 mL Erlenmeyer flask in a hot-water bath (on a hot plate) until the temperature in the water bath reaches approximately 30°C (±5°C) in the first elevated-temperature trial, and about 40°C (±5°C) in the second elevated-temperature trial. When the appropriate temperature has been reached, remove the flasks, mix the contents, and record the elapsed time for the blue color to appear. Record the final temperature of the reaction mixture as before. (If time permits, you can perform a low temperature run by preparing an ice/water bath that is between 10°C and 15°C.)

6. Dispose of the contents of the test tubes in the sink. Rinse the test tubes, and thermometer before returning into your locker or to the stockroom.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name(s)**	**Date**

6.1 Determining the Rate Law Using the Initial Rate Method

Reaction Mixture 1 (to confirm constancy between trials)

Trial	Elapsed Time, Δt (s)	Temperature (^oC)	Notes/Observations
1			
2			
3 (if needed)			
Average			

Data for Reaction Mixtures 1 through 4

Reaction Mixture	Elapsed time, Δt (s)	Temperature (^oC)	Notes/Observations
1 *(avg from above)*			
2			
3			
4			

6.2 Determining the Activation Energy, E_a, of the Reaction

Data for the Effect of Temperature on the Reaction Rate (Reaction Mixture 1)

	Low Temp	Room Temp	About 30^{o}C	About 40^{o}C
Temp of Reaction Mixture (^{o}C)				
Temp of Reaction Mixture (K)				
Elapsed time, Δt (s)				

7.0 CALCULATIONS

7.1 Initial Concentrations of the Reactants After Mixing

Use the dilution equation, $M_1V_1 = M_2V_2$ to calculate the initial concentration of each reagent after preparing the reaction mixture.

Reaction Mixture	$[I^-]_o$ (M)	$[S_2O_3^{2-}]_o$ (M)	$[BrO_3^-]_o$ (M)	$[H^+]_o$ (M)
1				
2				
3				
4				

Show your calculations below for each reagent in one of the reaction mixtures.

7.2 Relative Rates of Reaction

1. Based on the stoichiometry of the reaction in Equation 4, fill in the values of a and b in the relative rate expression:

$$-\frac{1}{a}\frac{\Delta\left[BrO_3^-\right]}{\Delta t} = -\frac{1}{b}\frac{\Delta\left[S_2O_3^{2-}\right]}{\Delta t}$$

Relative rate expression: ______________________________________

2. Use your concentration and time (elapsed time) data from the previous section (Part 7.1) to calculate the rate that $S_2O_3^{2-}$ is consumed in each of your four mixtures. Then, use the relative rate expression above to determine the corresponding initial rate of reaction in terms of the change in concentration of BrO_3^- reacted.

Relative rates of reaction

Reaction Mixture	$-\frac{\Delta[S_2O_3^{2-}]}{\Delta t}\,(M/s)$	$-\frac{\Delta[BrO_3^-]}{\Delta t}\,(M/s)$
1		
2		
3		
4		

Show your calculations below for one of the reaction mixtures.

7.3 Order of the Reaction with Respect to Each Reactant and the Rate Law

1. Copy (from Parts 7.1 and 7.2) the appropriate values of the initial reactant concentrations and reaction rates in the table below.

Reaction Mixture	$[I^-]_o$ (M)	$[BrO_3^-]_o$ (M)	$[H^+]_o$ (M)	Rate (M/s)
1				
2				
3				
4				

2. Using the initial rate method and the relevant data in the table above, determine the order of the reaction with respect to each reactant, and then write the experimentally determined rate law for the reaction (Equation 2). Show all your calculations below, noting the reaction mixtures used for your data.

Reaction order with respect to I^-:

Reaction order with respect to BrO_3^-:

Reaction order with respect to H^+:

Rate law: ____

7.4 Value of the Rate Constant, *k*

Using your experimentally determined rate law from Part 7.3, determine the value of the rate constant, *k*, for each reaction mixture and then, calculate its average. Observe proper significant figures and units.

Reaction Mixture	Value of *k*	Average value of *k*
1		
2		
3		
4		

Show your calculations below illustrating how you determined the value of *k* for one of the reaction mixtures.

7.5 Determining the Activation Energy, E_a, for the Reaction

1. Determine the value of the rate constant, *k*, for each temperature below and give the appropriate units.

	Low Temp	Room Temp	About 30ºC	About 40ºC
$-\frac{\Delta[S_2O_3^{2-}]}{\Delta t}\,(M/s)$				
$-\frac{\Delta[BrO_3^{-}]}{\Delta t}\,(M/s)$				
Rate constant, *k*				
ln *k* (no units)				
1/T (K^{-1})				

Show your calculations for one of the reaction temperatures below:

2. Use Excel to create a graph of "ln *k* versus 1/T". Your graph should have an appropriate title and labeled axes with an appropriate scale. Using the Excel trendline function, add a best-fit line to your plotted data and display the equation of this line and its R^2 value on your graph. **Submit this graph with your report.**

Use this graph to determine the value of the activation energy, E_a, and the frequency factor, *A*, for this reaction (Be certain to include the proper units for each!). Show your calculations below.

8.0 POST-LAB QUESTIONS

1. The volume of each the reagents listed in the table (Part 5.2) are varied one at a time in turn except for the volume of sodium thiosulfate, which is the same in each trial. Why do we keep the volume of sodium thiosulfate the same in each trial?

2. Based on your calculations in Part 7.4, even though the concentrations of the reactants are changed in each reaction mixture, the experimentally determined values of the rate constant, *k*, should be fairly similar. Why is this?

3. Discuss the trend you observed from your graph in Part 7.5. Provide a physical explanation to explain this? What is occurring on the molecular level?

2604 Chemical Equilibrium

1.0 OBJECTIVES

- Students will investigate a reaction between calcium nitrate and potassium iodate to determine if the reaction goes to completion or reaches a state of equilibrium.
- Students will gain experience producing a precipitate and filtering it out of its solution.

1.1 INTRODUCTION

While many reactions go to completion, many do not. The reactants begin to form products, but the products then begin to collide with one another and reform the reactants. While the initial forward reaction is the fastest with the higher concentration of reactants, as the concentration of products begins to increase, the rate of the reverse reaction begins to increase as well. Eventually, the forward reaction and the reverse reaction are moving at the same rate and a state of equilibrium is reached. There are some reactants changing into products and some products changing into reactants all the time, but the concentrations of the reactants and products become steady.

In this experiment, the reaction between calcium nitrate, $Ca(NO_3)_2$, and potassium iodate, KIO_3, will be examined to determine if it is a reaction that goes to completion or if it is a reaction that reaches a state of equilibrium.

$Ca^{+2} + 2IO_3^- \rightarrow Ca(IO_3)_2$ (s) Equation 1

$Ca^{+2} + 2IO_3^- \leftrightarrow Ca(IO_3)_2$ (s) Equation 2

The difference between Equations 1 and 2 is the arrow: → means a reaction that goes to completion and ↔ means a reaction that reaches equilibrium, implying it goes forwards and backwards.

If the reaction goes to completion, there may still be excess reagent, either the calcium or the iodate, if we did not mix the chemicals in their stoichiometric ratio: 1 calcium to 2 iodates. In an equilibrium situation, there would always be some of all the chemicals present, calcium and iodate and solid calcium iodate.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical/Solution	**Concentration**	**Amount**	**Notes**
$Ca(NO_3)_2$	0.200M	40 mLs	
KIO_3	0.100M	55 mLs	
Na_3PO_4	0.3M	5 mLs	
HCl	1.0M	5 mLs	
Solid KI	xxxx	0.25 inch on spatula (0.5 cm^3)	
Laboratory water	xxxx	25 mLs	

4.0 GLASSWARE AND APPARTUS

2 100- OR 150-mL beakers	2 filter funnels
2 250-mL beakers	2 Erlenmeyer flasks
2 graduated cylinders	Several small test tubes
Several disposable pipettes	Small spatula
2 filter papers	2 watch glasses

5.0 INVESTIGATION

5.1 Interaction of Calcium Nitrate and Potassium Iodate

1. Wash and label two 100 or 150 mL beakers. Obtain about 40 mL of 0.2 M calcium nitrate, $Ca(NO_3)_2$, and 55 mL of 0.1 M potassium iodate, KIO_3, stock solutions in the clean labeled beakers – one solution in each beaker. Record the molarity of each stock solution.

2. Wash and label two 250 mL beakers. Prepare the following two solutions – one in each beaker – using a graduated cylinder and **using the calcium nitrate and potassium iodate solutions you obtained previously and put into your 100 or 150 mL beakers**.

Solution 1: 25 mL $Ca(NO_3)_2$ + 25 mL KIO_3 (50 mL total)

Solution 2: 15 mL distilled H_2O + 10 mL $Ca(NO_3)_2$ + 25 mL KIO_3 (50 mL total)

3. Mix each solution with a separate stirring rod, rubbing the tip of the rod against the bottom or sides of the beakers. This is a process called **scratching**. Scratching produces an unpleasant sound something like imperfections on the surface of the beaker and provides a surface for precipitation of a solid to occur thus speeding up the precipitation process. Scratch the beaker with the glass rod frequently until a solid has begun to form. Then let the two solutions stand for 20-30 minutes. Begin filling in the data in Part 6.0 and do the calculations for the theoretical yield.

4. Label and weigh two pieces of filter paper and record their masses in your laboratory notebook.

5. Fold the weighed filter paper as illustrated by your instructor. Insert the folded filter paper into a filter funnel. Place the filter funnel in a clean Erlenmeyer flask. Seat (or seal) the filter to the funnel surface by filtering approximately 5 mL of distilled water through the filter.

6. Gently swirl your solution to suspend the solids. Quickly and carefully pour the solution into the filter funnel. Make sure you pour the solution from beaker 1 into filter 1. Use a stirring rod and rubber policemen to transfer (push) as much of the precipitate as possible into the filter funnel. Use a wash bottle of distilled water to wash any remaining precipitate out of your beaker and into the filter funnel.

7. Wash the precipitate and the surface of your filter with approximately 2 mL of distilled water from a wash bottle and allow the water to completely drain from the filter. Repeat this washing step at least 4 more times. Gently remove the filter paper from the funnel, place it on a watch glass and allow it to air dry. **Save your filtrates for further analyses.**

5.2 Tests of the Supernatant/Filtrate Identities

A **supernatant** is the solution standing above a settled precipitate in the bottom of a container.

1. Measure and transfer about 5 mL of 0. 3 M Na_3PO_4 solution into a small test tube. Measure about 5 mL of 1 M HCl into a separate small test tube.

2. Place about 1 mL of the stock 0.2 M $Ca(NO_3)_2$ solution from your 100 or 150 mL beaker in a test tube. Place about 1 mL of 0.1 M KIO_3 stock solution from your 100 or 150 mL beaker in a separate test tube. Note that 20 drops is equal to about 1 mL.

3. Calcium cation, Ca^{2+}, test: Add about 10 drops (0.5 mL) of 0.3 M Na_3PO_4 solution from your small test tube to the test tube containing 1 mL of stock 0.2 M calcium nitrate solution. Note what happens.

4. Iodate, IO_3^-, test: Dissolve about 0.5 cm^3 (about 1/4 inch on a spatula) of KI in the 1 mL of stock 0.1 M potassium iodate solution in your test tube; then add about 1 mL of 1 M HCl to the test tube. Note what happens.

5. Repeat the calcium cation test using 1 mL of solution 1 filtrate rather than 1 mL of 0.1 M $Ca(NO_3)_2$ stock solution. Repeat the test again using 1 mL of solution 2 filtrate. Compare your results for the calcium cation test using solution 1 and solution 2 filtrates to each other and to the results using the 0.2 M $Ca(NO_3)_2$ stock solution. Note what happens to each.

6. Repeat the iodate test using 1 mL of solution 1 filtrate rather than 1 mL of 0.1 M KIO_3 stock solution. Repeat the test again using 1 mL of solution 2 filtrate. Compare your results for the iodate test using solution 1 and solution 2 filtrates to each other and to the results using the 0.1 M KIO_3 stock solution. Note what happens to each.

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

Molarity of stock $Ca(NO_3)_2$ solution ____________ M (moles/L)

Molarity of stock KIO_3 solution________________ M

			Solution 1					Solution 2
L $Ca(NO_3)_2$ solution:			__________					__________
L KIO_3 solution:			__________					__________
ass of filter paper:			__________					__________
ass of filter & precipitate:			__________					__________
ass of precipitate:			__________					__________
oles of $Ca(NO_3)_2$:			__________					__________
oles KIO_3:			__________					__________
xpected mass of precipitate, $Ca(IO_3)_2$: r theoretical yield)			__________					__________
yield:			__________					__________
bservations of tests of filtrates: alcium test date Test								

7.0 POST-LAB QUESTIONS

1. How do your actual precipitate masses compare to the masses you predicted by calculation, in other words were your percent yields high or low? If your percent yields were low make a list of at least 4 possible explanations for the low yield.

2. Are your test results for the presence of Ca^{2+} and IO_3^- in the supernatants consistent with the limiting reagent and excess reagent? Or are they consistent with an equilibrium reaction? Give an explanation for your answer.

2605 Le Chatelier's Principle

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- Perturb chemical reactions at equilibrium and observe how they respond.
- Explain these observations using Le Chatelier’s Principle.
- Relate Le Chatelier’s Principle to the concept of coupled reactions.

1.2 Background

All chemical reactions eventually reach a state in which the rate of the reaction in the forward direction is equal to the rate of the reaction in the reverse direction. When a reaction reaches this state, it is said to be at chemical equilibrium.

The concentrations of reactants and products at equilibrium are constant as a function of time. Thus, for a homogeneous aqueous system of the form

$$aA_{(aq)} + bB_{(aq)} \rightleftharpoons cC_{(aq)} + dD_{(aq)} \quad (1)$$

we can express the equilibrium-constant expression for this reaction as,

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b} \quad (2)$$

where the values of [A], [B], [C] and [D] correspond to the equilibrium concentrations (or equilibrium positions) of all the aqueous chemical components, and a, b, c and d are their respective stoichiometric coefficients. Note that for a heterogeneous system including pure solids or liquids of the form

$$aA_{(aq)} + bB_{(s)} \rightleftharpoons cC_{(aq)} + dD_{(l)} \quad (3)$$

the pure liquids and solids do not appear in the equilibrium-constant expression:

$$K_c = \frac{[C]^c}{[A]^a} \quad (4)$$

It has been observed that when a reaction at equilibrium is perturbed by applying a stress, the reaction will respond by shifting its equilibrium position to counteract the effect of the perturbation/stress. In other words, the concentrations of the reactants and products will shift so that the relationship described by Equation (2) is again satisfied. This idea was first proposed by Henri-Louis Le Châtelier and has since been referred to as, “Le Châtelier’s Principle”.

Note that when a reaction makes more products as a response to the perturbation, we call it a right shift. When a reaction makes more reactants in response to the perturbation, we call it a left shift. We often designate these respective shifts by drawing right and left arrows below the chemical equation.

For chemical reactions at equilibrium in aqueous solution, the most common types of perturbations include changing the concentration of one of the aqueous solutes, changing the concentrations of all aqueous solutes by changing the total solution volume, or changing the temperature. The general responses of an aqueous system to these particular perturbations are tabulated below.

Perturbation	Effect on Equilibrium Position	Effect on K_c
Increase in concentration of a single reactant, or, decrease in concentration of a single product.	Shift to the right	None
Decrease in concentration of a single reactant, or, increase in concentration of a single product.	Shift to the left	None
Decrease in all aqueous concentrations due to an increase in solution volume resulting from the addition of solvent	Shift towards side with more solute particles	None
Increase in all aqueous concentrations due to a decrease in solution volume resulting from the removal of solvent (evaporation)	Shift towards side with less solute particles	None
Increase temperature of an exothermic reaction	Shift to the left	Decrease
Decrease temperature of an exothermic reaction	Shift to the right	Increase
Increase temperature of an endothermic reaction	Shift to the right	Increase
Decrease temperature of an endothermic reaction	Shift to the left	Decrease
Addition of an inert substance, catalyst, pure liquid, or pure solid	None	None

Notice that only a temperature change can affect the value of K_c; in all other cases, the value of K_c remains constant.

In this experiment, you will perturb reactions that have attained equilibrium. You will then observe how each reaction responds to that perturbation in order to restore equilibrium. In your report you describe these changes in terms of Le Chatelier’s Principle.

1.3 Specific Experimental Examples

1.3.1 Acid-Base Equilibrium

Here you will use **coupled equilibria** to change the equilibrium position of an acid-base

reaction. In order to understand how coupled equilibria work, consider the reactions described by the chemical equations below:

$A(aq) \rightleftharpoons B(aq)$ (5)

$B(aq) + C(aq) \rightleftharpoons D(aq)$ (6)

Notice that the species B*(aq)* is common to both reactions. The presence of this common

species couples these two reactions.

We can perturb the equilibrium position of Reaction (6) by the addition of some C*(aq)*. The addition of C*(aq)* will cause the equilibrium position of Reaction (6) to shift right in accordance with Le Chatelier’s Principle. This right shift in the equilibrium position of Reaction (6) will also result a corresponding decrease in the concentration of B*(aq)*. Because B*(aq)* is also present in Reaction (5), the decrease in the concentration of B*(aq)* will in turn result in a right shift in the equilibrium position of Reaction (5). Thus, the addition of C*(aq)* to Reaction (6) actually results in a right shift in the equilibrium position of Reaction (5) because the equilibria are coupled.

In Part A we will observe the effect of various solutes on an acid-base indicator (a weak acid) at equilibrium. The equilibrium system can be written in the general form

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$ (7)

The equilibrium constant expression for this reaction is

$$K_a = \frac{[H^+][A^-]}{[HA]} \quad (8)$$

where we denote the equilibrium constant, K, with a subscript a for acid. In this experiment, HA and A^-, are the acidic and basic forms of the indicator bromothymol blue. Since the two forms are different colors, you will be able to determine which form is predominant in the equilibrium mixture. In other words, you will be able to determine whether the equilibrium position lies to the left (more reactants and less products) or whether the equilibrium lies to the right (more products and less reactants).

Your goal will be to find a reagent that will shift the position of this equilibrium to the opposite side, and then another reagent that will shift it back towards its original position. Instead of directly adding HA or A^- to the system, you will cause these shifts by adding H^+ or OH^-. Note that in order to determine the effect of OH^- we must consider a second chemical reaction that shares a common species with the Reaction (7). The second reaction is the autoionization of water, which can be described by the equation

$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$ (9)

The equilibrium constant for this reaction is denoted by K_w, where the subscript w stands for water, and the associated equilibrium constant expression is

$$K_w = [H^+][OH^-] \quad (10)$$

Because Reactions (7) and (9) share a common chemical species (H^+), you can use the concept of coupled equilibria to shift the equilibrium position of Reaction (7) by increasing or decreasing the concentration of $OH^-(aq)$.

1.3.2 Solubility Equilibrium

Here you will test the effects of changing temperature and volume on the solubility of a slightly soluble salt at equilibrium. Some examples of slightly soluble salts are AgCl, $Cu(OH)_2$, $PbCl_2$, and Fe_2S_3, which you should recall are, “insoluble in water,” according to the solubility rules you learned in Chem 25. In fact, a very small amount of each of these substances does dissolve in aqueous solution, but the amount is so small that we often classify each of these compounds as “insoluble”.

This type of equilibrium is often called a **solubility equilibrium** because it is written in the direction of the dissolution of the solid, as shown in the following example:

$A_xB_y(s) \rightleftharpoons xA^+(aq) + yB^-(aq)$ (11)

The equilibrium constant expression for Reaction (11) is

$$K_{sp} = [A^+]^x [B^-]^y \quad (12)$$

where we denote the equilibrium constant, K, with a subscript sp for **solubility product**.

Now let's consider the process of precipitation. In a typical precipitation reaction two aqueous salt solutions are mixed together resulting in the production of an insoluble salt. Notice that this process corresponds to a left shift of Reaction (11), and so Equation (12) can also be used to examine the conditions required for the precipitation of a solid to occur. We can denote the product $[A^+]^x[B^-]^y$ under arbitrary conditions (not necessarily at equilibrium) as,

$$Q_{sp} = [A^+]^x [B^-]^y \quad (13)$$

where Q_{sp} is called the solubility product **reaction quotient**. Note that, upon mixing two solutions, one containing A and the other containing B , if $Q_{sp} < K_{sp}$ the system is not at equilibrium, but since no solid A_xB_y is present, the reaction cannot shift to the right and therefore no reaction will be observed. In contrast, if $Q_{sp} > K_{sp}$ the solution contains an excess of aqueous species, and Reaction (11) will shift left, forming the solid precipitate A_xB_y until the system reaches a state of equilibrium where $Q_{sp} = K_{sp}$. Thus, we can use the values of Q_{sp} and K_{sp} to predict the conditions under which a precipitation reaction will occur.

In Part B, we will study the solubility equilibrium of $PbCl_2$*(s)*. We will observe the effect on this solubility equilibrium of changes in solution volume (quantity of solvent) and temperature. We will express these changes in terms of the respective values of Q_{sp} and K_{sp}.

1.3.3 Complex Ion Equilibrium

Certain metal ions, most often transition metals, exist in solution as complex ions in combination with other ions or molecules, called ligands. Common ligands include H_2O, NH_3, Cl^- and OH^-. Many of these complex ions exhibit vibrant colors in solution. For example, the $Co(H_2O)_6^{2+}$*(aq)* complex ion is pink and the $CoCl_4^{2-}$*(aq)* complex ion is blue. In Part C, you will study the following complex ion formation reaction:

$$Co(H_2O)_6^{2+}(aq) + 4\ Cl^-(aq) \rightleftharpoons CoCl_4^{2-}(aq) + 6\ H_2O\ (l) \quad (14)$$

The equilibrium constant expression for Reaction (14) is

$$K_f = \frac{[CoCl_4^{2-}]}{[Co(H_2O)_6^{2+}][Cl^-]^4} \quad (15)$$

where we denote the equilibrium constant, K, with a subscript f for complex ion formation.

Your goal in Part C is to observe how Reaction (14) shifts from its equilibrium position as the result of various perturbations.

1.3.4 Dissolving Insoluble Solids

In Part D, you will use coupled equilibria to affect the solubility equilibrium of $Zn(OH)_2(s)$. The solubility equilibrium can be described by the equation

$Zn(OH)_2(s) \rightleftharpoons Zn^{2+}(aq) + 2\ OH^{-}(aq) \quad K_{sp} = 5 \times 10^{-17}$ (16)

Notice that $K_{sp} << 1$ for this reaction, demonstrating that $Zn(OH)_2(s)$ is only very slightly soluble in aqueous solution.

Now consider the reactions described by the following chemical equations, each of which shares a common species with the Reaction (16):

$H_2O(l) \rightleftharpoons H^{+}(aq) + OH^{-}(aq) \quad K_w = 1 \times 10^{-14}$ (17)

$Zn^{2+}(aq) + 4\ OH^{-}(aq) \rightleftharpoons Zn(OH)_4^{2-}(aq) \quad K_f = 3 \times 10^{15}$ (18)

$Zn^{2+}(aq) + 4\ NH_3(aq) \rightleftharpoons Zn(NH_3)_4^{2+}(aq) \quad K_f = 1 \times 10^{9}$ (19)

Because Reactions (17), (18), and (19) each share a common species with Reaction (16) they can be coupled together. In Part D of this experiment, you will observe the effect of coupling each of these equilibria on the solubility of $Zn(OH)_2(s)$.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

12-M HCl is extremely caustic and great care must be taken to avoid contact with eyes or skin. The bottle should be kept in a plastic tray and not removed from the fume hood. Should any of this solution enter your eyes rinse immediately in the emergency eyewash. Should any of this solution come in contact with your skin rinse with copious amounts of water and apply saturated sodium bicarbonate to the affected area from the stock bottle located on the sink.

The solutions you will use in Part B contain lead. Be certain that all of these lead-containing solutions are disposed of in the proper waste container and rinse your hands following this procedure.

Many of the chemicals used in this lab are hazardous to the environment. All metal ion waste (lead, zinc, magnesium, cobalt ions) must be disposed of in the hazardous-waste container in the fume hood. Rinse all glassware directly into the waste container twice using a small squirt bottle to be certain all hazardous waste ends up in the waste container.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Laboratory water	Pure	25 mL	
Bromothymol blue		several drops	
$NaOH(aq)$	6 M	3 mL	
$HCl(aq)$	6 M	3 mL	
$Pb(NO_3)_2(aq)$	0.3 M	5 mL	
$HCl(aq)$	0.3 M	5 mL	
$HCl(aq)$	12 M	2 mL	
$CoCl_2 \cdot 6\ H_2O(s)$		tip of the scoopula	
$Zn(NO_3)_2(aq)$	0.1 M	6 mL	
$Mg(NO_3)_2(aq)$	0.1 M	6 mL	
$NH_3(aq)$	6 M	3 mL	(often labeled as NH_4OH)

4.0 GLASSWARE AND APPARATUS

6 medium test tubes	test tube rack
10 mL graduated cylinder	stirring rod
20 mL graduated cylinder	scoopula
400 mL beaker (waste container)	

Bucket of ice from Stockroom

5.0 PROCEDURE

5.1 Acid-Base Equilibrium

1. Add approximately 5 mL of laboratory water to a medium test tube. Add 3 drops of the bromothymol blue indicator solution. Record the color of your solution on your data recording sheet.
2. Your solution from Step 1 currently contains one form of bromothymol blue (see background). Now predict which of the two 6 M reagents you obtained, the strong acid or the strong base, will cause a color change in your solution by making the bromothymol blue indicator shift to its other form. Add the 6 M reagent of your choice drop-by-drop and if your solution changes color, write the color of the solution and formula of the reagent on your data recording sheet. If the addition of your reagent does not result in a color change, try other reagents until you are successful.
3. Since equilibrium systems are reversible, it is possible to shift a reaction left or right repeatedly by changing the conditions. Now find another 6 M reagent that will cause your solution from Step

2 to revert back to its original color. Add the 6 M reagent of your choice drop-by-drop and if your solution changes color, write the formula of the reagent on the data recording sheet. If the addition of your second reagent does not result in a color change, try other reagents until you are successful.

5.2 Solubility Equilibrium

1. Set up a hot-water bath by filling a 400-mL beaker about half-full with tap water and heating the water on a hot plate while you work on the following steps. The hot-water bath will be used in Step 6 (and also in 5.3).
2. Measure 5.0 mL of the 0.3 M $Pb(NO_3)_2$*(aq)* solution into a medium test tube using a 10-mL graduated cylinder.
3. Thoroughly rinse the graduated cylinder with laboratory water, then measure 5.0 mL of the 0.3 M HCl*(aq)* solution using the 10-mL graduated cylinder.
4. Add about 1 mL of the 0.3 M HCl*(aq)* solution to the $Pb(NO_3)_2$*(aq)* solution in the medium test tube. Stir the mixture gently using your stirring-rod and record your observations on your data recording sheet.
5. Continue to add the 0.3 M HCl*(aq)* solution to the $Pb(NO_3)_2$*(aq)* solution in the medium test tube in roughly 1 mL increments until you just begin to see white $PbCl_2$ solid appear in your test tube. To confirm that the solid is present, let the test tube sit on the bench for about 3 minutes, allowing all solid to settle to the bottom where it is easier to see. On your data recording sheet, record the total volume of 0.3 M HCl*(aq)* needed to produce the solid.
6. Put the test tube containing the solid into your hot-water bath from Step 1. Stir the contents of the test tube gently for a few seconds using the glass stirring rod and record your observations on your data recording sheet. Continue heating and stirring until a change is observed (about 50^oC).
7. Make a cold-water bath by filling a 400-mL beaker half full of tap water and ice. Cool the test tube down by placing it into this cold-water bath. Observe what happens and record your observations. Save this cold-water bath for use in 5.3.
8. Dispose of the remainder of the 0.3 M HCl*(aq)* solution in your 10-mL graduated cylinder by pouring it into the appropriate waste container and then rinse the cylinder using laboratory water. Next measure 5.0 mL of laboratory water using this graduated cylinder.
9. Add about 1 mL of deionized water to the medium test tube containing the $PbCl_2$ solid. Stir the mixture and observe what happens. Continue to add the laboratory water in 1 mL increments until the white $PbCl_2$ solid just dissolves (or disappears). On your data recording sheet, record the volume of water needed to dissolve the solid.
10. Pour the contents of the medium test tube into your large graduated cylinder and measure the total volume of the solution. Record this volume on your data recording sheet.
11. Dispose of all solutions used in Part B in the proper waste container. Rinse all glassware twice using a wash bottle of laboratory water to ensure that all of the lead solution has been removed and transferred to the waste container. Save the hot- and cold-water baths for 5.3.

5.3 Complex Ion Equilibrium

1. Reheat your hot-water bath from 5.2 to a near boil. The hot-water bath will be used in Step 4.

2. Put a few small crystals (an amount that fits at the tip of the scoopula) of $CoCl_2 \cdot 6H_2O$*(s)* in a medium test tube. Working in the fume hood, carefully add 2 mL of 12 M HCl*(aq)* (this acid is extremely caustic; see cautions in the safety section of this experiment). Stir to dissolve the crystals. Record the color of this solution.
3. Using a 10-mL graduated cylinder, add laboratory water to the solution in your test tube in approximately 2 mL increments, stirring after each addition, until no further (obvious) color change occurs (make sure you are not just diluting the solution!). Record the new color.
4. Place the test tube into the hot-water bath from Step 1 and record any color change.
5. Cool the solution down in the cold-water bath and record any color change.

5.4 Dissolving Insoluble Solids

1. Label three medium test tubes A, B, and C. Add about 2 mL of 0.1 M $Zn(NO_3)_2$*(aq)* solution to each test tube. Add one drop of 6 M NaOH*(aq)* solution to each test tube. Stir each solution and record your observations.
2. Into test tube A, add at least 20 drops of 6 M HCl*(aq)* drop-by-drop while stirring. Record all observations.
3. Into test tube B, add at least 20 drops of 6 M NaOH*(aq)* drop-by-drop while stirring. Record all observations.
4. Into test tube C, add at least 20 drops of 6 M NH_3*(aq)* drop-by-drop while stirring. Record all observations.
5. Label three additional medium test tubes D, E, and F. Add about 2 mL of 0.1 M $Mg(NO_3)_2$*(aq)* solution to each test tube. Add one drop of 6 M NaOH*(aq)* solution to each test tube. Stir each solution and record your observations.
6. Into test tube D, add at least 20 drops of 6 M HCl*(aq)* drop-by-drop while stirring. Record all observations.
7. Into test tube E, add at least 20 drops of 6 M NaOH*(aq)* drop-by-drop while stirring. Record all observations.
8. Into test tube F, add at least 20 drops of 6 M NH_3*(aq)* drop-by-drop while stirring. Record all observations.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name*(s)***	**Date**

6.1 Acid-Base Equilibrium

Equilibrium system: HA*(aq)* $\rightleftharpoons$ H^+*(aq)* + A^-*(aq)*

Record your observations for each of the following steps in the table below.

Step 1	Color of bromothymol blue in distilled water	
Step 2	Name of reagent “A” causing color change when added	
Step 3	Name of reagent “B” causing a return to original color	

The acidic form of the bromothymol blue indicator, HA*(aq)*, is ______________ in color.

The basic form of the bromothymol blue indicator, A^-*(aq)*, is _______________ in color.

Explain why reagent A (in Step 2) caused the color change observed.

Explain why reagent B (in Step 3) caused the color change observed.

6.2 Solubility Equilibrium

Equilibrium system: $PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\ Cl^{-}(aq)$

Record your observations for each of the following steps in the table below.

Step 4	Observations upon addition of just 1.0 mL of HCl*(aq)* to the $Pb(NO_3)_3$*(aq)* solution	
Step 5	Total volume of HCl*(aq)* required for noticeable precipitation (mL)	
Step 6	Observations upon placing the test tube with precipitate in hot water	
Step 7	Observations upon placing the test tube with precipitate in cold water	
Step 9	Volume of water added to just dissolve $PbCl_2$ precipitate (mL)	
Step 10	Total solution volume upon completion (mL)	

1. Why didn't any solid $PbCl_2$ form immediately upon addition of 1 mL of HCl*(aq)* in Step 4? What condition must be met by $[Pb^{2+}]$ and $[Cl^{-}]$ if solid $PbCl_2$ is to form?

2. Consider your observation in hot water in Step 6:

- In which direction did the equilibrium shift? ________________
- Did the value of K_{sp} get smaller or larger? ________________
- Is the dissolution of $PbCl_2$ *(s)* exothermic or endothermic? ________________

Explain below.

3. Explain why the solid $PbCl_2$ dissolved when water was added to it in Step 9. What was the effect of this water on $[Pb^{2+}]$, $[Cl^{-}]$, and Q_{sp}? In which direction would such a change drive the equilibrium system?

4. The point at which the $PbCl_2$ precipitate just dissolves in Step 9 can be used to determine the value of K_{sp} for this equilibrium system, where $K_{sp} = [Pb^{2+}][Cl^{–}]^2$. Calculate $[Pb^{2+}]$ and $[Cl^{–}]$ in the final solution (consider the “dilution effect”). Then use these equilibrium concentrations to determine the value of K_{sp} for this system. Show all work below.

6.3 Complex Ion Equilibrium

Equilibrium system: $Co(H_2O)_6^{2+}$*(aq)* + 4 Cl^-*(aq)* ⇌ $CoCl_4^{2-}$*(aq)* + 6 H_2O*(l)*

pink blue

Record your observations for each of the following steps in the table below.

Step 2	Color of solution in 12 M HCl*(aq)*	
Step 3	Color of solution upon addition of water	
Step 4	Color of solution in hot water	
Step 5	Color of solution in cold water	

1. What form of the complex ion, $Co(H_2O)_6^{2+}$*(aq)* or $CoCl_4^{2-}$*(aq)*, is predominant in:

the 12 M HCl*(aq)* __________________

the diluted solution __________________

the heated solution __________________

2. Explain why you obtained the observed color in 12 M HCl*(aq)* (Step 2).

3. Explain the observed color change that occurred when water was added to the solution in Step 3. Consider how water affects the ion concentrations and Q in this system.

4. Consider your observations in the hot water bath in Step 4.

In which direction did the equilibrium shift? ________________

Did the value of K get smaller or larger? ________________

Is the reaction (as written) exothermic or endothermic? ________________

Explain.

6.4 Dissolving Insoluble Solids

Equilibrium system: $Zn(OH)_2$*(s)* $\rightleftharpoons$ Zn^{2+}*(aq)* + 2 OH^-*(aq)* $K_{sp} << 1$

Record your observations for each of the following steps in the table below.

Step 1	Adding 1 drop of NaOH*(aq)* to $Zn(NO_3)_2$*(aq)*	
Step 2	Tube A: Effect when HCl*(aq)* is added	
Step 3	Tube B: Effect when NaOH*(aq)* is added	
Step 4	Tube C: Effect when NH_3*(aq)* is added	

1. Explain your observation upon addition of HCl*(aq)* to the precipitate in Tube A. You must consider the various equilibria that are occurring in solution and the effect of HCl*(aq)* on $[OH^-]$.

2. Explain your observations upon addition of NaOH*(aq)* to the precipitate in Tube B. Consider the various equilibria that are occurring in solution and remember that Zn^{2+} forms stable complex ions with OH^- at sufficiently high concentrations.

3. Explain your observations upon addition of NH_3*(aq)* to the precipitate in Tube C. Consider the various equilibria that are occurring in solution and remember that Zn^{2+} forms stable complex ions with NH_3.

Equilibrium system: $Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\ OH^-(aq)$ $K_{sp} << 1$

Record your observations for each of the following steps in the table below.

Step 5	Adding 1 drop of NaOH*(aq)* to $Mg(NO_3)_2$*(aq)*	
Step 6	Tube D: Effect when HCl*(aq)* is added	
Step 7	Tube E: Effect when NaOH*(aq)* is added	
Step 8	Tube F: Effect when NH_3 *(aq)*is added	

1. Explain your observation upon addition of HCl*(aq)* to the precipitate in Tube D.

2. Based on your observations in Steps 7 and 8 do you think that Mg^{2+} forms stable complex ions? Explain your reasoning.

2606 Properties of weak and strong acids

1.0 INTRODUCTION

According to the Arrhenius definition, acids ionize in water to produce a hydronium ion (H_3O^+), and bases dissociate in water to produce hydroxide ion (OH^-). An aqueous (water) solution that has a lot of hydronium ions but very few hydroxide ions is considered to be very acidic. If instead an aqueous solution has a lot of hydroxide ions but very few hydronium ions, it is considered to be very basic. Acids and bases are measured on a scale called pH. The pH is the negative of the log of the hydronium ion concentration.

$pH = -\log[H_3O^+]$

pH ranges from less than 1 to 14. It lets us quickly tell if something is very acidic, a little acidic, neutral (neither acidic nor basic), a little basic or very basic. A pH of 1 is highly acidic, a pH of 14 is highly basic, and a pH of 7 is neutral.

pH indicators, litmus paper, and pH paper can be used to determine whether something is an acid or a base and the strength of its acidity or basicity. An indicator is a substance that turns a different color at a certain pH. Litmus paper is a form of an indicator. It is made by coating paper with the indicator litmus. Litmus is known to change color at a pH of about 7. Either red or blue litmus paper can be purchased. Blue litmus paper remains blue when dipped in a base, but it turns red when an acid touches it. Red litmus paper stays red when dipped in an acid, but turns blue when a base touches it.

Another way to more specifically determine an acid or base is through the use of pH paper. pH paper allows us to determine to what degree a substance is acidic or basic. When a substance is placed on pH paper, a color appears. The color is compared to a color chart showing the color the pH paper will turn at different pH values.

The use of a pH meter provides a more precise measurement of pH. The pH meter converts voltage readings from a pH electrode, an electrode constructed from a special composition glass which senses the hydrogen ion concentration. This thin, fragile glass is typically composed of alkali metal ions. The alkali metal ions of the glass and the hydrogen ions in solution undergo an ion exchange reaction, generating a potential difference.

1.1 Objectives

After completing this experiment, the student will be able to:

- Classify substances as weak acids or strong acids.
- Determine the pH values of several acid solutions using pH paper and a pH meter.
- Explain the effect on pH when adding a common ion to a weak acid solution.

2.0 SAFETY PROCEDURES AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
HCl solution	Between 1 M and 6 M	5 mL	
HNO_3 solution	Between 1 M and 6 M	5 mL	
H_2SO_4 solution	Between 1 M and 6 M	5 mL	
Acetic acid $HC_2H_3O_2$ solution	Between 1 M and 6 M	5 mL	
Acetic acid $HC_2H_3O_2$ solution, standardized	1 M	25 mL	Record exact concentration
Oxalic acid dihydrate $H_2C_2O_4 \cdot 2H_2O$	solid	1 g	
Potassium hydrogen phthalate (KHP, $HKC_8H_4O_4$)	solid	1 g	
Sodium acetate $NaC_2H_3O_2$	solid	2 g	

4.0 GLASSWARE AND APPARATUS

pH meter (fragile!)	10 mL volumetric pipet and bulb
Several beakers, any size 150 mL or smaller	Two 100 mL volumetric flasks with stoppers
10 mL graduated cylinder	pH paper
50 mL graduated cylinder	

5.0 PROCEDURE

5.1 Preparation and pH Measurement of Dilute Acid Solutions

1. Prepare 100 mL of the following solutions at approximately 0.1 M. You will need to use the dilution equation $M_1V_1 = M_2V_2$ to calculate how much volume to transfer from each stock solution provided. For the acids that are solid, use its molar mass to calculate how much mass to weigh. Use 150 mL or larger beakers.
2. Show your calculation for one of your solutions below.

Acid	**Concentration of stock solution**	**Amount measured out for a 0.1 M solution**
HCl solution		
HNO_3 solution		
H_2SO_4 solution		
Acetic acid $HC_2H_3O_2$ solution, standardized		
Oxalic acid dihydrate $H_2C_2O_4 \cdot 2H_2O$	N/A	
Potassium hydrogen phthalate (KHP, $HKC_8H_4O_4$)	N/A	

3. Use pH paper to estimate the pH of each solution and record in Table 1 found in 6.0 DATA RECORDING SHEET below.

5.2 Measuring How pH Changes with Concentration

1. In a clean, dry beaker, obtain approximately 25 mL of the 1.0 M Standardized Acetic Acid solution. Label the beaker “Stock Solution”.
2. Fill a 10 mL volumetric pipet with “Stock Solution” and drain into the sink. Using the rinsed volumetric pipet, transfer 10 mL of “Stock Solution” to a 100 mL volumetric flask, fill to the mark with laboratory water, and mix well. Label the flask “Dilution #1”.
3. Fill the 10 mL volumetric pipet with “Dilution #1” and drain into the sink. Using the rinsed volumetric pipet, transfer 10 mL of “Dilution #1” to a second 100 mL volumetric flask, fill to the mark with laboratory water, and mix well. Label this flask “Dilution #2”.
4. Record the concentration of “Stock Solution” in Table 2 found in 6.0 DATA RECORDING SHEET below, then measure and record the pH of each solution using a pH meter (pour some of each solution into a beaker to make the pH measurements).
5. The procedure below describes how to use a portable Flinn pH meter (model AP8673). If the pH meter is not measuring properly (unstable, or inaccurate), refer to the manual for troubleshooting. This same procedure may be applied to other brands of pH meter.

Warning: The pH meter probe (end or tip) is extremely fragile! Do not allow the tip to touch the flask or other solid object (except filter paper).

1. Check out a portable pH meter from the stockroom.
2. Remove the protective cap on the electrode. Clean any salt build-up off by rinsing with laboratory water.
3. Press the ON/OFF button once.
4. Rinse the electrode with laboratory water and blot dry with filter paper.

5. Immerse the electrode in the flask containing the analyte solution. Once the display stabilizes (approx. 1 min.), record the exact pH.
6. Remove the pH meter from the solution.
7. Repeat steps (d) – (g) after a certain volume of the titrant is addedif you are titrating.
8. When finished, rinse the pH meter electrode with laboratory water and blot dry with filter paper. Replace the cap and return the pH meter to the stockroom at the end of lab.

5.3 Measuring How pH Changes with Addition of a Common Ion

1. Use a 50 mL graduate cylinder to transfer 50 mL of “Dilution #1” to a clean dry beaker.
2. Measure its pH with the pH meter and record in Table 3 found in 6.0 DATA RECORDING SHEET below.
3. Dissolve 0.5 g solid sodium acetate in your 50 mL solution. In Table 3 below record the exact mass of sodium acetate that was added.
4. Measure its pH with the pH meter and record in Table 3 below.
5. Dissolve an additional 1.5 g solid sodium acetate to your 50 mL solution. In Table 3 below record the exact mass of sodium acetate that was added.
6. Measure its pH with the pH meter and record in Table 3 below.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name(s)**	**Date**

Use the colors of the pH paper to estimate the pH of each solution from “5.1 Preparation and pH measurement of dilute acid solutions”. Then use your estimates to calculate the approximate H^+ concentrations.

Table 1

Acid solution	**Color of pH paper**	**pH**	**[H^+]**
HCl solution			
HNO_3 solution			
H_2SO_4 solution			
Acetic acid $HC_2H_3O_2$ solution, standardized			
Oxalic acid dihydrate $H_2C_2O_4 \cdot 2H_2O$			
Potassium hydrogen phthalate (KHP, $HKC_8H_4O_4$)			

1. Show your calculation of [H^+] for one of your solutions below.

2. Given a concentration of 0.1 M, what is the predicted pH value for a monoprotic strong acid?
3. Analyze and comment on your data. For example, since all the acid solutions were prepared at the same concentration, do they all have the same pH value? If they do not, propose several reasons why not (do not include experimental error as a reason). Which acids had a pH close to the predicted value? Which acids did not? For each acid, propose reason(s) for their measured pH. Choose one acid and write a balanced chemical equation to show what happens when that acid dissolves in water.

Table 2

	Concentration of acetic acid	**pH**	**[H^+]**	**Actual [$C_2H_3O_2^-$]**	**Actual [$HC_2H_3O_2$]**
“Stock Solution”					
“Dilution #1”					
“Dilution #2”					

1. Use the concentration and pH data to calculate the missing values in the data table above. Hint: Try using an ICE table.
2. In the space below, show the calculations for every value (except pH) in the row of the Table for “Dilution #1”.
3. Use your values from Table 2 to calculate the equilibrium constant for each solution and enter the results below.

	Calculated Equilibrium Constant
“Stock Solution”	
“Dilution #1”	
“Dilution #2”	

4. Discuss how closely these constants agree with each other (Are they constant?)

Table 3

Concentration of acetic acid in “Dilution #1”	**Additional grams of sodium acetate**	**pH**	**[H^+]**	**Actual [$C_2H_3O_2^-$]**	**Actual [$HC_2H_3O_2$]**
	0				

1. Calculate the missing values in Table 3. When sodium acetate is present, assume the “Actual [$C_2H_3O_2^-$]” is provided exclusively by sodium acetate. Remember to add together the two portions of sodium acetate (0.5 g + 1.5 g) when calculating the “Actual [$C_2H_3O_2^-$]” of the last solution!

2. Is the pH the same or different for all three solutions? Propose reasons for your observations.

3. Use your values from Table 3 to calculate the equilibrium constant for each solution and enter the results below.

	Calculated Equilibrium Constant
"Dilution #1"	
"Dilution #1" With 0.5 g	
"Dilution #2" With another 1.5 g	

4. Discuss how closely these constants agree with each other (Are they constant?)

5. Do these equilibrium constants from Table 3 agree with the equilibrium constants calculated from Table 2? Should they agree?

2607 Buffers

1.0 INTRODUCTION

In this lab you will explore the concept of pH, a measure of acidity, by using acid-base indicators. You will also prepare and experiment with buffer solutions, solutions that resist changes to pH.

1.1 OBJECTIVES

- Determine the pH of various solutions using pH indicators
- Create and study the properties of buffer solutions.

1.2 BACKGROUND

In the first part of today's lab you will use five indicators to determine the pH of several solutions to within one pH unit. An acid-base indicator is a chemical species that changes color at a specific pH as the pH (or acidity) of the solution varies. Acid-base indicators are themselves weak acids where the color of the aqueous acid is different than the color of the corresponding conjugate base.

The five indicators you will use in this experiment, their color transitions, and their respective values of pK_a are given in Table 1 below.

We can use the values in Table 1 to determine the approximate pH of a solution. For example, suppose we have a solution in which methyl violet is violet. This tells us that the pH of our unknown solution is greater than or equal to 2 because methyl violet turns violet at pH values of 2 or greater. Now suppose we add some congo red to a fresh sample of our solution and find that the color is violet. This tells us that the pH of our solution is less than or equal to 3 because congo red turns violet at pH values of 3 or less. From these two tests we know that the pH range of our solution must be between 2 and 3. Thus, we have determined the pH of our solution to within one pH unit. Proceeding in a similar manner, you will use the acid-base indicators in Table 1 to determine the pH range of four solutions to within one pH unit.

Table 1: Acid-Base Indicators

Indicator	pKa	pH and color transition (0 1 2 3 4 5 6 7 8 9 10 11 12 13 14)
phenolphthalein	9.4	clear, pink
bromothymol blue	7.1	yellow, green, blue
congo red	4.1	dark purple, red
bromocresol green	4.9	yellow, green, blue
methyl orange	3.4	orange, yellow
methyl red	4.95	red, yellow
thymol blue*	1.7 & 8.8	red, yellow, dark blue
universal indicator**	N/A	orange, green, purple

The actual colors in solution vary somewhat from those shown here depending on the concentration.
*Thymol blue has two pKa values.
**Universal indicator is a mixture of multiple indicators that spans a larger pH range

Figure 1. This shows an example of six of the indicators from about pH 4 – pH 10. From left to right: universal indicator, phenolphthalein, bromothymol blue, congo red, bromocresol green, and methyl orange.

In the second part of lab, you will prepare and experiment with buffer solutions: solutions that resist changes to pH.

The dissociation of a weak acid in water yields H^+ ion and the conjugate base of the acid, where the extent of dissociation is defined by the acid equilibrium constant, K_a:

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$ $K_a = [H^+][A^-]/[HA]$ (1)

The anion (A^-) in the above reaction, referred to as the conjugate base of HA, gives rise to the following equilibrium when an ionic salt (i.e. NaA) of this ion is dissolved in water:

$A^-(aq) + H_2O(l) \rightleftharpoons HA(aq) + OH^-(aq)$

Note that the anion accepts H^+ from water, thereby acting as a base in this equation, consistent with its name: conjugate *base*.

When a weak acid is dissolved in a solution containing its conjugate base, the result is a mixture that is highly resistant to changes in $[H^+]$ upon addition of either strong acid or strong base. Added OH^- ions consume H^+ ions already present which are replaced by further dissociation of HA; while added H^+ ions are consumed as the equilibrium shifts to the left producing more HA. These solutions are known as buffer systems. They are used to control the $[H^+]$ as is required in a wide variety of practical applications, including life! Our blood is buffered by bicarbonate ions.

For buffer solutions, it is convenient to rearrange the equilibrium expression (1) into the following form:

$[H^+] = K_a[HA]/[A^-]$

Given that $[H^+]$ is usually expressed in units of pH (pH = -log $[H^+]$), the above equation can be conveniently rewritten by taking (– log) of both sides, and substituting:

$-\log[H^+] = pH$ and $-\log K_a = pK_a$

This results in the Henderson-Hasselbalch equation:

$pH = pK_a + \log([A^-]/[HA])$ (2)

In this lab you will use the Henderson-Hasselbalch equation in two ways. In one experiment you will measure the pH of a mixture of known concentrations of a weak acid ([HA]) and its conjugate base ($[A^-]$) in order to determine the pK_a. In another experiment you will use the Henderson-Hasselbalch equation in order to design and prepare a buffer solution of a specific pH.

2.0 SAFETY PROCEDURES AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	Approximate Amount	Notes
0.1 M HCl	5 mL	
0.1 M NaH_2PO_4	5 mL	
0.1 M CH_3COOH	5 - 25 mL	
0.1 M $ZnSO_4$	5 mL	
Methyl red solution	Several drops	
Thymol blue solution	Several drops	
Bromothymol Blue	Several drops	
Methyl orange solution	Several drops	
Congo red solution	Several drops	
Phenolphthalein	Several drops	
Bromocresol green	Several drops	
Universal indicator solution	Several drops	optional
0.1 M $NaC_2H_3O_2$	20 mL	
0.1 M NH_4Cl	20 mL	
0.1 M NH_3	20 mL	
0.1 M $NaHCO_3$	20 mL	
0.1 M Na_2CO_3	20 mL	
0.1 M NaOH	10 mL	

4.0 GLASSWARE AND APPARATUS

pH meter and electrode (fragile!)	5 small test tubes (or a well-plate)
6 beakers, any sizes 50 - 150 mL	Glass stir rod
10 mL graduated cylinder	Disposable pipets
50 mL graduated cylinder	

5.0 PROCEDURE

5.1 Determination of pH using Acid-Base Indicators

1. Rinse several small test tubes using laboratory water (there is no need to dry these). To each test tubes add approximately 1 mL of 0.1 M HCl(aq). (Alternatively, if using a well-plate, place 2-

3 drops of HCl to each well.)

2. Into each of these test tubes add one of the eight indicators listed in Table 1. (On a well-plate, use only 1 drop of the indicator.) Record the color of each solution in the appropriate data table below.
3. Repeat steps 1-2 for the following solutions and record your color observations in the data table:

0.1 M sodium hydrogen phosphate, $NaH_2PO_4(aq)$

0.1 M acetic acid, $CH_3COOH(aq)$

0.1 M zinc sulfate, $ZnSO_4(aq)$

5.2 Determination of K_a

1. From among the reagents provided by the stockroom, select one buffer system that you can prepare from two of the provided solutions. (You need to understand the definition of buffers in order to do this step correctly! See 1.1 BACKGROUND) Use a graduated cylinder to transfer 20 mL of your chosen acid component of your buffer into a 150 mL beaker and then transfer 20 mL of your chosen base component of your buffer into the same 150 mL beaker. Stir this mixture well using a glass stir rod. Record on your data chart the buffer system you have selected.
2. Calibrate a pH meter as directed by your instructor. Rinse the pH electrode well with laboratory water prior to any measurements. ***Do not allow the tip of the pH electrode to touch any solid object—it is very fragile!***
3. Measure the pH of your buffer solution and record it in the data table below.
4. Fill a clean beaker with 30 mL laboratory water. Measure and record the pH of pure water.
5. Pour the 30 mL laboratory water into your buffer solution. Stir well. Measure and record the pH of your diluted buffer.
6. Divide your diluted buffer solution into two portions in separate beakers.
7. Add 1-2 mL 0.1 M HCl to one of the portions, and 3 mL 0.1 M NaOH to the other portion. Stir each solution well, then measure and record the pH of each solution.
8. Dispose of your buffer solutions, then clean and rinse the beakers well.
9. Fill a clean beaker with 30 mL laboratory water, then add 1-2 mL 0.1 M HCl. Measure and record the pH.
10. Fill a clean beaker with 30 mL laboratory water, then add 3 mL 0.1 M NaOH. Measure and record the pH.
11. In a clean beaker prepare a new buffer of the same system; however, this time use 2 mL of your selected acid and 20 mL of your selected base. Measure and record the pH of this 2:20 buffer solution.
12. Add 3 mL 0.1 M NaOH to the 2:20 buffer solution. Measure and record the pH.

5.3 Designing and Preparing a Buffer of a Specific pH:

1. You will be assigned a specific pH by your instructor which will not be the same as any of your classmates. Record the pH you have been assigned (your “target pH”). Determine which buffer system available would be appropriate to design a buffer of the pH you were assigned. Check

with your instructor to make sure you've selected the correct system for your target pH before proceeding further.

2. Showing your work in 7.0 DATA ANALYSIS, calculate the volume of each stock solution you need to prepare 30 – 50 mL buffer solution of your target pH. (Hint, choose either acid or base to begin with 20 mL, and determine the volume you will need of the other solution to have the correct base/acid ratio for your target pH).
3. Prepare the buffer solution, then measure and record the pH.
4. Allow for some small experimental error between your target pH and your measured pH. However, if the experimental error is large, you should recalculate and prepare another buffer solution.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name(s)**	**Date**

Determination of pH using Acid-Base Indicators

Record the colors of each solution tested. Then after lab ends, use these colors and Table 1 to estimate the pH range of each solution (for example, pH = 3 - 4).

Indicator	**0.1 M HCl**	**0.1 M NaH_2PO_4**	**0.1 M CH_3COOH**	**0.1 M $ZnSO_4$**
Methyl red				
Thymol blue				
Methyl orange				
Congo red				
Phenolphthalein				
Bromocresol green				
Bromothymol blue				
Universal indicator (optional)				
pH range =				

Determination of K_a

List the buffer system you chose:

Acid component = ______________________

Base component = ______________________

Record your pH measurements in the Table below. After lab ends, use your data and the Henderson-Hasselbalch equation as discussed in 1.1 BACKGROUND to calculate the pKa and the Ka, showing your work in 7.0 DATA ANALYSIS.

	Buffer System		**Pure Water**
	Acid/Base = 1:1	Acid/Base = 2:20	(pure water)
Observed pH			
pH after dilution with water		N/A	N/A
pH after addition of HCl		N/A	
pH after addition of NaOH			
Calculated pK_a			N/A
Calculated K_a			N/A

7.0 DATA ANALYSIS

7.1 Determination of pH using Acid-Base Indicators

1. Choose one of the four solutions (0.1 M HCl, 0.1 M NaH_2PO_4, 0.1 M CH_3COOH, or 0.1 M $ZnSO_4$), describe the most helpful observations of that solution with the indicators, and explain how you used Table 1 and the colors of the indicators to determine the pH range for your chosen solution.

2. Consider your results for the solutions of 0.1 M HCl and 0.1 M CH_3COOH. Notice that the concentrations are the same. Which has the lower pH and why is its pH lower?

3. Consider your results for the 0.1 M NaH_2PO_4 solution.
1. Is the solution acidic or basic?
2. Which ion, Na^+ or $H_2PO_4^-$, is causing the observed acidity or basicity?
3. Write the net ionic equation below that shows why this ion is acidic or basic.

4. Consider your results for the 0.1 M $ZnSO_4$ solution.
1. Is the solution acidic or basic?
2. Which ion, Zn^{2+} or SO_4^{2-}, is causing the observed acidity or basicity?
3. Write the net ionic equation below that shows why this ion is acidic or basic.

2. Determination of K_a
1. Show your work for the calculations of pK_a and K_a (see 1.1 BACKGROUND). Record your answers in the Table found in 6.0 DATA RECORDING SHEET.

2. Compare your experimental value of pK_a with the literature value of the pK_a of the weak acid. Propose reasons for any differences.

3. Consider the addition of 3 mL of NaOH to the 1:1 buffer and to the 2:20 buffer. Did the pH change by the same amount when NaOH was added to each buffer? Propose a reason for why or why not the pH changes were the same.

4. Consider the addition of 30 mL of pure water to the 1:1 buffer. Did the pH of the buffer change upon dilution? Propose a reason for why or why not.

5. Consider the addition of HCl to the 1:1 buffer and to pure water. Was the pH different? Propose a reason for why or why not. Include chemical reaction(s) in your explanation.

7.3 Designing and Preparing a Buffer of a Specific pH

Target pH _________________

List the buffer system you chose:

Acid component = ____________________

Base component = ____________________

pK_a of acid component (look it up) = ____________________

1. Show your work and calculate the volume of each stock solution you need to prepare 30 – 50 mL buffer solution of your target pH. (Hint, choose either acid or base to begin with 20 mL, and determine the volume you will need of the other solution to have the correct base/acid ratio for your target pH. See 1.1 BACKGROUND for more info).

Volume of Acid component = ____________________

Volume of Base component = ____________________

Measured pH of your buffer solution _________________

Percent error = (measured pH – target pH)/target pH x 100 = _________________

2. Discuss your results and propose reasons for your percent error.

2608 Evaluating Antacids

1.

INTRODUCTION

A titration is a procedure for determining the concentration of a solution (the *analyte*) by allowing a carefully measured volume of this solution to react with another solution whose concentration is known (the *titrant*). The point in the titration where enough of the titrant has been added to react exactly with the analyte is called the *equivalence point*, and it occurs when moles of titrant equal moles of analyte according to the balanced equation between the analyte and titrant. There are many types of titrations. In this experiment, you will be performing an acid-base titration.

Actually, today you will be performing a *back-titration*. Back titration is also titration. It is called back-titration because it is not carried out with the solution whose concentration is required to be known (analyte) as in the case of normal or forward titration, but with the excess volume of reactant which has been left over after completing reaction with the analyte. Back-titration works in the following manner.

In today's lab you will analyze antacids which are common medications used to neutralize stomach acid. Antacids are weak bases, and the goal is to measure how much acid can be neutralized by a particular antacid (in other words, you will determine the number of moles of H^+ neutralized per gram of each antacid.) But because antacids are insoluble in water, it is problematic to perform a normal titration. Instead, you will react the antacids with a known amount of HCl such that the products of this reaction will be soluble in water. After the reaction, the solution will contain left over HCl, and you will perform a titration (a *back-titration*) to calculate how much HCl is left over. The difference between the total amount of HCl and the amount of left over HCl (measured by the back-titration) will yield the amount of HCl required to neutralize the antacid.

To obtain accurate results, the titration must be stopped precisely at the equivalence point. If too much titrant is added accidentally during a normal titration, the results usually are discarded and the titration must be started over. However, during a *back-titration*, it is possible to rescue a bad titration by adding more HCl to the failed titration, resuming the titration, and stopping at the new equivalence point. Afterwards, simply recalculate the total amount of HCl used, and the difference between the total amount of HCl and the amount of HCl for the back-titrations will yield the amount of HCl required to neutralize the antacid. Please realize that a second back-titration is only necessary if too much titrant (NaOH) is added during the first back-titration and the equivalence point is exceeded.

1.1 OBJECTIVES

After completing this experiment, the student will be able to:

- Determine the number of moles of H^+ neutralized per gram of each antacid.
- Calculate the cost effectiveness of each antacid.

>>> Get cost data from the instructor during lab!! <<<

1.2 BACKGROUND

Acid indigestion is a common ailment caused by the overproduction of stomach acid, HCl. Over-the-counter antacids provide some relief from the symptoms of acid indigestion. They are generally made up of some mixture of weak bases such as $Mg(OH)_2$, $Al(OH)_3$, and $CaCO_3$ that can react with HCl as shown in these net ionic equations:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l) \quad \text{(Equation 1)}$$

$$2\,H^+(aq) + CO_3^{2-}(aq) \rightarrow H_2O(l) + CO_2(g) \quad \text{(Equation 2)}$$

In this experiment, the method of titration will be used to determine the number of moles of H^+ neutralized per gram of antacid. In the "back-titration," a portion of antacid will be mixed with an excess of HCl. The H^+ that has not reacted with the antacid is then titrated with standardized NaOH in the presence of the indicator bromophenol blue to a blue end point.

The *end point* is defined as the volume of OH^- needed to see a color change. Because only the tiniest excess of OH^- over H^+ can cause the color change of an indicator, the end point is a close approximation of the *equivalence point*. (The difference between the end point and the equivalence point is known as the *titration error*.) At the equivalence point, the number of moles of OH^- added is equal to the number of moles of excess H^+ that had not been neutralized by the antacid. By knowing the total moles of HCl added, one can then calculate the number of moles of H^+ neutralized by the antacid.

Total moles of H^+= moles of H^+ neutralized by antacid + moles of H^+ neutralized by NaOH

Because the antacid may include both OH^- and CO_3^{2-}, it is not possible to calculate the number of moles of each of these ion species independently. Instead, the number of H^+ neutralized by the antacid is found. The amount of antacid required to neutralize one mole of H^+ neutralized is said to be one "*equivalent*."

Total equivalent of antacid = Total moles of H^+ neutralized

The more cost-effective antacid is the one that costs fewer dollars per equivalent.

References and further reading

Technique G: Buret Use

2. SAFETY PROCEDURES AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Antacid tablets *(Preferably tablets that are white and without added flavor)*		2 tablets (each a different brand)	1 tablet from one brand, and 1 tablet from a different brand
HCl solution, standardized	About 0.1 M	100 mL	
NaOH solution, standardized	About 0.1 M	100 mL	
Bromothymol blue solution		6 - 8 drops	

4.0 GLASSWARE AND APPARATUS

Mortar and pestle	2 burets (each 50 mL)
125 mL or 250 mL Erlenmeyer flasks	Buret clamp
100 mL or 250 mL beakers	Ring stand
Bunsen burner	Funnel (to fill buret)
Wire gauze	

5.0 PROCEDURE

5.1 Preparation of Antacid Sample

1. Choose an antacid and record its name on the data sheet.
2. With a mortar and pestle, crush one tablet of antacid to as fine a powder as possible.
3. Weigh out between 0.3 – 0.4 g of the powdered antacid into a pre-weighed Erlenmeyer flask. Record the mass of the antacid sample to 0.0001g.

5.2 Preparation of Burets

1. Obtain a buret, label it “NaOH” and make sure that it is clean and does not leak. If necessary, clean the buret with a buret brush and soapy water and rinse with laboratory water.
2. Rinse the buret several times with a few milliliters of your NaOH solution (transfer the NaOH to the buret from a beaker using a funnel to minimize spillage), making sure that the stopcock and buret tip are also thoroughly rinsed with NaOH. (Why rinse with NaOH? To remove water that would otherwise dilute and change the concentration of your NaOH solution!)
3. Fill the buret with your NaOH solution making sure that there are no air bubbles and no leaks in the stopcock or tip. Record the initial volume of the buret (reading the meniscus to 2 decimal places).

4. Obtain a second buret, label it “HCl” and make sure that it is clean and does not leak. If necessary, clean the buret with a buret brush and soapy water and rinse with laboratory water.
5. Rinse the buret several times with a few milliliters of your HCl solution, and fill the buret with your HCl solution making sure that there are no air bubbles and no leaks in the stopcock or tip. Record the initial volume of the buret (reading the meniscus to 2 decimal places).
6. Record the concentrations of NaOH and HCl on your data sheet.
7. The buret does not need to be cleaned or rinsed between titrations. Simply refill the buret with the same solution prior to beginning a new titration. Be sure to record the *new* initial volume of the buret (reading the meniscus to 2 decimal places).

5.3 Addition of Excess HCl to the Antacid

1. Using the buret, add approximately 40 mL of standardized HCl to the prepared antacid sample. Record the final volume of HCl in the buret to 0.01 mL.
2. In order to dissolve as much as of the antacid as possible and to drive off as much dissolved CO_2 as possible, gently boil the mixture of antacid + HCl for about two minutes.
3. Cool the mixture to room temperature.
4. There may be a significant amount of substances that are not dissolved in your mixture. Because the active ingredients of an antacid are quite water soluble, the solids will not affect the results. (The solids are a mixture of inactive ingredients such as the coating and some binding compounds).
5. Add 6 - 8 drops of bromophenol blue indicator solution. Swirl to mix.
6. Check that the solution is yellow because there should be excess H^+ in solution. If the solution is green, blue, or a mixture of both green and blue, then there is excess base (antacid) in solution and more HCl must be added. *Helpful tip: In a separate container, mix about 25 water with about 4 mL HCl and add 2-3 drops of bromophenol blue indicator solution. This solution will be yellow, and you can use it for comparison with your antacid solution.*
7. If more HCl must be added, add about 20 ml of HCl from the buret, again recording to 0.01 mL. Make sure there is enough HCl in the buret. Remember, it is not possible to measure the volume of HCl if the level drops below the 50.00 mL mark!
8. Assess the color of the solution. If it is not yellow, repeat the addition of HCl until it is.

5.4 Titration of Excess HCl (Back-titration)

1. Begin the titration by carefully opening and closing the buret stopcock to allow the NaOH solution to drain into the antacid solution while swirling the flask. Stop the flow of NaOH frequently to assess the titration by observing its color.
2. You may wash the sides of the flask with your squirt bottle containing laboratory water during the titration.
3. As the endpoint approaches, the indicator will change color from yellow to green and finally to blue. However, the solution may change from blue back to green as you swirl the flask. Slowly continue the titration if the blue does not persist.
4. As the endpoint gets closer, add NaOH one drop at a time while swirling the reaction mixture well before adding another drop. Stop the addition of NaOH as soon as one drop causes the solution to change permanently to blue (about 30 seconds) — this is the endpoint!

5. Record the final volume of the buret (reading the meniscus to 2 decimal places).

Tip: Identifying the endpoint is not easy. When you suspect that you have reached the endpoint, proceed to record the final buret volume. Then, add one more drop of NaOH and assess the titration. If the color does not change or if the color becomes darker blue, then the endpoint was reached before that drop was added and your recorded volume was correct. But if the color changes to blue, then the true endpoint has now been reached—cross out the previously recorded buret volume and write in the new final volume of the buret. You may repeat this process if you are still uncertain of the endpoint.

5.5 Second Back-titration – only necessary if too much NaOH was added

1. If your solution is intensely blue – you've added too much NaOH! Don't panic, this error can be corrected.
2. If too much NaOH was added, add enough of the standardized HCl to make the solution yellow. (About 5 - 8 mL) Record the new total volume of HCl.
3. Again from the buret, add NaOH slowly (as described earlier) until the solution turns blue.
4. Record the final volume of the buret (reading the meniscus to 2 decimal places).

Repeat the procedure with another brand of antacid. Make every effort to reach an endpoint that is equivalent in its "blueness."

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

Mass of Antacid

1. Antacid name	
2. Mass of empty flask	
3. Mass of flask + antacid	
4. Mass of antacid	

Addition of HCl

5. Molarity of HCl	
6. Initial buret reading	
7. Final buret reading	
8. Volume of HCl added	
9. Moles of HCl added	

Addition of more HCl – only if needed in Section 5.3
(i.e. if solution is blue after addition of bromophenol blue)

10. Initial buret reading	
11. Final buret reading	
12. Volume of extra HCl added	
13. Moles of extra HCl added	

Second Back Titration with more HCl – only if needed in Section 5.5
(i.e. if you pass the endpoint when titrating with NaOH and
need to add more HCl.)

14. Initial buret reading	
15. Final buret reading	
16. Volume of extra HCl added	
17. Moles of extra HCl added	
18. Total moles of HCl added	

Titration with NaOH

19. Molarity of NaOH	
20. Initial buret reading	
21. Final buret reading	
22. Volume of NaOH added	
23. Total moles of NaOH added to neutralize excess HCl	

*Additional Data – to be provided by your instructor (*see note on next page)*

24. Cost of antacid per package	
25. Mass of antacid per package	
26. Cost of antacid per gram	

*Note regarding the *"Additional Data – to be provided by your instructor"*

During lab, be sure to obtain from your instructor the "Cost of antacid per package" and "Mass of antacid per package" for each brand of antacid that you analyzed. If necessary, contact your instructor or classmates after lab if this data was not obtained earlier. If you do not have enough time to obtain this data from the instructor or classmates, then it is suggested that you conduct an internet search at a website such as Amazon, CVS, Walgreens, or other online store in order to find prices and weights to obtain the "Cost of antacid per package" and "Mass of antacid per package" for each brand of antacid that you analyzed. Cite the website and the data taken from the website. It is at the discretion of the instructor to accept (or not) this alternate data.

7.0 DATA ANALYSIS

Calculations

1. Moles of H+ neutralized by the antacid sample	
2. Equivalents of antacid in the sample analyzed	
3. Equivalents per gram of antacid (in eq/gram)	
4. Cost of antacid per equivalent (in $/eq)	

Below, show your work for each of these calculations for the first antacid that you analyzed.

1. Moles of H+ neutralized by the antacid sample
2. Equivalents of antacid in sample analyzed

3. Equivalents per gram of antacid (in eq/gram)

4. Cost of antacid per equivalent (in $/eq)

Conclusions

Which antacid is the better buy? Explain.

1.

INTRODUCTION

A titration is a procedure for determining the concentration of a solution (the *analyte*) by allowing a carefully measured volume of this solution to react with another solution whose concentration is known (the *titrant*). In this experiment, the analyte is NaOH, and the titrant is an acid called KHP (potassium hydrogen phthalate). The point in the titration where enough of the titrant has been added to react exactly with the analyte is called the *equivalence point*, and it occurs when moles of titrant equal moles of analyte according to the balanced equation between the analyte and titrant. There are many types of titrations. In this experiment, you will be performing an acid-base titration.

1.1 Objectives

After completing this experiment, the student will be able to:

- determine the concentration of an NaOH solution using data from titrations involving (visual) indicators and a pH meter.
- determine which indicator(s) provides the best titration data.

1.2 Background

In general, an acid, HA, and a base such as sodium hydroxide react to produce a salt and water by transferring a proton (H^+):

$$HA(aq) + NaOH(aq) \rightarrow NaA(aq) + H_2O(l) \quad \text{(Equation 1)}$$

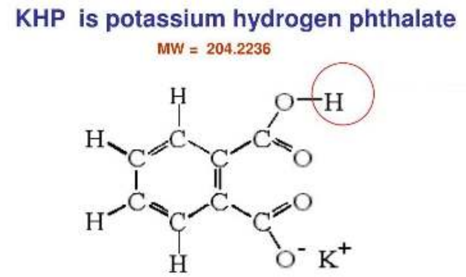
Because sodium hydroxide is hygroscopic, it draws water from its surroundings. This means one cannot simply weigh out a sample of sodium hydroxide, dissolve it in water, and determine the number of moles of sodium hydroxide present using the mass recorded, since any sample of sodium hydroxide is likely to be a mixture of sodium hydroxide and water. Thus, the most common way to determine the concentration

of any sodium hydroxide solution is by titration. Determining the precise concentration of NaOH using a primary standard is called standardization.

To find the precise concentration of the NaOH, it must be titrated against a primary standard, an acid that dissolves completely in water, has a high molar mass, that remains pure upon standing, and is not hygroscopic (tending to attract water from the air).

In this experiment you will be given the acid named potassium hydrogen phthalate (KHP). *Warning: the "P" in KHP is phthalate, not phosphorus!* This acid is available as a very pure solid, and therefore, it is very convenient for use in titrations because the number of moles of KHP can be accurately calculated from careful measurement of its weight.

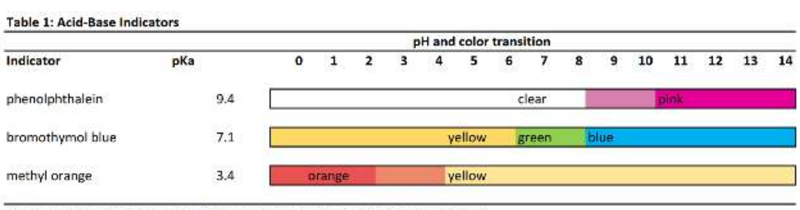
The structure of KHP (the acidic hydrogen is circled).



After preparing your own dilute solution of NaOH, you will use it to perform several titrations with KHP. Although the dilution equation, $M_1V_1 = M_2V_2$, can approximate the concentration (molarity) of your NaOH from your original preparation (it will be approximate because the concentration of NaOH stock solution has only one or two significant figures), one goal of this experiment is to determine more precisely the concentration of your NaOH solution using an appropriate set of data obtained from titrations using three different acid-base (visual) indicator solutions and a pH meter.

Visual indicators change color over a relatively narrow pH range, known as the **endpoint**. When using a visual pH indicator, it is important to match the endpoint of the indicator with the expected pH of the equivalence point of the titration being observed. Recall from the 'Introduction' (Part 1.0) that the equivalence point is the point in a titration when the moles of acid and base present in the reaction match the stoichiometry of an appropriately balanced chemical equation. The endpoint is the pH when the visual indicator changes color. *Not all of the indicators in this experiment have endpoints that match the equivalence point!*

The following three indicators will be used in this experiment. The color changes to expect when going from an acidic to a basic solution (i.e., increasing pH) are:



When using a **pH meter**, during a titration, the pH of the analyte solution will be recorded after each increment of a specific volume of the titrant solution is added. After which, a graph of the pH of the analyte solution vs the volume of NaOH added will be created. This is called a **titration curve**. (Figure 1) To find the exact volume of NaOH needed to reach the equivalence point, a tangent line that touches the steepest part of the titration curve is drawn. The equivalence point is exactly halfway between the two points where the titration curve deviates from the tangent line.

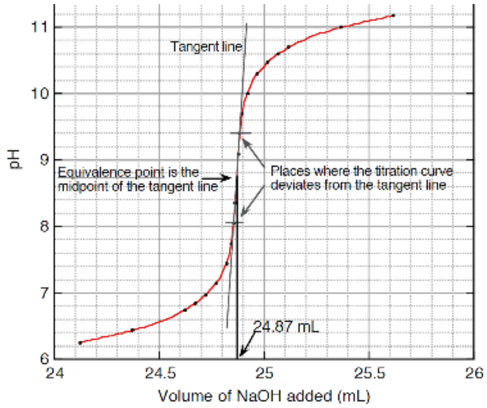


Figure 1. Titration Curve: pH of analyte vs volume of NaOH added during titration. *(Warning: this graph is for illustration purposes only. Your actual data may differ.)*

References and further reading

Technique G: Buret Use

Experiment 2501 Using Excel for Graphical Analysis of Data of the laboratory manual

2. SAFETY PROCEDURES AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	**Concentration**	**Approximate Amount**	**Notes**
Potassium hydrogen phthalate (KHP)	Pure	2.5 g	
NaOH solution	6 M	10 mL	
Phenolphthalein solution		Several drops	
Bromothymol blue solution		Several drops	
Methyl red solution		Several drops	

4.0 GLASSWARE AND APPARATUS

pH meter (fragile!)	50 mL buret
125 mL or 250 mL Erlenmeyer flask	Buret clamp
10 mL graduated cylinder	Ring stand
600 mL beaker	Funnel (to fill buret)

5.0 PROCEDURE

5.1 PREPARATION OF DILUTE NaOH SOLUTION

Into a clean 600 mL beaker (or another container), add approximately 500 mL of laboratory water. Then add about 10 mL of 6M NaOH stock solution and mix thoroughly. Note that it is not necessary to accurately measure volumes at this time because the final concentration of your dilute NaOH solution will be calculated from titration data. Label and cover the beaker with a watch glass.

Be careful not to contaminate your dilute NaOH solution at any time during the lab as this may change its concentration.

5.2.0 TITRATION OVERVIEW WITH HELPFUL TIPS

In this experiment, you will collect six (6) sets of titration data:

- two (2) titrations using phenolphthalein
- two (2) titrations using bromothymol blue
- one (1) titration using methyl red
- one (1) titration using a pH meter

Each titration requires a different amount of KHP; however, the choice of indicator and/or use of the pH meter can be in any order. Lab partners should alternate so that every student has an opportunity to use the buret to complete at least one titration.

Tip #1: To save time, you may use any one indicator and the pH meter at the same time to collect both sets of data simultaneously. Discuss this option with your instructor.

Tip #2: If pH meters are in short supply, some lab groups should use a pH meter at the beginning of the experiment whereas other groups should plan to use a pH meter at a later time.

Tip #3: To speed up your titrations, initially add large amounts (1-2 mL at a time) of solution from the buret. Then, slow down when you are 2-3 mL before the equivalence point and add solution from the buret dropwise. Estimate the equivalence point of each titration by using the balanced chemical equation, the mass of KHP, and an estimate of the concentration of your NaOH solution obtained from the dilution equation (see Part 7.0 DATA ANALYSIS *below for a step-by-step process). Note: Different amounts of KHP are used for each titration which means the equivalence point will be different for each titration!*

Warning: Do not share titration data with other lab groups because each group has an NaOH solution that is slightly different from everyone else. Remember that the goal is to accurately determine the *concentration of NaOH solution in your group!*

5.2.1 TITRATION PROCEDURE WITH INDICATORS

1. Obtain a buret and make sure that it is clean and does not leak. If necessary, clean the buret with a buret brush and soapy water and rinse with laboratory water. Then, rinse the buret several times with a few milliliters of your NaOH solution, making sure that the stopcock and buret tip are also thoroughly rinsed with NaOH. (Why rinse with NaOH? To remove water that would otherwise dilute and change the concentration of your NaOH solution!) Fill the buret with your NaOH solution making sure that there are no air bubbles and no leaks in the stopcock or tip. Record the initial volume of the buret (reading the meniscus to 2 decimal places).
2. The buret does not need to be cleaned or rinsed between titrations. Simply refill the buret with NaOH prior to beginning a new titration. Be sure to record the *new* initial volume of the buret (reading the meniscus to 2 decimal places).
3. Clean a 125 mL or 250 mL Erlenmeyer flask and rinse well with laboratory water. The flask may remain wet.
4. On a weighing paper, weigh about 0.5 g KHP and record the exact mass in the column of the data table for Phenolphthalein #1. Completely transfer the KHP to the flask, and then add approximately 50 mL of laboratory water and 2-3 drops of the phenolphthalein indicator. Swirl to mix thoroughly and dissolve the KHP completely.
5. (Optional) Obtain an estimate of the equivalence point by completing the data table in 7.0 DATA ANALYSIS below. If you know where the endpoint is, you will have a better chance of not passing it during the titration. *This step (Step 5) is optional if you want to devote more lab time to careful titrations. If you choose to skip this step now, be sure to complete these calculations after the lab in order to earn all points on the lab write-up.*
6. Begin the titration by carefully opening and closing the buret stopcock to allow the NaOH solution to drain into the KHP/phenolphthalein solution while swirling the flask. Stop the flow of NaOH frequently to assess the titration by observing its color. You may wash the sides of the flask with your squirt bottle containing laboratory water during the titration. As the endpoint approaches, the indicator may change color from colorless to light pink initially as NaOH is added. However, the solution may change back to colorless as you swirl the flask. Slowly continue the titration if the new color does not persist. *Tip: place a sheet of white paper under the titration flask to help you detect the faintest of pink color.* As the endpoint gets closer, add NaOH one drop at a time while swirling the reaction mixture well before adding another drop. Stop the addition of NaOH as soon as one drop causes the solution to change permanently to the new color (about 30 seconds) — this is the endpoint! Record the final volume of the buret (reading the meniscus to 2 decimal places).

Tip: Identifying the endpoint is not easy. When you suspect that you have reached the endpoint, proceed to record the final buret volume. Then, add one more drop of NaOH and assess the titration. If the color does not change or if the color becomes darker without changing hue, then the endpoint was reached before that drop was added and your recorded volume was correct. But if the color changes, then the true endpoint has now been reached—cross out the previously recorded buret volume and write in the new final volume of the buret. You may repeat this process if you are still uncertain of the endpoint.

7. Repeat the titration procedure (steps 2-6) to acquire all sets of titration data with the other indicators, using different quantities of KHP as follows:

Phenolphthalein #2: Use about 0.7 g KHP (record exact amount) and 2-3 drops of phenolphthalein. The color change for this indicator is colorless → pink. Stop the titration when the solution turns very light pink.

Bromothymol blue #1: Use about 0.3 g KHP (record exact amount) and 2-3 drops of bromothymol blue. The color change for this indicator is yellow → green → blue. Stop the titration when the solution turns green.

Bromothymol blue #2: Use about 0.6 g KHP (record exact amount) and 2-3 drops of bromothymol blue. The color change for this indicator is yellow → green → blue. Stop the titration when the solution turns green.

Methyl red: Use about 0.4 g KHP (record exact amount) and 2-3 drops of methyl red. The color change for this indicator is red → orange → yellow. Stop the titration when the solution turns orange.

Note: If you think that you overshot the endpoint of any titration, you may repeat that titration at the end of the lab, if time permits.

5.2.2 TITRATION PROCEDURE WITH pH METER

Warning: The pH meter probe (end or tip) is extremely fragile! Do not allow the tip to touch the flask or other solid object.

Complete the titration as described in Part 5.2.1 steps 2-6 with the following modifications:

1. Use a pH meter instead of an indicator (or in addition to an indicator, as mentioned in Part 5.2.0)
2. Use an Erlenmeyer flask with an opening large enough for both the pH meter and the buret tip.
3. Weigh about 0.4 g KHP and record the exact mass.
4. The procedure below describes how to use a portable Flinn pH meter (model AP8673). If the pH meter is not measuring properly (unstable, or inaccurate), refer to the manual for troubleshooting. This same procedure may be applied to other brands of pH meter.
    1. Check out a portable pH meter from the stockroom.
    2. Remove the protective cap on the electrode. Clean any salt build-up off by rinsing with laboratory water.
    3. Press the ON/OFF button once.
    4. Rinse the electrode with laboratory water and blot dry with filter paper.
    5. Immerse the electrode in the flask containing the analyte solution. Once the display stabilizes (approx. 1 min.), record the exact pH.
    6. Remove the pH meter from the solution.
    7. Repeat steps (d) – (g) after a certain volume of the titrant is added.
5. During the titration, record both the buret volume and the pH measurement periodically: every 1 mL increment when the pH is lower than 8.0, every 1-2 drops when the pH is between 8 and 10, and every 1 mL increment when the pH is higher than 10.
6. Continue the titration after the endpoint, recording both the buret volume and the pH measurement in 1 mL increments until the pH is higher than 12.5 or the NaOH solution in the

buret reaches the 50 mL mark.

7. When finished, rinse the pH meter electrode with laboratory water and blot dry with filter paper. Replace cap and return the pH meter to the stockroom.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name(s)**	**Date**

Table 1. Titrations with Indicators

	Phenolphthalein #1	Phenolphthalein #2	Bromothymol blue #1	Bromothymol blue #2	Methyl Red
Mass of weighing paper					
Mass of KHP + weighing paper					
Mass of KHP					
Moles of KHP					
Initial buret reading					
Final buret reading					
Total NaOH volume					

Table 2. Titration with pH Meter

Mass of weighing paper	
Mass of KHP + weighing paper	
Mass of KHP	
Moles of KHP	

Volume of NaOH added (Buret reading)	pH reading

Volume of NaOH added (Buret reading)	pH reading

1. DATA ANALYSIS

1. Estimate the titration equivalence point by completing the following data table using the balanced chemical equation, the mass of KHP, and an estimate of the concentration of your

NaOH solution obtained from the dilution equation (see Parts 1.0 Introduction and 1.2 Background).

	Phenolphthalein #1
Mass of KHP	
Moles of KHP (molar mass = 204.2236 g/mol)	
Moles of NaOH required to react in a 1:1 ratio with KHP	
Approximate concentration of your NaOH solution *Use the dilution equation $M_1V_1 = M_2V_2$ and find the unknown M_2 when M_1 = 6 M (or use the concentration written on the bottle of 6 M NaOH), V_1 = 10 mL, and V_2 = 500 mL.*	
Total NaOH volume in liters required for titration (also the estimate for the equivalence point). *Divide* **Moles of NaOH** *by the* **Approximate concentration of your NaOH solution**	
Convert liters to mL by multiplying **total NaOH volume in liters** by 1000	

2. Using a spreadsheet program such as Microsoft Excel or Google Sheets, construct a graph of pH (y-axis) vs. volume of NaOH added in mL (x-axis).
- Attach the graph to your lab report.
- Determine the equivalence point (in mL of NaOH) of this pH titration.
3. Using a spreadsheet program such as Microsoft Excel or Google Sheets, construct a graph of volume of NaOH solution in mL at the endpoint of each titration (y-axis) vs. mass of KHP for each corresponding titration (x-axis).
- Attach the graph to your lab report.
- Determine the line of best fit (linear regression line) and in the space below, write the equation of this line in the form y = mx + b.

4. In the space below, describe how well your data points match the line of best fit (from #3 above). Do one or more points appear to be outliers? For these outliers, propose a reason why they do not match the line of best fit: experimental error/inaccurate use of the buret or perhaps the indicator's endpoint poorly matched the titration equivalence point?

5. What are the advantages and disadvantages of using a pH meter instead of a visual indicator?

6. Calculate the concentration of your NaOH solution using each set of titration data.

- Fill in the following table with your results.
- For one set of titration data, show the calculations and/or explain how you obtained each of the values you entered in that row.
- Comment on the quality of all your titrations: Do they all give reliable results, or should one or more titrations not be included in the average concentration of NaOH? Explain.

Table 2: NaOH Concentration Calculation

Titration data from:	Moles of KHP	Moles of H^+	Moles of NaOH	Liters of NaOH solution	Concentration of NaOH
Phenolphthalein #1					
Phenolphthalein #2					
Bromothymol blue #1					
Bromothymol blue #2					
Methyl Red					
pH meter					

Table note: Remember, KHP is the ***monoprotic*** *acid we are using to standardize the NaOH. Litres of NaOH should be the same in this table as total volume added of NaOH from Table 1.*

8.0 POST-LAB QUESTIONS

1. Draw the balanced chemical equation for the titration reaction using Lewis Dot structures (for example, instead of “NaOH” in your balanced chemical equation, draw “Na^+ $^-$O-H” and include 3 lone pairs of electrons around the oxygen atom in your drawing). *Do not include indicator molecules.*

2. What volume of 0.812 M HCl is required to titrate 1.33 g of KOH to the endpoint? Show your work (attach a separate sheet of paper, if necessary).

3. What volume of 1.346 M H_2SO_4 is required to titrate 1.54 g of KOH to the endpoint? Show your work, including the balanced chemical equation and a short explanation on how you used the balanced equation when calculating for the endpoint.

1.16: New Page

2610 Determination of a Formation Constant

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- determine the value of the equilibrium constant for formation of $FeSCN^{2+}$ by using the visible light absorption of the complex ion.
- confirm the stoichiometry of the reaction.

1.2 Background

While some reactions go to completion – and we can calculate the amount of reactants required, products produced, and limiting reagents –, most reactions do not behave this way. Instead, after mixing the reactants, reactions reach a state where a stable mixture of reactants and products is produced. This mixture is called the *equilibrium state*; at this point, chemical reactions occur in both directions at equal rates. Therefore, once the equilibrium state has been reached, no further change occurs in the concentrations of reactants and products.

We can still determine quantities of chemicals present and the equilibrium constant, K_{eq}, is used to do this. The expression for the equilibrium constant for a reaction is determined by examining the balanced chemical equation. For a reaction involving aqueous reactants and products, the equilibrium constant is expressed as a ratio between reactant and product concentrations, where each term is raised to the power of its reaction coefficient (Equation 1). The ratio chosen gives products over reactants rather than the alternative. When the quantities of reactants and products are given in molar concentrations, the equilibrium constant is referred to as K_C. The value of this constant at equilibrium is always the same, regardless of the initial reactant and product concentrations, at a given temperature. Equilibrium constants change with temperature, but at a given temperature, whether the reactants are mixed in their exact stoichiometric ratios, or one reactant is initially present in large excess, the ratio described by the equilibrium constant expression will be achieved once the reaction composition stops changing.

$$aA(aq) + bB(aq) \rightleftharpoons cC(aq) + dD(aq) \quad K_C = \frac{[C]^c[D]^d}{[A]^a[B]^b} \qquad \text{(Equation 1)}$$

The reaction to be studied involves the formation of the reddish orange iron(III) thiocyanate complex ion, $Fe(H_2O)_5SCN^{2+}$. The actual reaction involves the displacement of a water ligand by thiocyanate ligand, SCN^-.

$$Fe(H_2O)_6^{3+}(aq) + SCN^-(aq) \rightleftharpoons Fe(H_2O)_5SCN^{2+}(aq) + H_2O(l)$$

For simplicity, and because water ligands do not change the net charge of the species, water can be omitted from the formulas of $Fe(H_2O)_6^{3+}$ and $Fe(H_2O)_5SCN^{2+}$. Thus, $Fe(H_2O)_6^{3+}$ is usually written as Fe^{3+} and $Fe(H_2O)_5SCN^{2+}$ is written as $FeSCN^{2+}$. Also, because the concentration of liquid water is essentially unchanged in aqueous solution, we can write a simpler expression for K_C that expresses the equilibrium condition only in terms of species with variable concentrations.

$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$ $K_c = \frac{[FeSCN^{2+}]}{[Fe^{3+}][SCN^{-}]}$ (Equation 2)

In this experiment, students will prepare several different aqueous mixtures of Fe^{3+} and SCN^{-}. Since the reaction reaches equilibrium nearly instantly, these mixtures turn reddish orange very quickly due to the formation of the product, $FeSCN^{2+}(aq)$. The intensity of the color of the mixtures is proportional to the concentration of the product formed at equilibrium. As long as all mixtures are measured at the same temperature, the ratio described in Equation 2 will be the same.

1.2 Measurement of [$FeSCN^{2+}$] Concentration

Since the complex ion product, $FeSCN^{2+}$, is the only strongly colored species in the system, its concentration can be determined by measuring the intensity of the orange color in equilibrium systems of the Fe^{3+}, SCN^{-}, and $FeSCN^{2+}$ ions. You will be using a spectrophotometer and applying Beer's Law (Equation 3) to do this. The absorbance, A, is directly proportional to two parameters: path length, b (the length of the sample through which the light travels) and C (the molar concentration of the complex ion). Molar absorptivity ε, is a constant that expresses the absorbing ability of a chemical species at a certain wavelength. The absorbance, A, is roughly correlated with the color intensity observed visually; the more intense the color, the higher the absorbance.

$A = \varepsilon bC$ (Equation 3)

Solutions containing $FeSCN^{2+}$ are placed into the spectrophotometer and their absorbances at 447 nm are measured. In this method, the path length, b, is the same for all measurements. The value of the combined constants, εb, is determined by plotting the absorbances, A, vs. molar concentrations, C, for several solutions with known concentration of $FeSCN^{2+}$. The slope of this calibration curve is equal to εb and is then used to find unknown concentrations of $FeSCN^{2+}$ from their measured absorbances.

1.3 Calculations

In order to determine the value of K_C, the equilibrium values of [Fe^{3+}], [SCN^{-}], and [$FeSCN^{2+}$] must be known. The equilibrium value of [$FeSCN^{2+}$] is determined spectroscopically. Its initial value will be zero, since no $FeSCN^{2+}$ will be added to the solution.

The equilibrium values of [Fe^{3+}] and [SCN^{-}] can be determined from a reaction table (ICE table) as shown in Table 1. The initial concentrations of the reactants—that is, [Fe^{3+}] and [SCN^{-}] prior to any reaction—can be found by a dilution calculation based on the values from Table 2 found in the procedure. Once the reaction reaches equilibrium, we assume that the reaction has shifted forward by an amount, *x*. The equilibrium concentrations of the reactants, Fe^{3+} and $SCN^{–}$, are found by subtracting the equilibrium [$FeSCN^{2+}$] from the initial values. Once all the equilibrium values are known, they can be applied to Equation 2 to determine the value of K_C.

Table 1: ICE Table for Determination of Equilibrium Concentrations

	[Fe^{+3}]	**[SCN^{-}]**	**[$FeSCN^{2+}$]**
Initial	$[Fe^{+3}]_0$ (after mixing)	$[SCN^{-}]_0$ (after mixing)	0
Change	- *x*	- *x*	+ *x*
Equilibrium	$[Fe^{+3}]_0 – x =$ $[Fe^{+3}]_0 – [FeSCN^{2+}]_S$	$[SCN^{-}] – x =$ $[SCN-] – [FeSCN^{2+}]_S$	$x = [FeSCN^{2+}]_S$

$[FeSCN^{2+}]_S$ measured spectrophotometrically = $A\varepsilon b$

1.4 Standard Solutions of $FeSCN^{2+}$

In order to find the equilibrium [$FeSCN^{2+}$], this experiment requires the preparation of standard solutions with known [$FeSCN^{2+}$]. These are prepared by mixing a small amount of dilute KSCN solution with a more concentrated solution of $Fe(NO_3)_3$. The solution has an overwhelming excess of Fe^{3+}, driving the equilibrium position far towards products. As a result, the equilibrium [Fe^{3+}] is very high due to its large excess, and therefore the equilibrium [$SCN^{–}$] must be very small. In other words, we can assume that ~100% of the $SCN^{–}$ is reacted, meaning that $SCN^{–}$ is a limiting reactant resulting in the production of an equal amount of $FeSCN^{2+}$ product. The equilibrium [$FeSCN^{2+}$] can be approximated as the initial concentration of SCN^{-}, $[SCN^{–}]_0$.

Reference and further reading

Technique I: Use of Spectrophotometer of the laboratory manual

2.0 SAFETY PROCEDURES AND WASTE DISPOSAL

!!Wear your safety goggles!!

The iron(III) nitrate solutions contain nitric acid. Avoid contact with skin and eyes; wash hands frequently during the lab and wash hands and all glassware thoroughly after the experiment.

Collect all your solutions during the lab and dispose of them in the proper waste container.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Iron (III) nitrate	2.00×10^{-3} M in 1 M HNO_3	30-40 mL	
Iron (III) nitrate	0.200 M in 1 M HNO_3	25-30 mL	
KSCN	2.00×10^{-3} M	15 mL	
Laboratory water		25-30 mL	

4.0 GLASSWARE AND APPARATUS

Test tubes, large (1) and medium (5)	Spectrophotometer
Stirring rod	Cuvettes (2)
10-mL graduated pipet	5-mL graduated pipet
50-mL or 100-mL beakers (4)	

5.0 PROCEDURE

5.1 STOCK Solution, $FeSCN^{2+}$, PREPARATION FOR STANDARDIZATION

1. Prepare a stock solution with known concentration of $FeSCN^{2+}$. Obtain 15 mL of 0.200 M $Fe(NO_3)_3$ in 1 M HNO_3. (Note the actual concentration of this solution.) Rinse your graduated pipet with a few mL of this solution and add 10.00 mL of this solution into a clean and dry small beaker.
2. Using your graduated pipet, add 8.00 mL laboratory water to the small beaker.
3. Rinse your graduated pipet with a few mL of 2.00×10^{-3} M KSCN. Then add 2.00 mL of the KSCN solution to the small beaker.
4. Mix the solution with a clean and dry stirring rod until a uniform dark orange solution is obtained. A few dilutions of the stock solution will be used to prepare four standard solutions for your calibration curve. Calculate the concentrations of these diluted solutions and record on your data recording sheet. In addition, one 'blank' solution containing only $Fe(NO_3)_3$ will be used to zero the spectrophotometer (Table 2).

Table 2: Amount of Solutions Needed to Prepare Standards

Standard solutions	Composition of solution
Blank	0.200 M $Fe(NO_3)_3$ in 1 M HNO_3
1	stock solution
2	4.00 mL stock solution plus 1.00 mL water
3	4.00 mL stock solution plus 2.00 mL water
4	4.00 mL stock solution plus 3.00 mL water

5.2 PREPARATION OF THE EQUILIBRIUM MIXTURES

1. Label two clean and dry 50-mL beaker. Transfer 30-40 mL of 2.00 x 10^{-3} M $Fe(NO_3)_3$ in 1 M HNO_3 into one beaker and 25-30 mL of 2.00 x 10^{-3} M KSCN into the other beaker.
2. Label five clean and dry medium test tubes to be used for the five test mixtures you will make. Using your graduated pipet, add 5.00 mL of 2.00 x 10^{-3} M $Fe(NO_3)_3$ in 1 M HNO_3 solution into each of the five test tubes.
3. Using your graduated pipet, add the corresponding amount of KSCN solution to each of the labeled test tubes, according to Table 3 below.
4. Rinse your graduated pipet several times with laboratory water, and then use it to add the appropriate amount of laboratory water into each of the labeled test tubes.
5. Stir each solution thoroughly with your stirring rod until a uniform orange color is obtained. To avoid contaminating the solutions, rinse and dry your stirring rod between each solution preparation.

Table 3: Mixtures of Equilibrium Solutions

Sample	Amount of 2.00 x 10^{-3} M $Fe(NO_3)_3$ in 1 M HNO_3	Amount of 2.00 x 10^{-3} M KSCN	Amount of laboratory water
1	5.00 mL	5.00 mL	0.00 mL
2	5.00 mL	4.00 mL	1.00 mL
3	5.00 mL	3.00 mL	2.00 mL
4	5.00 mL	2.00 mL	3.00 mL
5	5.00 mL	1.00 mL	4.00 mL

5.3 SPECTROPHOTOMETRIC DETERMINATION OF [$FeSCN^{2+}$]

Review Technique I: Use of Spectrophotometer of the laboratory manual.

1. Fill a cuvette with the blank solution and carefully wipe off the outside with a tissue. Insert the cuvette and make sure it is oriented correctly by aligning the mark on cuvette towards the front.

Close the lid. Your instructor will show you how to zero the spectrophotometer with this solution. Set aside this solution for later disposal.

2. For each standard solution in Table 2, rinse your cuvette three times with a small amount (~0.5 mL) of the standard solution to be measured, disposing the rinse solution each time. Then fill the cuvette with the standard, insert the cuvette as before and record the absorbance reading.
3. Finally, repeat this same procedure for the five mixtures with unknown [$FeSCN^{2+}$] prepared in Part 5.2. The [$FeSCN^{2+}$] in these solutions will be determined from the calibration curve.

6.0 DATA RECORDING SHEET

Last Name	First Name		Partner Name(s)	Date

1. Absorbances for Standard Solutions from Table 2

Standard solution	[$FeSCN^{+2}$] (= diluted [SCN^-])	Absorbance
1		
2		
3		
4		

Show the calculation of [$FeSCN^{+2}$] (= diluted [SCN^-]) for one standard solution below.

2. Initial Reactant Concentrations for Equilibrium Mixture from Table 3

Sample	2.00 x 10^{-3} M $Fe(NO_3)_3$	2.00 x 10^{-3} M KSCN	Laboratory Water	Initial Concentrations (M)	
				Fe^{3+}	SCN^-
1	5.00 mL	5.00 mL	0.00 mL		
2	5.00 mL	4.00 mL	1.00 mL		
3	5.00 mL	3.00 mL	2.00 mL		
4	5.00 mL	2.00 mL	3.00 mL		
5	5.00 mL	1.00 mL	4.00 mL		

Show a dilution calculation for $[Fe^{3+}]_{initial}$ in Sample #1 only.

3. Absorbances for the Equilibrium Mixtures from Table 3.

Sample	Absorbance
1	
2	
3	
4	
5	

7.0 DATA ANALYSIS

1. Using Microsoft Excel or Google Sheets, plot Absorbance vs. [$FeSCN^{2+}$] for the standard solutions. Obtain an equation for the line (best-fit). This equation should be in the form, $y = mx + b$ (A = m·[$FeSCN^{2+}$] + constant) and will be used to determine the [$FeSCN^{2+}$] in the equilibrium mixtures. **Attach your graph to this report.**
2. Use the equation you obtained from your graph in #1 to determine the [$FeSCN^{2+}$] in the equilibrium mixtures ($[FeSCN^{2+}]_{equil}$).

Sample	$[FeSCN^{2+}]_{equil}$ (M)
1	
2	
3	
4	
5	

Show the calculation of $[FeSCN^{+2}]_{equil}$ for one sample below.

3. Combine the selected information you gathered and complete the table below for all the equilibrium concentrations (see Table 1) and the value of K_c.

Sample	Equilibrium Concentrations (M)			K_C
	Fe^{3+} ($[Fe^{3+}]_0$ - $[FeSCN^{2+}]$)	**SCN^-** ($[SCN\text{-}]_0$ - $[FeSCN^{2+}]$)	**$FeSCN^{2+}$**	
1				
2				
3				
4				
5				

Average value of K_C ____________________

(Use reasonable number of significant digits, based on the distribution of your K_C values.)

8.0 POST-LAB QUESTIONS

Problem: Is an alternative reaction stoichiometry supported?

Suppose that instead of forming $FeSCN^{2+}$, the reaction between Fe^{3+} and SCN^- resulted in the formation of $Fe(SCN)_2^+$. The reaction analogous to the one in Equation 2 would be:

$$Fe^{3+}(aq) + 2\ SCN^-(aq) \rightleftharpoons Fe(SCN)_2^+(aq)$$

That would mean two moles of SCN^- displace two moles of H_2O in $Fe(H_2O)_6^{2+}$, making $Fe(SCN)_2(H_2O)_4^+$.

1. Write the equilibrium constant expression for this reaction.

2. Using the same absorbances for the standard solutions but setting the concentration of the $Fe(SCN)_2^+$ equal to ½ of the diluted concentration of SCN^- (see Part 6.0 #1), construct a new graph for the calibration curve. Obtain the new equation of the line (best-fit) and provide it below.

(This is because the moles of $SCN^–$ assumed to be equal to the moles of complex ion product, $FeSCN^{2+}$, in our standard solutions. In this alternative reaction, the moles of product, $Fe(SCN)_2^+$, equal ½ the number of moles of $SCN^–$ added in the standard solutions.)

New equation of the line: ____________________

3. Use this new equation of the line to determine the equilibrium concentration of $Fe(SCN)_2^+$ ($[Fe(SCN)_2^+]_{equil}$) in each of the sample mixtures and fill in the table below. Remember that since 2 moles of SCN^- react in the alternative reaction, $[SCN^-]$ at equilibrium is $[SCN^-]_{initial}$ – $2[FeSCN^{2+}]_{equil}$. The initial concentrations of the Fe^{3+} and SCN^- are the same as before (see Part 6.0 #2).

Sample	**Equilibrium Concentrations (M)**			K_C
	Fe^{3+} $[Fe^{3+}]_{initial}$ – $[Fe(SCN)_2^+]$	**$SCN^–$** $[SCN^-]_{initial}$ – $2[Fe(SCN)_2^+]$	**$Fe(SCN)_2^+$**	
1				
2				
3				
4				
5				

4. Based on the values you got for the K_C in this alternative reaction compared to what you calculated assuming only on SCN- reacted, which reaction appears to be correct: one SCN^- or two SCN^-? Explain your reasoning.

2611 Thermodynamics of Borax solubility

Thermodynamics of Borax Solubility

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- Determine the solubility product constant.
- Generalize the relationship of the solubility product constant and temperature.
- Determine the values of ΔH°, ΔS°, and ΔG° for a reaction.

1.2 Background

In this experiment, you will determine the values of ΔH° and ΔS° for the reaction which occurs when borax (sodium tetraborate octahydrate) dissolves in water.

In previous experiments, you have determined ΔH° values directly, by measuring temperature changes when the reaction occurred. However, in many cases, this technique is not practicable. For example, the reaction may not go to completion, or it may give off such a small amount of heat that the temperature change is too small to measure. In addition, there is no direct method for measuring ΔS° for a reaction. It is therefore useful to be able to determine ΔH° and ΔS° indirectly, by using their relationship to the equilibrium constant of a reaction.

The equilibrium constant of any reaction can be related to the free energy change of the reaction:

$\Delta G° = -RT \ln K$

The free energy change is also related to the enthalpy and entropy changes during the

reaction:

$\Delta G° = \Delta H° - T\,\Delta S°$

Combining these two equations gives the general relationship between K, ΔH°, and ΔS°:

$-RT \ln K = \Delta H° - T\,\Delta S°$

Dividing both sides by –RT gives a particularly useful form of this relationship:

$$\ln K = -\frac{\Delta H°}{R}\left(\frac{1}{T}\right) + \frac{\Delta S°}{R}$$

This represents a linear equation of the form y=mx+b. In this case, y=ln K and x= 1/T; a plot of ln K against 1/T will therefore be linear. In addition, the slope of this line (m) will equal –(ΔH°/R), and its y-intercept (b) will equal (ΔS°/R). It is therefore possible to determine ΔH° and ΔS° by simply measuring the equilibrium constant at two different temperatures, graphing ln K against 1/T, and measuring the slope and intercept of the resulting line. In practice, K is measured at several temperatures, so that the effect of any experimental errors in one measurement will be minimized.

The reaction you will study is the dissolution of borax (sodium tetraborate octahydrate) in water. “Borax” is a naturally occurring compound; it is in fact the most important source of the element boron. Borax has been used for many years as a water softening agent. Borax is a rather complicated ionic salt which has the chemical formula $Na_2B_4O_5(OH)_4{\cdot}8\ H_2O$. When it dissolves, it dissociates as follows:

$Na_2B_4O_5(OH)_4.8\ H_2O\ (s) \rightleftharpoons 2\ Na^+\ (aq) + B_4O_5(OH)_4^{2-}\ (aq) + 8\ H_2O\ (l)$ (1)

Notice that the products of this reaction are two sodium ions and one other ion (this ion is called "tetraborate"), along with the eight molecules of water. Since water does not appear in equilibrium constant expressions, the K expression for this reaction is:

$K = [Na^+]^2[B_4O_5(OH)_4^{2-}]$

You will measure K by analyzing a saturated solution of borax (i.e. a solution in which Reaction (1) has come to equilibrium!) for the tetraborate ion. Tetraborate is a weak base, so it can be titrated with a strong acid. It may surprise you that tetraborate can react with only two hydrogen ions -- not four! -- and that in this reaction, the tetraborate ion "falls apart", producing four molecules of boric acid:

$B_4O_5(OH)_4^{2-}\ (aq) + 2\ H_3O^+\ (aq) + H_2O\ (l) \rightarrow 4\ H_3BO_3\ (aq)$ (2)

Once you know the number of moles of tetraborate in the solution, you can calculate the number of moles of sodium ion by using the stoichiometry of Reaction (1). Then, you can calculate the molar concentrations of the two ions and, finally, the value of K.

References and further reading

Technique G: Buret Use

Experiment 2501 Using Excel for Graphical Analysis of Data of the laboratory manual

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Borax, $Na_2B_4O_5(OH)_4$	Use as is	25 – 30 g	
Hydrochloric acid, HCl	0.5 M	150 mL	Standardized solution, refer to reagent bottle for exact concentration
Bromocresol green		drops	Visually impaired students might also use Congo red is an alternative

4.0 GLASSWARE AND APPARATUS

250 mL beaker (2)	Large beaker for water bath
150 mL beaker	125/250 mL Erlenmeyer flask (4)
10 mL graduate cylinder	50 mL buret
Glass stir rod	Buret clamp
Thermometer	Hot plate
Ring stand	Beaker tongs

Note on Cleaning Glassware: Do not use soap to wash glassware for this experiment or you will be titrating the soap residue as well as the borax (soap is also a base). You should rinse all glassware several times with deionized water before beginning this experiment.

5.0 PROCEDURE

1. PREPARATION OF SATURATED BORAX SAMPLES

It is important that the borax solution be saturated in order to achieve equilibrium between the solid and dissolved phases. If you can see solid borax crystals at the bottom of the beaker you are at equilibrium; if not you should add some additional solid borax until you can see white crystals at the bottom of the solution.

1. Obtain a sample of solid borax. You will need enough borax to reach the 40 mL line in a small beaker (50 or 100-mL). Pour the borax into a 150–mL beaker and add around 80 mL of deionized water.
2. Using your glass stirring-rod, stir the borax solution you have made and place it into an ice-water bath. Insert your thermometer into the solution. Continue to stir the solution gently as it cools. Allow the solution to cool to about 10°C.
3. While your solution is cooling,
1. Label a 250-mL beaker “hot water rinse.” Fill the beaker about half full of deionized water and heat on a hot plate. Keep the temperature of this water between about 60°C to 80°C. You will need this hot deionized water throughout the experiment. Add more deionized water to this beaker as needed.
2. Label four Erlenmeyer flasks “10°C,” “30°C,” “50°C,” and “70°C.”
4. When the saturated borax solution in the 150-mL beaker reaches about 10°C, stop stirring the solution and allow the borax crystals in the solution to settle. This takes about 2 minutes. After the solid has settled measure the temperature in the flask to ±0.1 °C and record this temperature on your data sheet.
5. Pour between 7 and 9 mL of the solution into a 10 mL graduated cylinder. Try to avoid pouring any solid. Read and record the volume of solution you poured into the graduated cylinder to ±0.01 mL. Then pour the solution into the Erlenmeyer flask labeled “10°C.”
6. Using your beaker tongs to grasp the hot water rinse beaker, fill the graduated cylinder with hot deionized water to dissolve any borax crystals that formed on the inside the graduated cylinder

and pour this into the Erlenmeyer flask labeled “10°C.” Repeat this rinse once more to be certain that all the borax from the graduated cylinder has been transferred to the Erlenmeyer flask.

7. Bring the volume of water in the Erlenmeyer flasks up to about the 50-mL mark by adding room temperature deionized water. This amount does not need to be exact because adding more or less deionized water will not change the number of borax molecules in the flask.
8. Prepare a second hot-water bath. This will not be used for a rinse, so you may use tap water to fill this second bath. Place the 150-mL beaker containing the borax solution into this hot-water. You may need to clamp the 150-mL beaker in place to keep it from turning over in the bath. Allow the borax solution in the 150-mL beaker to warm to about 30°C while stirring. (Do not use the hot deionized water bath for this or you may contaminate the deionized rinse water with borax from the outside of the 150- mL beaker). Be certain there are still visible crystals at the bottom of the 150-mL beaker so that you know the borax solution is in equilibrium with the solid phase and thus still saturated. If necessary, add more solid borax to the solution so that white crystals are visible at the bottom of the beaker at all times.
9. Be careful not to overheat the borax solution. If you allow the borax to heat past the desired temperature and then cool back down, you may get supersaturation of the borax. This could result in significant errors in your experimental results. Should you overheat the solution, either collect your data at higher temperatures, or discard the borax solution in the proper waste container and prepare a fresh one, being careful not to exceed the temperature desired the second time. (The later trials for this experiment do not need to be at exactly 30°C, 50°C, and 70°C, so long as they are somewhere around these values).
10. When the saturated borax solution reaches about 30°C, stop stirring the solution and allow the borax crystals in the solution to settle as before. Then record the exact temperature of the solution to ±0.1 °C on your data sheet.
11. Using the 10-mL graduated cylinder, transfer 7 to 9 mL of the approximately 30°C borax solution to the 30°C Erlenmeyer flasks recording the actual volume to ±0.01 mL on your data table. Once again make sure no solid borax from the bottom of the 150-mL beaker is transferred to graduated cylinder during this process. Rinse the graduated cylinder twice with the hot deionized rinse water and bring the volume of water in the Erlenmeyer flasks up to about the 50-mL mark by adding room temperature deionized water as before.
12. Using the hot water bath, heat the borax solution to about 50°C and then to 70°C, collecting 7 to 9 mL samples at each temperature and rinsing the graduated cylinder with the hot deionized rinse as before.
13. You may need to add additional borax to the solution in the 150-mL beaker as you heat it to maintain equilibrium. Be certain that for each trial you can see crystals of solid borax at the bottom of the solution in the 150-mL beaker. Also be sure to allow the solution to settle for at least 2 minutes before collecting your 5 samples. Again, be careful not to overheat the borax solution between trials.

5.2 TITRATING SATURATED BORAX SAMPLES

1. Rinse a 250-mL beaker labeled “standardized HCl” with deionized water and then rinse with about 5-mL of the 0.5 M standardized HCl solution. Fill this beaker with about 150 mL of the standardized HCl solution. Record the exact concentration of the standardized 0.5 M HCl

solution (as given on the reagent label) on your data sheet. You will use this as your standardized HCl solution throughout the titrations. Should you require additional standardized HCl solution, be sure to check that the concentration is the same as the one on your beaker. If it is different you will need to rinse and refill your buret with this new solution and repeat any titrations underway making note of the new concentration for your calculations.

2. Rinse a 50-mL buret with deionized water and then using about 5-mL of the 0.5 M standardized HCl solution. Fill the buret with standardized HCl. Be certain that there are no small air bubbles in the buret tips and that the tips are pressed in firmly and do not leak. Clamp the buret to a buret clamp on a ring stand, making sure that it is vertical.
3. Add four drops of bromocresol green indicator to each of the Erlenmeyer flasks. The solutions in each should turn a light blue color.
4. Titrate the samples with the standardized HCl solution. Record the initial and final buret reading to ±0.01 mL. Bromocresol green is blue in basic solutions and yellow in acidic solutions, so the color of the endpoint should be green.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name(s)**	**Date**

Molarity of standardized 0.5 M HCl solution: _____________________

Table 1. Titration Data

Approximate temperature	10°C	30°C	50°C	70°C
Measured temperature (to ±0.1 °C)				
Volume of $B_4O_5(OH)_4^{2-}$ sample (to ±0.01 mL)				
Initial Buret Volume (to ±0.01 mL)				
Final Buret Volume (to ±0.01 mL)				
Volume of HCl added (Final-Initial volumes)				
Mole of HCl added				
Mole of $B_4O_5(OH)_4^{2-}$ titrated				
Molarity $B_4O_5(OH)_4^{2-}$				
Molarity Na^+				

* *remember to use stoichiometry from the equations in the introduction to calculate moles in the table!*

Show a sample calculation below for each of the calculated entries in your table. Remember to consider the stoichiometry of the acid-base titration reaction in your calculations; it is not 1:1.

7.0 CALCULATIONS

1. Use the borate ion concentration to determine K_{sp} and ln K at each temperature. Convert Celsius temperatures to inverse Kelvin temperatures (1/T). Show a sample calculation below for each of the calculated entries in your table.

Table 2. Data Analysis

Trial	$[B_4O_5(OH)_4^{2-}]$	$[Na^+]$	K_{sp}	ln K	Measured T (K)	$\frac{1}{T}(K^{-1})$
10°C						
30°C						
50°C						
70°C						

2. Complete the data analysis table by collecting data from other students.
3. Use Excel to create a graph of ln K versus 1/T for your data. Your graph should have an appropriate title and labeled axes with an appropriate scale. Using the Excel trendline function, add a best-fit line to your plotted data and have Excel display the equation of this line and its R^2 value on your graph. Submit the graph with your lab and provide the equation here.

y = ____

4. Use this graph to determine the values of ΔH°_{rxn} and ΔS°_{rxn} for the dissolution of borax (chemical equation 1). Be certain to include the proper units for each! Show your calculations. (Check the introduction for help with calculations!)

5. Use the values of ΔH°_{rxn} and ΔS°_{rxn} you obtained to determine the value of ΔG°_{rxn} at 298K for the dissolution of borax. Show your calculations.

6. The literature values for enthalpy and entropy of the dissolution of borax in water are 110 kJ/mol and 380 J/mol-K, respectively. Determine the percent error in your experimentally determined values.

8.0 POST-LAB QUESTIONS

1. Does the solubility of borax increase or decrease with increasing temperature? Use both your physical observations and your K_{sp} measurements to justify your response.

2. Does your value of $\Delta H_{rxn}°$ suggest that the reaction for the dissolution of borax is exothermic or endothermic? Explain why this result is consistent with your answer to Question 1 above.

3. The value of ΔG°_{rxn} that you determined for the dissolution of borax at 25°C should be positive (if not you should check your work). Given that $\Delta G^{\circ}_{rxn} > 0$, how do you explain that some of the borax actually dissolved at 25°C?

4. A geologist removes a sample of water from a lake in Nevada where the water level has been slowly dropping over time. The sample is found to be 0.02 M in borax. The lake is circular in shape and is currently 500 meters in diameter and 50 meters deep. Use your experimental data to predict the level of the lake when borax will begin to precipitate on the shore if the mean temperature of the lake water is 15°C. Assume that the lake is equally deep at all points. (helpful formulas: circumference of a circle = $2\pi r$, volume of a cylinder = $2\pi rh$)

EXPERIMENT 2612 - THERMODYNAMICS OF KNO_3 SOLUBILITY

1.0 INTRODUCTION

1.1 OBJECTIVES

- Students will interpret data on a graph.
- Students will use the data to make several calculations regarding the solubility of the salt.
- Students will graph their data to further determine the thermodynamics of the solubility of the salt.

1.2 BACKGROUND

Equilibrium constants are connected to the thermodynamics of a reaction through the equation:

$\Delta G^0 = -RT\ln K$

The ΔG of the reaction, and the equilibrium constant, change with temperature.

$\Delta G = \Delta H - T\Delta S$

However, enthalpy and entropy change very little over a wide temperature range. We can connect the equilibrium constant, measured at different temperatures, to the enthalpy and entropy as follows:

$-RT\ln K = \Delta G^0 = \Delta H^0 - T\Delta S^0$

$\ln K = -\Delta G^0/RT = -\Delta H^0/RT + T\Delta S^0/RT = -\Delta H^0/RT + \Delta S^0/R = (-\Delta H^0/R)1/T + \Delta S^0/R$

$y \quad = \quad m\,x \; + \; b$

If we plot lnK for a reaction (y-axis) versus 1/T (x-axis), the slope of the line is equal to $(-\Delta H^0/R)$ and the y-intercept is equal to $\Delta S^0/R$. We can then get ΔG^0 either from the lnK at 298K or by solving $\Delta G^0 = \Delta H^0 - T\Delta S^0$.

The reaction to be studied is the solubility of $KNO_3(s)$.

$KNO_3(s) \leftrightarrow K^+(aq) + NO_3^-(aq)$

Let s dissolve +s +s

$K = K_{sp} = [K^+][NO_3^-] = s^2$

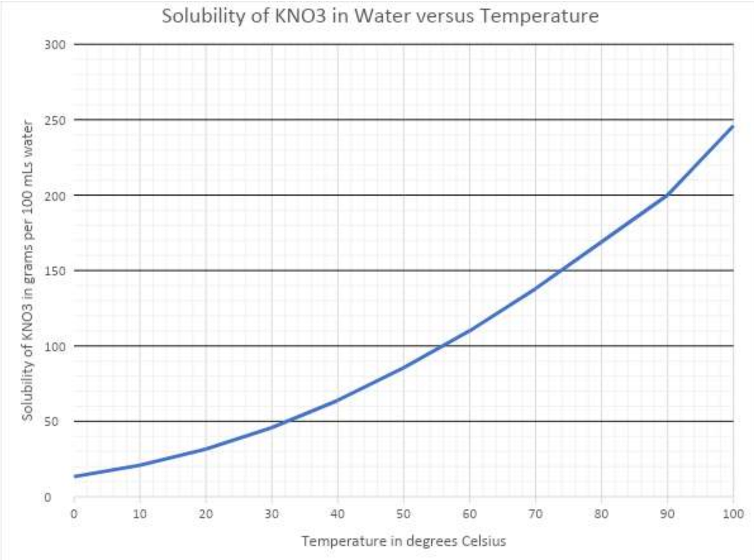
The source of your data is the solubility graph below. (You will need to convert the solubility units given in the graph to molarity for the calculation of the equilibrium constant. Grams of solute per 100 grams water = grams solute per 100mLs water = grams solute per 0.100L. Just change grams of solute to moles of solute and complete the calculation.)

2.0 DIRECTIONS

- Select 5 temperatures from the graph over the range given.
- Read the solubility and convert it to appropriate units.
- Calculate K_{sp}, $\ln K_{sp}$, 1/T.
- Graph lnK versus 1/T.
- Use the slope and y-intercept to determine ΔH^0, ΔS^0, ΔG^0.

- Answer the questions that follow.

SOLUBILITY OF KNO3 VERSUS TEMPERATURE (Data taken from IUPAC and NIST Tables)



3.0 DATA RECORDING TABLE

Temp (^{0}C)	Temp (K)	1/T(K)	Solubility (g solute per100 g water)	Solubility, s (M)	K_{sp} s^2	lnK_{sp}

4.0 ANALYSIS OF DATA

Make a graph of lnK versus 1/T and paste it here. Use it to answer the questions which follow.

5.0 CONCLUSIONS

Show your work.

1. What is the slope of the line?
2. What is the ΔH^0 of the reaction?
3. What is the y-intercept of the line?
4. What is the ΔS^0 of the reaction?
5. What is the ΔG^0 of the reaction from the graph?
6. What is the ΔG^0 of the reaction from the ΔH^0 and ΔS^0?

6.0 POST-LAB QUESTION

Explain why the signs (+ or -) of the thermodynamic properties make sense.

ΔG^0:

ΔH^0:

ΔS^0:

2613 Electrolytic Cells

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- Construct a galvanic cell and an electrolytic cell.
- Determine the charge of one mole of electrons (one Faraday).
- Identify half-reactions occurring in an electrochemical cell.

1.2 Background

1.2.1 Part A: A Galvanic Cell

The reaction you study in this part is:

$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

That reaction is spontaneous, so it can act as an energy source. If the two reactants contact each other directly, the only energy given off is heat. If they are not allowed to contact one another directly, but are made part of an electrical cell, so that they can transfer electrons through a wire, the energy given off does electrical work.

1.2.2 Part B: Quantitative Measurement: Determination of the Faraday

In this part of the experiment you will move Pb from one electrode to another by means of an electric current pushed by a DC power supply. Since a power supply is necessary, it is classified as an electrolysis, although the net reaction is neither spontaneous or non-spontaneous:

$Pb + Pb^{2+} \rightarrow Pb + Pb^{2+}$

You will use a solution of 0.1 M $Pb(NO_3)_2$, with a copper strip as the cathode (–),

and a weighed strip of lead as the anode (+). The half reactions are:

ANODE: $Pb \rightarrow Pb^{2+} + 2\ e^–$

CATHODE: $Pb^{2+} + 2\ e^– \rightarrow Pb$

Note that since there is actually no net reaction (no net change), since Pb^{2+} ions are being replaced at the same rate as they are being used up, there is no change in the concentration of Pb^{2+} ions in solution as the process goes on.

The Pb^{2+} ions that form from oxidation of the lead anode are dissolved in the solution as they are created; therefore, the anode loses weight as the electrolysis proceeds. The weight loss is directly proportional to the number of coulombs of charge passed through the solution; i.e., the more electrons that are transferred, the more Pb from the anode goes into solution. As the 2 to 1 ratio in the balanced equation for the half-reaction shows, there are twice as many moles of electrons transferred as moles of Pb removed from the weighed anode. Because of this proportionality, the loss in weight of the anode can be used to determine the charge of a mole of electrons. This is the value of the Faraday.

1.2.3 Part C: Electrolysis Reactions

Electrolysis reactions involve the use of a DC power supply to “push” the electrons in a non-spontaneous direction. The connections for the wires on the power supply are marked (+) and (–) according to whether they are pulling or pushing electrons. Electrons come out of the (–) and into the (+). The wires that you will connect to the power supply have “alligator clips” to connect to the metal strip or graphite rod that you will be using as electrodes. Whichever electrode you connect to the (–) connection of the power supply will be the (–) charged electrode, which supplies electrons for reduction, so it is the cathode. The cathode, in electrochemistry, is defined as the electrode where reduction takes place. The other electrode, which will be (+) charged by being connected to the (+) connection of the power supply, accepts electrons from an oxidation, so is the anode. The anode, in electrochemistry, is defined as the electrode where oxidation takes place.

With the power supply, you will have an ammeter, which reads the intensity of the current flow, in amperes. The power supply has a knob which you can turn that varies the voltage. If the voltage is less than a certain minimum amount (needed for the reaction to be observed), no current flow will show on the ammeter. Once the voltage is turned up to the level necessary for the reaction, the ammeter needle will suddenly jump up to a certain current flow. If the voltage is turned up even higher, the current will increase also (and the reaction will go faster, as more electrons are being transferred per second). **However, you should avoid turning the voltage up to a value much greater than the minimum for the reaction, because you could complicate the experiment by allowing other reactions to start happening.** The variable voltage knob should be turned to zero (or the whole power supply unit unplugged) except when you are doing an electrolysis experiment. Start the electrolysis process, once the cell is assembled, by turning the variable voltage knob up from zero until the ammeter shows that a current is flowing. When you are finished, turn it back to zero again.

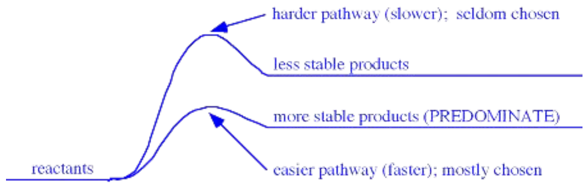
1.2.4 Connecting Electrochemistry and Reaction Pathways

Another factor besides E° value actually determines the which half-reaction you

will see: the activation energy and thus the rate. If the reaction that required the smallest voltage is very very slow, you will not have seen any appreciable current-flow when you have turned the voltage up to the necessary voltage for that easiest reaction. So, you will have continued to increase the voltage until it is high enough for another reaction, which goes fast enough to see. Actually at that voltage, both reactions are happening, but you only see evidence of the fast one, as products of the slow one won’t show up unless you wait a very long time. Thus you should keep in mind that E° values, like ΔG° values, can only tell you information about whether a reaction is spontaneous or not (“will go” or not) -- but they can’t tell you how fast it will go.

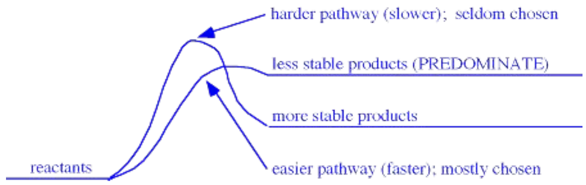
This is an important idea, that when there is a choice to be made between two or more reactions, the determining factor is not necessarily which products are most stable; often what matters is the difference in activation energy; i.e., which pathway is easier, so which reaction is faster.

Often the two pathways are parallel, so that the more stable products form via the easiest pathway:



This situation makes you think that product stability determines the choice of which reaction predominates.

However, sometimes the two pathways cross, so that the less stable products form via the easiest pathway. Then the reaction which can predominate is the faster one, even though its ΔG or E value is less favorable:



2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

!!Wear your safety goggles!!

Lead and its compounds are poisonous; use caution in handling them and wash your hands with plenty water immediately after handling them. In setting up the electrolysis experiment, be careful to follow the directions exactly; do not connect both terminals of an ammeter directly to a power supply.

WASTE DISPOSAL: All liquid waste from this experiment should be poured into the INORGANIC WASTE containers in the fume hood. The electrodes should be rinsed off and returned to the containers where you found them.

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Laboratory water (deionized water)	Pure	mL	Be sure to use laboratory water
Bromothymol blue		Several drops	
$CuSO_4$	?	3 mL	Copper (II) Sulfate
HNO_3	6 M	mL	
Powdered zinc		A small scoop	
Strip of lead		1 piece	
$Pb(NO_3)_2$	0.1 M	200 mL	
Strips of copper		2 pieces	
KI	0.1 M	300 mL	
Starch solution		Several drops	Used to test for I_2

4.0 GLASSWARE AND APPARATUS

6 large test tubes	Test tube rack
thermometer	Stirring rod
250 mL beaker	scoopula
400 mL beaker (waste container)	Power supply
Wires to connect to power supply	Voltmeter

5.0 INVESTIGATION

5.1 Part A Investigation: A Galvanic Cell

1. Data Collection
1. Observing the direct reaction:
    1. Put about 2 or 3 mL of the copper (II) sulfate solution into a test tube.
    2. Put in a thermometer and measure and record the temperature.
    3. Leave the thermometer in the solution and put some powdered zinc metal (enough to be about one-fourth of the solution volume) into the solution; stir gently with the thermometer. Record all evidence of reaction.
2. Now set up a galvanic cell ("battery") using the same spontaneous oxidation-reduction reaction of Zn with Cu^{2+}. Separate the two half-reactions so that the electrons must be transferred by traveling through the wires attached to the electrodes. Attach the wires to a voltmeter, or to a little electric fan, or other device as evidence that your battery is producing electric current. (See Figure 2613.1.)

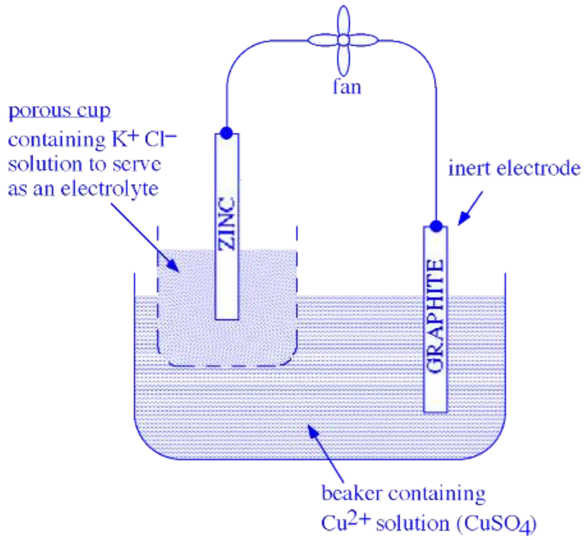
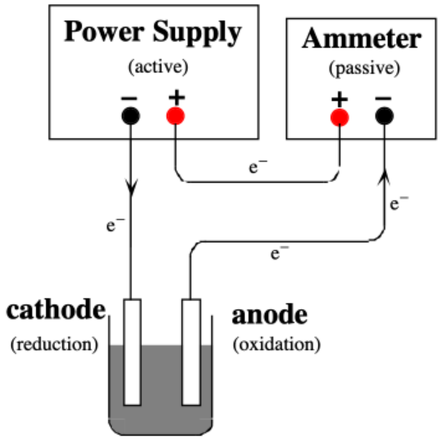


Figure 2613.1: Setup for Part A

5.2 Part B Investigation: Quantitative Measurement: Determination of the Faraday

1. Data Collection
1. Take a strip of lead and clean it by covering it with 6 M HNO_3 in a beaker or graduated cylinder. Let it sit in the HNO_3 solution for a few minutes. A slow evolution of H_2 gas should be seen. (The acid first cleans off any lead oxide coating, then starts to react with the lead metal.)
2. Pour the nitric acid solution off, rinse the lead with water, and then dry it thoroughly with a paper towel. Rub off any surface coating on the lead with the towel until the lead is shiny. When the lead is dry, weigh it as accurately as possible. Record its mass.
3. Fill a 250 mL beaker about two-thirds full of 0.1 M $Pb(NO_3)_2$ solution. Obtain a power supply, two wires, and a strip of copper. Attach one wire from the negative terminal to the copper strip; attach another wire from the positive terminal to your weighed strip of lead. Suspend the two electrodes at opposite sides of the beaker. Refer to Figure 2613.2.

The power supply is an active component in this circuit: it pushes electrons out of its (–) terminal and pulls electrons into its (+) terminal. The ammeter is a passive component in this circuit: it counts electrons as they flow into its (–) terminal and out of its (+) terminal. Some of the power supplies and ammeters have color coded terminals instead of (+) and (–) labels. The red terminals are (+) and the



black terminals are (–).

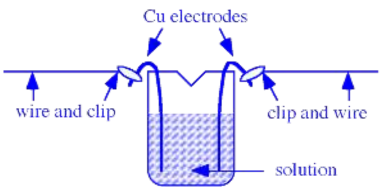
Figure 2613.2: Setup for Part B.

4. Turn on the power supply, recording the exact time it was turned on, and adjust the current flow so that it has a steady value between 0.5 and 1.0 amps. Watch the dial and adjust it if necessary, maintaining the same amperage as closely as possible, for about 20 minutes.
5. ***NOTE****: As the electrolysis proceeds, lead crystals will form at the cathode, and they tend to stick out toward the anode, shortening the path of current flow, and thus lowering the resistance of the cell; that may affect the current, so that is one reason why you need to monitor and adjust the current. If the crystals of lead extend far enough, they could even make electrical contact with the anode and "short out" the electrolysis (bypass the ion-flow and ion-formation process), so it is a good idea to push the lead crystals downward periodically using a glass rod.*
6. Be sure there are no gas bubbles forming at the anode. This would mean that the voltage is too high, such that electrons are being removed from water molecules, forming oxygen gas. If you see gas bubbles, turn down the voltage.
7. When the electrolysis has gone for about 20 minutes, turn off the power supply and record the exact time you turned it off. Remove the lead anode, rinse it with water, wipe it thoroughly dry with a paper towel, and weigh it again. **Record its new weight and calculate the weight lost during electrolysis.**

5.3 Part C Investigation: Electrolysis Reactions

1. Data Collection

1. Use two strips of copper as the electrodes; bend them so they will hang on the edge of a beaker (150 mL or 250 mL).
2. Attach a wire with an alligator clip to each strip of copper. Arrange them so that they are on the far sides of the beaker; if the electrodes touch each other, you'll get a short circuit! (A mild version, since this is a low voltage.) *Also be sure the alligator clip does not touch the solution; only the copper electrode should touch the solution; that is because the reaction of copper is one of the possibilities for the reaction at the electrode; if the alligator clip touches the solution, reaction of one or more of the metals in the alligator clip would become a possibility, and you don't know what metals are in the alligator clip.*



3. Put some 0.1 M KI solution into the beaker. Connect the wires to the power supply and turn up the voltage until you just barely get noticeable current flow. *(That way, you will cause only one noticeable reaction; if you turn the voltage up higher and higher, you will make more and more different reactions possible, and it will get very complicated to figure out what is happening!)* ***Record your observations.***
4. Now see if you can figure out **what half reaction is happening at each electrode**. The possible reactants present in the solution are: K+ (aq), I– (aq), H_2O (l), Cu (s). As you try the following tests, it will help to turn off the power supply when you are not making observations. This will keep the product concentrations small so that you will be able to see what is being produced at each electrode.
5. **Which of the possible half reactions should affect the acid / base balance of the solution?** Add an acid-base indicator by the appropriate electrode. ***Record your observations.***
6. You can use starch solution to test for elemental iodine (I_2). Starch forms a characteristic blue-black color with iodine, even when the concentration of iodine present is still so low that without the starch it would only be a pale yellow. Put a few drops of starch solution by the appropriate electrode. ***Record your observations.***

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name*(s)***	**Date**

6.1 Part A: A Galvanic Cell

1. Record the temperature and all evidence of reaction.

2. Sketch a diagram of the galvanic cell that you set up, similar to Figure 14.1.

3. Label the anode and cathode and write the half-reactions that are occurring at each electrode.
4. Which way are the electrons moving through the wire? Show the direction of e– flow on the diagram.
5. Which way are the positive ions moving through the solution? Why? Show the movement of the Cu^{2+} ions, Zn^{2+} ions, and K^{+} ions on the diagram.
6. Which way are negative ions moving through the solution? Why? Show the movement of the Cl^{-} ions and SO_4^{2-} ions on the diagram.
7. In this part of the experiment, you observed the same product-favored reaction in two different set-ups. What happened to the energy released by the reaction in each case?

6.2 Part B: Quantitative Measurement: Determination of the Faraday

1. Write the net ionic equation for lead oxide reacting with acid.

2. Write the net ionic equation for lead metal reacting with acid.

3. Write the net ionic equation for the reaction that occurs during this electrolysis.

4. Calculate the mass of the lead that was lost from the anode during electrolysis. Convert this to moles of lead lost, and finally to moles of electrons lost from the lead.

5. From the total elapsed time (seconds) of the electrolysis, and the average current (amperes), calculate how many coulombs of charge were transferred during the process.

coulombs=amperes × seconds= coulombssecond × seconds

(If the current could not be held constant, estimate its average value as best you can.)

6. Calculate the value of the Faraday (coulombs / mole e–) from your data.

7. Calculate the percent error in your value of the Faraday. The accepted value is 96485 coulombs / mole e–.

6.3 Part C: Electrolysis Reactions

Pre-lab questions: Complete before the experiment.

In this part of the experiment, you will. try to figure out what reaction occurs when electrical current is passed through an aqueous solution of potassium iodide. You will consider the half-reactions at the anode and at the cathode separately. The only substances available as reactants are: K+ (aq), I– (aq), H_2O (l), Cu (s), and electrons (of course). You can either figure out the possible half-reactions using common sense, or you can select appropriate half-reactions from the table of standard reduction potentials in your textbook.

1. List 3 possibilities for the oxidation half-reaction.

2. List 2 possibilities for the reduction half-reaction.

3. For each of these half reactions, what would you expect to see if it were occurring? What products would form? Consider the possibilities of a color change, formation of a solid, formation of a gas (bubbles), a shift in the acid / base balance of the solution (which you could detect using an acid-base indicator), and the formation of elemental iodine (which you could detect using starch solution).

During the lab, please record your observations below.

Procedure Step	Observations
3. Record your observations.	
4. what half reaction is happening at each electrode?	
5. Which of the possible half reactions should affect the acid / base balance of the solution?	
6. Record your observations.	

Further Analysis for Electrolysis of Potassium Iodide Solution

1. Interpret the results of your test with the acid-base indicator. Is an acid or a base being formed in the reaction?

2. Interpret the results of your test with the starch solution. Is elemental iodine (I_2) being formed?

3. Which of the oxidation and reduction reactions that you listed in the pre-lab question is actually occurring?

4. Look up the E° values for each half reaction. Is the reaction that you observed the one that requires the smallest voltage to make it happen?

2614 Electrochemistry and Galvanic Cells

1. INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- set up galvanic cells from several combinations of half-cells to determine anode, cathode, direction of electron flow, and electrochemical potential.
- vary concentrations of one solution in a half-cell to determine its effect on the electromotive potential.

1.2 Background

Spontaneous chemical reactions that involve the transfer of electrons can produce an electrical current in a galvanic cell. These oxidation-reduction reactions, called redox reactions, must physically separate the oxidation process of losing electrons from the reduction process of gaining electrons in two half-cells. The electrons generated then travel through an external wire from the electrode at the site of the oxidation, a piece of metal called the anode, to the electrode at the site of the reduction, another piece of metal called the cathode. To complete the electrical circuit, a salt bridge with a non-reacting ionic solution connects the two half-cells. Ions are able to flow out of the salt bridge to maintain electrical neutrality in each half-cell. See Figure 1 below.

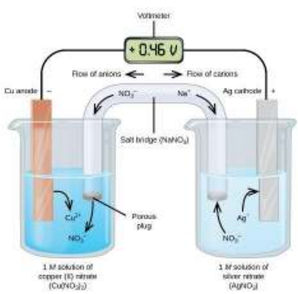


Figure 1. A sample galvanic cell with Cu, Cu^{2+}, Ag, and Ag^{+} reacting

Typically, the anode is displayed on the left and the cathode on the right, so the flow of electrons in the external wire is from left to right. In the figure, both metals are part of the overall reaction, but an inert electrode, usually a metal that is very non-reactive like platinum, may be used when the reactants are all dissolved species. The electromotive force, E, is measured with a voltmeter through which the electrons flow in the circuit.

The reaction at the anode is shown below.

$Cu \rightarrow Cu^{2+} + 2e^{-}$

The cathode reaction is shown below.

$Ag^+ + e^- \rightarrow Ag$

The overall reaction is balanced, which means that the electrons must be equal on both sides and therefore cancel.

$Cu + 2\,Ag^+ \rightarrow Cu^{2+} + 2Ag$

Because the anode is where electrons are generated, it is the negative electrode. Electrons are drawn to the cathode so it is the positive electrode. The electrons flow from anode to cathode spontaneously, but it is possible to connect the wires to the voltmeter incorrectly. A negative voltage simply means the wires have been attached incorrectly to the voltmeter.

While the electromotive force (emf) of a half-cell cannot be measured directly because it is an incomplete circuit, arbitrary values can be assigned to half-cells as long as the combinations which represent actual complete cells give the correct total electromotive force. Tables of these half-cell potentials are available. For the sake of consistency, these values are given for reduction potentials (E_{red}). The values listed are those for standard conditions: 298 K, 1 atm, and 1 M. Look up the standard half-cell potentials for the three half cells you will be studying and put their values on your data recording sheet (Table 1).

The standard electromotive force for a complete cell (Equation 1) is the difference in the reduction potentials for the half-cells.

$E^0_{cell} = E^0_{cathode} - E^0_{anode}$ (Equation 1)

It is common to set up half-cells with concentrations that are not 1M. The value of E is then not the standard one but can be found from the Nernst equation (Equation 2).

$E_{cell} = E^0_{cell} - \mathrm{RTnF} \ln Q$ (Equation 2)

n is the moles of electrons (electrons transferred in the reaction)

F is the Faraday's constant = 96487 C/mol e^-

R is the gas constant = 8.314 J/mol-K

T is the temperature in K

Q is the reaction quotient or equilibrium constant expression (but not necessarily the value of K_{eq})

In this experiment, several galvanic cells will be set up and their electromotive forces measured in volts. Also, the electromotive force of one cell with varying concentrations of the solution will be measured to confirm the Nernst equation.

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
Zn, Cu, Fe metal	N/A	1 strip each	If they appear tarnished, clean with sandpaper until shiny
$Zn(NO_3)_2$	0.100 M	50 mL	
$Cu(NO_3)_2$	0.100 M	50 mL	
$Fe(NH_4)_2(SO_4)_2$ (ferrous ammonium sulfate)	0.100 M	50 mL	Keep bottle closed! Sensitive to air

4.0 GLASSWARE AND APPARATUS

100-mL beakers (3)	voltmeter
100-mL volumetric flask	ceramic cup
10.00-mL pipette	extra clean dry beakers

5.0 PROCEDURE

5.1 Measuring the Potential of a Galvanic Cell

1. Collect one strip each of Zn (dark silver metal), Cu (orange metal), and Fe metal (the Fe metal may be a nail). Make sure that these metal strips are clean by rubbing with sandpaper, if necessary.
2. Collect about 50 mL of each of the three metal solutions: $Zn(NO_3)_2$, $Cu(NO_3)_2$, and $Fe(NH_4)_2(SO_4)_2$ in clean and dry labelled 100-mL beakers.
3. You will not be using a salt bridge. You will be setting up your half cells in a slightly different manner (Figure 2). Pour a small amount of the $Zn(NO_3)_2$ solution inside the ceramic cup. Pour a small amount of the $Cu(NO_3)_2$ solution inside a 250 ml beaker (or other container of appropriate size). The solution level cannot be higher than the height of the ceramic cup. Place the ceramic cup inside the 250 ml beaker, taking care to keep the two solutions separated. The ceramic cup is porous, and allows for contact between the two solutions, but no appreciable mixing.

Figure 2. Set up for galvanic cell without a salt bridge

4. Using the red and black cables, attach the Zn and Cu strips to the voltmeter. Set the voltmeter to direct-current, and the voltage between 0 and 5 volts, or as appropriate to the meter. Dip the Zn electrode in the $Zn(NO_3)_2$ solution in the ceramic cup and the Cu electrode in the $Cu(NO_3)_2$ solution outside the cup. If the electrodes touch each other, a short circuit will occur, and there will be no voltage observed.
5. If the voltage reading is negative, you need to switch the clips between electrodes. (Because the voltmeter measures electrons flowing in a specific direction, one way shows a negative reading and the other positive. We know this is a spontaneous reaction however, so the Ecell is by definition going to be positive. Therefore, if we see a negative reading, we know we just need to switch the electrode clips to tell the voltmeter which is the correct cathode and which is the correct anode.) The reading may jump around a bit as you hold the clips, so take the highest reading and do it quickly. (The readings will decrease over time. Why?)
6. Record your readings on your data recording sheet (Table 2). Note which electrode is the negative one (anode) and record that also.
7. ***Clean the ceramic cell and beaker thoroughly with laboratory water*** and repeat steps 3 to 6 to set up the Zn-Fe cell combination and the Cu-Fe cell combination. It does not matter which solution is in the ceramic cup as long as the appropriate electrode is used with the appropriate solution.
8. Compare the theoretical values for the voltages to your experimental ones and give the percent errors on your data recording sheet (Table 2).

5.2 Nernst Equation and the Effect of Concentration on E_{cell}

1. Using your 10.00-mL pipette, take 10 mL of the Cu solution and place it in a 100-mL volumetric flask. Fill with laboratory water to the mark. Put on the stopper and thoroughly mix. Pour the solution into a clean dry beaker and label it as 'Dilution 1'.
2. Thoroughly rinse the volumetric flask with laboratory water. It needs to be clean but not necessarily dry. Then, using a pipette, take 10-mL of the new diluted Cu solution (Dilution 1) and

place them into the volumetric flask. Add laboratory water to the mark, mix well, and pour into a new clean and dry beaker labelled 'Dilution 2'.

3. Repeat step 2 to make a third, fourth, and fifth dilution and label as Dilution 3, Dilution 4, and Dilution 5, respectively.
4. When all the diluted solutions are prepared, use each one of them with the Cu electrode as half-cell to connect with the Zn half-cell. Set up the galvanic cell as before (Part 5.1). Be sure to clean the ceramic cup thoroughly between measurements. Record the voltage you measure for each combination of the original Zn half-cell with Dilutions 1 through 5 of the $Cu(NO_3)_2$ solution on your data recording sheet (Table 3).
5. On Table 3, record the concentration of the Cu^{2+} ions in each of the dilutions and the predicted value of the voltage when combined with the Zn half-cell according to the Nernst equation (Equation 2). Compare the results of experimental and theoretical values and determine the percent error. (Remember % error = |(observed value-theoretical)/(theoretical)| * 100%)

6.0 DATA RECORDING SHEET AND CALCULATIONS

Last Name	**First Name**		**Partner Name(s)**	**Date**

Table 1. Standard Value of Electromotive Forces

Half-cell Reaction	**E^0_{red}**
$Cu^{2+} + 2e^- \rightarrow Cu$	
$Fe^{2+} + 2e^- \rightarrow Fe$	
$Zn^{2+} + 2e^- \rightarrow Zn$	

What source did you use for these values? ________________________________

Table 2. Experimental and Theoretical Values of E^o_{cell}

Cell	**Which electrode is the anode?**	**Experimental E^o_{cell}**	**Theoretical E^o_{cell} ($E^o_{cathode} - E^o_{anode}$)**	**% Error**
Zn-Cu				
Zn-Fe				
Cu-Fe				

Show calculations for one cell below.

Table 3: Effect of Concentration on E_{cell}

Dilution	$[Cu^{+2}]$	Experimental E_{cell}	Theoretical E_{cell}	% Error
1				
2				
3				
4				
5				

Show calculations for one cell below.

7.0 POST-LAB QUESTIONS

1. It was indicated in the procedure that the voltage would drop over time, eventually reaching zero. Why?

2. All solutions used in this experiment were 0.100 M, which is not standard. Did that cause our experimental values to be different from the theoretical E^0_{cell}? Why or why not?

3. Could have any other factor caused our experimental values to be different from E^0_{cell}? Explain your answer briefly.

4. Based on your data in Part 6.0, what trend did you observe on the effect of concentration on the value of E_{cell}?

2615 Qualitative Analysis of Selected Cations

Qualitative Analysis of Selected Cations

1.0 INTRODUCTION

1.1 Objectives

After completing this experiment, the student will be able to:

- follow a classic analytical scheme to separate ions in a known mixture of cations.
- apply this scheme to identify the ions in an unknown mixture of cations.

1.2 Background

One common task in analytical chemistry is the identification of the various ions present in a particular sample. For example, if you are an environmental chemist your job may be to recover soil or water samples in order to determine the presence of toxic ions such as Pb^{2+} or Ba^{2+}. A common experimental method used to identify ions in a mixture is called qualitative analysis.

In qualitative analysis, the ions in a mixture are separated by selective precipitation. Selective precipitation involves the addition of a carefully selected reagent to an aqueous mixture of ions, resulting in the ***precipitation*** (when a substance is deposited in solid form from a solution) of one or more of the ions, while leaving the rest in solution. Cations are separated first into five groups, then within each group cations are separated selectively. Once each ion is isolated, its identity can be confirmed by using a chemical reaction specific to that ion.

Cations are typically divided into groups, where each group shares a common reagent that can be used for selective precipitation.

- Group I Cations (Ag^{+}, Hg_2^{2+} and Pb^{2+}) form insoluble chlorides
- Group II Cations (As^{3+}, Bi^{3+}, Cd^{2+}, Cu^{2+}, Hg^{2+}, Sb^{3+} and Sn^{2+}) for insoluble sulfides in acidic conditions
- Group III Cations (Al^{3+}, Co^{2+}, Cr^{3+}, Fe^{2+}, Fe^{3+}, Mn^{2+}, Ni^{2+} and Zn^{2+}) form insoluble sulfides or hydroxides in basic conditions
- Group IV Cations (Ca^{2+}, Sr^{2+} and Ba^{2+}) form insoluble carbonates or phosphates
- Group V Cations (Mg^{2+}, Na^{+}, K^{+} and NH_4^{+}) remain soluble in separation of groups I – IV ions

To illustrate quantitative analysis notice Ag^{+}, Hg_2^{2+} and Pb^{2+} are called the Group I cations since they are the first group separated from the larger mixture. Since these ions all form insoluble chlorides, their separation from all other ions may be accomplished by the addition of 6 M HCl (*aq*) resulting in the precipitation of AgCl (*s*), $PbCl_2$ (*s*) and Hg_2Cl_2 (*s*):

$Ag^{+}(aq) + Cl^{-}(aq) \longrightarrow AgCl(s)$

$Hg_2^{+2}(aq) + 2\ Cl^{-}(aq) \longrightarrow Hg_2Cl_2(s)$

$Pb^{2+}(aq) + 2\ Cl^{-}(aq) \longrightarrow PbCl_2(s)$

The sample must then be centrifuged (spun rapidly), which separates the solid precipitates from the ions still in solution. The solids settle to the bottom, and the solution containing the remaining ions (Groups 2 – 5) remains on top of the solid. This solution is called the ***supernatant*** and must be carefully decanted (poured off without disturbing the solid) and saved for further study. The Group I cations contained within the collected precipitate must then be separated from each other in order for the presence of each ion to be confirmed.

Lead (II) chloride can be separated from the other two chlorides based on its increased solubility at higher temperatures. This means that lead (II) chloride will dissolve in hot water, leaving the mercury (I) chloride and the silver chloride in solid form:

$PbCl_2(s) \longrightarrow Pb^{2+}(aq) + 2\ Cl^{-}\ (aq)$

The presence of Pb^{2+} in the aqueous solution can then be confirmed by the formation of a yellow precipitate of$PbCrO_4$ upon the addition of aqueous K_2CrO_4:

$Pb^{2+}(aq) + CrO_4{}^{2-}(aq) \longrightarrow PbCrO_4(s)$

Next, $Hg_2{}^{2+}$ and Ag^{+} cations can be separated by adding 6 M $NH_3(aq)$ to the solid mixture of the two chlorides. Silver chloride will dissolve since it forms a soluble complex ion with ammonia:

$AgCl(s) + 2NH_3(aq) \longrightarrow Ag(NH_3)_2{}^{+}(aq) + Cl^{-}(aq)$

However, mercury (I) chloride reacts with the ammonia yielding what appears to be a gray solid which is actually a mixture of black Hg (*l*) and white $HgNH_2Cl$ (*s*). The presence of this gray solid is confirmation of the presence of $Hg_2{}^{2+}$.

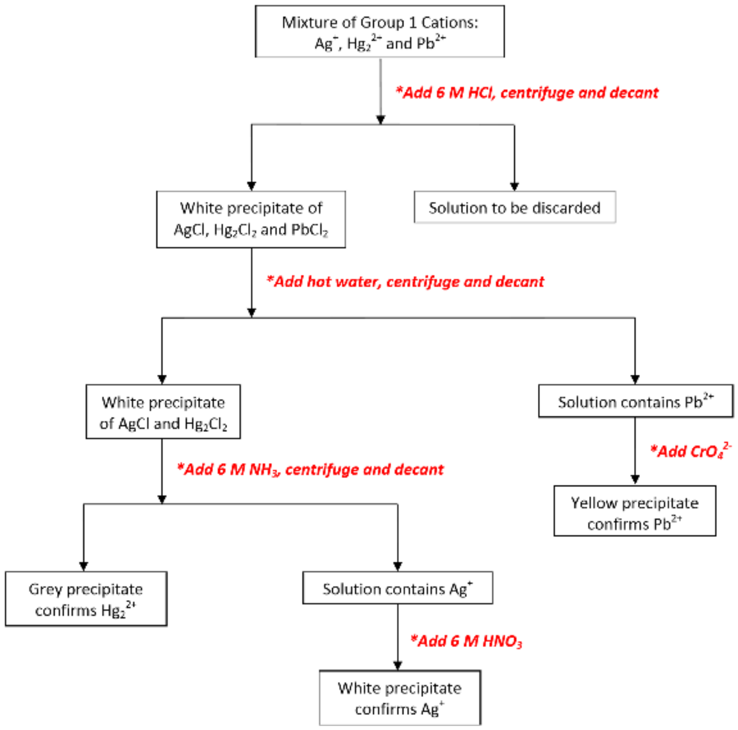
$Hg_2Cl_2(s) + 2NH_3(aq) \longrightarrow Hg(l) + HgNH_2Cl(s) + NH_4{}^{+}(aq) + Cl^{-}(aq)$

The presence of Ag^{+} can be confirmed by the appearance of a white precipitate upon adding 6 M HNO_3 (*aq*) to the solution. The nitric acid reacts with the ammonia and thus destroys the complex ion containing the silver cation. Once in solution again the silver cation precipitates with the chloride as indicated by the following reactions:

$Ag(NH_3)^{2+}(aq) + 2H^{+}(aq) \longrightarrow Ag^{+}(aq) + 2NH_4{}^{+}(aq)$

$Ag^{+}(aq) + Cl^{-}(aq) \longrightarrow AgCl(s)$

The analysis scheme described represented in abbreviated form using a flow chart:



In this experiment you are tasked with separation and identification of group 1 cations (Ag^+ and Pb^{2+}), group 3 cations (Ni^{2+} and Fe^{3+}), and group 4 cation (Ba^{2+}). After the selective precipitation and identification of Ag^+ and Pb^{2+}, the Ba^{2+} ions can be precipitated with addition of 3 M H_2SO_4.

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$$

Finally, 15 M NH_3 can be used to separate the Ni^{2+} from the Fe^{3+} ions. Once the solution has become basic, the excess ammonia will react with Ni^{2+} to form an aqueous complex ion while the Fe^{3+} will precipitate out as red $Fe(OH)_3$ since Fe^{3+} does not typically form complex ions with NH_3:

$$Ni^{2+}(aq) + 6NH_3(aq) \longrightarrow Ni(NH_3)_6^{2+}(aq)$$

$$Fe^{3+}(aq) + 3NH_3(aq) + 3H_2O(l) \longrightarrow Fe(OH)_3(s) + 3NH_4^+(aq)$$

The presence of Ni^{2+} in the aqueous solution is confirmed by adding dimethylglyoxime ($C_4H_8N_2O_2$) resulting in the formation of a rose red precipitate:

$Ni^{2+}(aq) + 2C_4H_8N_2O_2(aq) \longrightarrow Ni(C_4H_7N_2O_2)_2(s) + 2H^+(aq)$

In order to confirm the presence of Fe^{3+}, the red $Fe(OH)_3$ precipitate is dissolved in HCl, and then NH_4SCN is added to the solution. A positive result is the formation of a dark red solution indicating the presence of $FeSCN^{2+}(aq)$:

$Fe^{3+}(aq) + SCN^-(aq) \longrightarrow FeSCN^{2+}(aq)$

References and further reading

Technique J: Centrifuge Use

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	Concentration	Approximate Amount	Notes
$AgNO_3$	0.1 M	10 mL	
$Ba(NO_3)_2$	0.1 M	10 mL	
$Fe(NO_3)_3$	0.1 M	10 mL	
$Ni(NO_3)_2$	0.1 M	10 mL	
$Pb(NO_3)_2$	0.1 M	10 mL	
HCl	6 M	drops	
K_2CrO_4	2 M	drops	
$HC_2H_3O_2$	6 M	drops	
H_2SO_4	3 M	drops	
NaOH	6 M	drops	
NH_3	15 M	drops	Concentrated ammonia is stored and used in the hood. Labeled NH_4OH.
NH_4SCN	1 M	drops	
Bromocresol blue		drops	
Dimethylglyoxime	1%	drops	

4.0 GLASSWARE AND APPARATUS

Beakers, varying sizes (5+)	Large beaker for hot-water bath
Medium sized test tube (5+)	Large beaker for waste
Centrifuge tubes (4)	Benchtop centrifuge
Glass stir rod	Plastic pipets (5+)
Hot plate	Test tube clamp

5.0 PROCEDURE

To get correct results in this experiment, good organizational skills and techniques are essential. Be sure to label all tubes and solutions because these accumulate rapidly, and it is very easy to get tubes and/or solutions mixed up. It is also suggested that students keep a “waste” beaker near their work area in which to discard all solutions locally, and then pour their combined waste solutions into the large waste container for the class at the end of the lab.

Finally, be sure to clean out and thoroughly rinse all glassware (including stirring rods!) with deionized water between uses. Cross-contamination is one of the most common causes for false observations leading to incorrect conclusions.

5.1 CHEMICAL TESTS OF INDIVIDUAL CATIONS

1. Obtain about 10 mL of the following solutions in separate clean, dry, labeled beakers: 0.1 M $AgNO_3$, 0.1 M $Ba(NO_3)_2$, 0.1 M $Fe(NO_3)_3$, , 0.1 M $Ni(NO_3)_2$ and 0.1 M $Pb(NO_3)_2$.

5.1.1 Precipitation with Cl^-

2. Transfer 10 – 15 drops of each solution to separate clean, dry, labeled test tubes. Record observations of the solutions on data table 1.
3. To each test tube add 5 drops of 6M HCl. Mixing each solution by flicking the test tube with finger. Record observations in data table 1.
4. If a precipitate forms, transfer the mixture to a centrifuge tube then centrifuge the mixture. Make sure that you balance the centrifuge before you start. Balancing is done by placing centrifuge tubes of approximately equal weight opposite each other. If no precipitate forms, the mixture can be discarded in your waste container.
5. Pour the supernatant solution into your waste container and save the precipitate for further analysis.

5.1.2 Heat and precipitation with CrO_4^{2-}

6. Wash the precipitate from Step 5 to remove contaminates that may be trapped within the solid. This can be accomplished by adding 1.0 mL of deionized water, mixing, and then centrifuging. The wash supernatant can be discarded in your waste container. Wash the precipitate twice saving the precipitate.
7. Prepare a hot water bath by adding tap water to a 250 mL beaker. Add water until the beaker is about one half full. Heat the water on a hot plate until it is boiling.

8. Add 1.5 mL of deionized water to the centrifuge tubes containing the clean precipitate. Place it in the hot water bath. Periodically stir the mixture in the test tube for 3 – 4 minutes. Record observations in data table 1.
9. Remove the centrifuge tube from the hot water bath and immediately centrifuge the hot mixture. Carefully decant the hot supernatant solution into another fresh labeled test tube.
10. Add 1 drop of 2 M K_2CrO_4 to the hot liquid. Record observations in data table 1. Discard solutions and precipitates from your tests.

5.1.3 Precipitation with CrO_4^{2-}

11. To 10 – 15 drops of fresh solution from step 1 in separate clean, dry, labeled test tubes, add 1 drop of 6 M $HC_2H_3O_2$ to the supernatant solution. Next add 3-4 drops of 2 M K_2CrO_4. Record observations in data table 1. Discard solutions and precipitates from your tests.

5.1.4 Precipitation with SO_4^{2-}

12. To 10 – 15 drops of fresh solution from step 1 in separate clean, dry, labeled test tubes, add 5 drops of 3 M H_2SO_4. Record observations in data table 1. Discard solutions and precipitates from your tests.

5.1.5 Precipitation with OH^-

13. To 10 – 15 drops of fresh solution from step 1 in separate clean, dry, labeled test tubes, add 10 drops 6M NaOH. Record observations in data table 1.
14. If a precipitate forms, transfer the mixture to a centrifuge tube then centrifuge the mixture. The supernatant can be discarded in your waste container. Save the precipitate for further analysis. If no precipitate forms, the mixture can be discarded in your waste container.
15. Wash the precipitate once with deionized water.

5.1.6 Reaction of hydroxide precipitate with NH_3 and dimethylglyoxime

16. To the precipitate from Step 15 add 10 drops of 15M NH_3 (labeled NH_4OH) and mix vigorously for one minute. Record observations in data table 1.
17. Add eight drops of dimethylglyoxime (dmg) to the solutions. Record observations in data table 1. Discard solutions and precipitates from your tests.

5.1.7 Reaction of hydroxide precipitate with acid and SCN^-

18. For solutions that formed a hydroxide precipitate in Step 13, prepare fresh test tubes: 10 – 15 drops fresh solution from step 1 in separate clean, dry, labeled test tubes.
19. Obtain and wash hydroxide precipitates again by adding 10 drops 6M NaOH to the test tubes, transferring to centrifuge tubes, centrifuging, and washing the precipitate once with deionized water (repeating Steps 13 – 15).
20. To the precipitate from Step 18, add 5 drops of 6M HCl and mix vigorously. Record observations in data table 1.
21. Add two drops of 1M NH_4SCN and record observations in data table 1. Discard solutions and precipitates from your tests.

5.1.8 Reaction of fresh solution with NH_3

22. To 10 – 15 drops of fresh solution in separate clean, dry, labeled test tubes, add 10 drops 15M NH_3. Record observations in data table 1. Discard solutions and precipitates from your tests.

5.1.9 Reaction of fresh solution with SCN^-

23. To 10 – 15 drops of fresh solution in separate clean, dry, labeled test tubes, add two drops of 1M NH_4SCN. Record observations in data table 1. Discard solutions and precipitates from your tests.

All supernatant and precipitates from tests can be discarded in your waste container. Save original cations solutions and test reagents for further analysis. Be sure to clean out and thoroughly rinse all glassware (including stirring rods!) with deionized water before proceeding to the next section.

5.2 ANALYSIS OF A KNOWN MIXTURE OF CATIONS

As the section is preformed add observations to data table 2. Be mindful to include observations of the solution, precipitate and supernatant. Include interpretations of the observations. For example, ***what ions*** *are contributing to the observations observed.*

1. Prepare a mixture of cations by adding 1.0 mL of each of the following aqueous solutions to a small test tube: 0.1 M $AgNO_3$, 0.1 M $Ba(NO_3)_2$, 0.1 M $Fe(NO_3)_3$, , 0.1 M $Ni(NO_3)_2$ and 0.1 M $Pb(NO_3)_2$. Note that 1.0 mL is generally between 10-15 drops. Swirl to thoroughly mix.

5.2.1 Separation of Ag^+ and Pb^{2+} from Ni^{2+}, Fe^{3+} and Ba^{2+}

2. Transfer 1.0 mL of the mixture into a second small test tube and add 5 drops of 6 M HCl to this test tube.
3. Transfer the mixture to a centrifuge tube then centrifuge the mixture. Make sure that you balance the centrifuge before you start. Balancing is done by placing centrifuge tubes of approximately equal weight opposite each other.
4. Pour the supernatant solution into a fresh labeled test tube. Save the precipitate; it contains the Ag^+ and Pb^{2+} ions as AgCl(*s*) and $PbCl_2$(*s*).
5. To the supernatant solution, add one more drop of 6M HCl to ensure the precipitation is complete. If more precipitate is observed, centrifuge the mixture again. Save the supernatant solution; it contains the Ni^{2+}, Fe^{3+} and Ba^{2+} cations.

5.2.2 Separation and identification of Pb^{2+}

6. Wash the precipitate from Step 4 to remove contaminates that may be trapped within the solid. This can be accomplished by adding 1.0 mL of deionized water, mixing, and then centrifuging. The wash supernatant can be discarded in your waste container. Wash the precipitate twice saving the precipitate.
7. Prepare a hot water bath by adding tap water to a 250 mL beaker. Add water until the beaker is about one half full. Heat the water on a hot plate until it is boiling.
8. Add 1.5 mL of deionized water to the centrifuge tubes containing the clean precipitate. Place it in the hot water bath. Periodically stir the mixture in the test tube for 3 – 4 minutes.
9. Remove the centrifuge tube from the hot water bath and immediately centrifuge the hot mixture. Carefully decant the hot supernatant solution into another fresh labeled test tube; it contains the Pb^{2+} cations. Save the precipitate for further analysis; it contains Ag^+ ions as AgCl(*s*).

10. In order to confirm the presence of Pb^{2+}, add 1 drop of 6 M $HC_2H_3O_2$ to the supernatant solution. Next add 3 – 4 drops of 2 M K_2CrO_4. The formation of a bright yellow precipitate is confirmation of the presence of the Pb^{2+} ion.

5.2.3 Identification of Ag^+

11. Wash the precipitate from Step 9 containing the Ag^+ cations once with deionized water.
12. Add 10 – 15 drops of 15 M NH_3 (labeled NH_4OH) to dissolve the precipitate. It now contains the $Ag(NH_3)_2{}^+$ ion.
13. To the $Ag(NH_3)_2{}^+$ solution, slowly add 6M HCl until the solution is acidic. The acidity can be tested by dipping a stirring rod into the solution and then touching it (with a drop of solution) to a piece of blue litmus paper resting on a clean, dry watch glass. If the solution is acidic, it will turn blue litmus paper red. Formation of a white precipitate of AgCl in the acidic solution is confirmation of the presence of Ag^+.

5.2.4 Separation and identification of Ba^{2+} from Ni^{2+} and Fe^{3+}

14. To the supernatant solution containing the Ni^{2+}, Fe^{3+} and Ba^{2+} cations from Step 4, add 5 drops of 3 M H_2SO_4. Formation of a white precipitate of $BaSO_4$ in the solution is confirmation of the presence of Ba^{2+} as all of the Ag^+ and Pb^{2+} ions have been removed from the supernatant.
15. Transfer the mixture to a centrifuge tube then centrifuge the mixture.
16. Pour the supernatant solution into a fresh labeled test tube and add one more drop of 3 M H_2SO_4 to ensure the precipitation is complete. If more precipitate is observed, centrifuge the mixture again, decanting the supernatant to a fresh labeled test tube. Save the supernatant for further analysis; it contains the $Ni^{2+}$and Fe^{3+} cations.

5.2.5 Separation Ni^{2+} from Fe^{3+}

17. To the supernatant solution containing the $Ni^{2+}$and Fe^{3+} from Step 16, slowly add 15 M NH_3 to the solution dropwise until it is basic (test with red litmus paper). You should see a red-brown precipitate of $Fe(OH)_3$ appear when the solution turns basic.
18. Add an additional 10 drops of 15 M NH_3 and flick to mix well. At this point the nickel should be dissolved in the solution as $[Ni(NH_3)_6]^{2+}$.
19. Centrifuge the mixture and decant the supernatant solution into another fresh labeled test tube. Save both the precipitate and supernatant.

5.2.6 Identification of Fe^{3+}

20. Wash the precipitate from Step 19 once with deionized water.
21. To the $Fe(OH)_3$ precipitate add twelve drops of 6 M HCl. Next, add 4 mL of deionized water and stir well until the precipitate is completely dissolved. Finally, add two drops of 1 M NH_4SCN. The formation of a dark red solution confirms the presence of Fe^{3+} in the original solution.

5.2.7 Identification of Ni^{2+}

22. To the $[Ni(NH_3)_6]^{2+}$ solution from Step 19, add eight drops of dimethylglyoxime to the solutions. The formation of a red precipitate confirms the presence of Ni^{2+} in the original solution.

5.3 ANALYSIS AND IDENTIFICATION OF AN UNKNOWN SAMPLE

1. Obtain a test tube which contains a mixture of cations. Record the identification of the sample on your data sheet.
2. Transfer 1.0 mL of the unknown mixture into a test tube and repeat the procedure used in section 5.2.
3. On your data sheet, record observations and construct a flow chart similar to the one shown in the Background. Indicate on the flow chart whether the test for each ion is positive or negative. Then, in space provided, indicate which ions are present in your unknown sample.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name(s)**	**Date**

Table 1. Chemical Tests

Section Test and Reagents	Ag^{+}	Ba^{2+}	Fe^{3+}	Ni^{2+}	Pb^{2+}
Solution appearance					
5.1.1 Cl^{-}					
5.1.2 heat					
CrO_4^{2-}					
5.1.3 CrO_4^{2-}					
5.1.4 SO_4^{2-}					
5.1.5 OH^{-}					
5.1.6 NH_3					
dmg					
5.1.7 H^{+}					
SCN^{-}					
5.1.8 NH_3					
5.1.9 SCN^{-}					

Table 2. Separation of a known mixture

	Observations & Interpretations (Why?)
5.2.1 Separation of Ag^{+} & Pb^{2+}	
5.2.2 Separation & identification of Pb^{2+}	
5.2.3 Identification of Ag^{+}	
5.2.4 Separation & identification of Ba^{2+}	
5.2.5 Separation Ni^{2+}	
5.2.6 Identification of Fe^{3+}	
5.2.7 Identification of Ni^{2+}	

Analysis and Identification of Unknown

In the space provided below construct a flow cart for the analysis of your unknown. Indicate on the flow chart whether the test for each ion is positive or negative:

Unknown identifications: _________________

Ions present in your unknown: ___________________________________

2617 Quantitative Analysis of Iron

1.0 INTRODUCTION

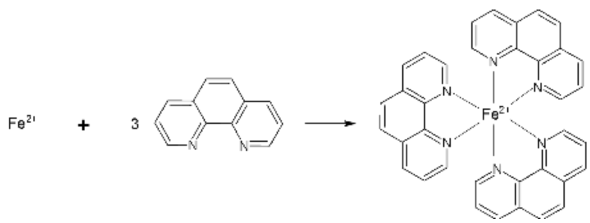
1. Objective

After completing this experiment, the student will be able to analyze the iron content in a sample using spectrophotometry.

1.2 Background

Common iron supplement tablets contain various amounts of Fe as ferrous sulfate. One example is 65 mg Fe in 325 mg ferrous sulfate. How reliable is this information? This experiment will enable you to check up on the manufacturer.

In this experiment, the presence of iron(II) ion will be detected by reacting the sample with 1,10-phenanthroline (Equation 1).



(Equation 1)

Three molecules of 1,10-phenanthroline will chelate with iron(II) to form a tris-1,10-phenanthrolineiron(II) orange-red complex. To ensure fast and optimal color development, this reaction will be maintained between a pH of 2.9 and 3.3 in the presence of excess phenanthroline. It is important to note that interferences to this reaction can come from strong oxidizing agents, phosphates, and other metal ions such as chromium, zinc, cobalt, copper, and nickel. The errors caused by these interferences can be minimized by using a larger excess of phenanthroline, initial boiling with acid, and adding excess hydroxylamine.

Spectroscopy is one of the most powerful analytical techniques in modern science. Before the advent of spectrophotometric techniques, a chemist interested in determining the amount of a particular substance present in a sample had to analyze the sample via a series of chemical reactions specific to that species and then carefully weigh the products (entire tomes exist detailing such analytical reactions). This process was extremely time consuming, prone to error, and generally impractical for measuring trace amounts. Today, most routine assaying is done quickly and efficiently by means of spectroscopy.

Spectroscopy works by correlating the concentration of a species in solution to the amount of light it absorbs. In this experiment, we will determine the quantity of iron (as Fe-phen) using visible absorption spectroscopy. Because the wavelengths of light we will use are in the visible portion of the electromagnetic spectrum, our solutions will all be colored. However, this technique can also be used in other regions of the spectrum where the wavelength is not visible to our eyes but can be measured using a photodetector.

1.3 Experimental Considerations

1.3.1 Choosing a Wavelength

It is important to choose a wavelength where the solution strongly absorbs light.

The stronger the absorption at a particular wavelength the more sensitive the instrument will be at that wavelength and the more accurate your results.

In practice, one first determines the best range of wavelengths at which to test a solution and then selects one "best" wavelength for the experiment by measuring the absorption of the solution in question across this range and choosing the wavelength that gives the greatest absorption. For example, for the orange-red complex described above, the experimenter might choose to measure the absorption of the sample at a range of wavelengths between 480 nm and 560 nm, and then narrow-in on the wavelength that gives the greatest numeric value of the absorbance.

Please note that while some spectrometers (like Spec 200 that will be used in this experiment) can do this automatically, others require changing the wavelength manually and each time the wavelength is adjusted these instruments need to be re-zeroed or they will not work properly.

1.3.2 Calibrating the Instrument

Once the specific wavelength that will be used for the experiment is chosen the experimenter needs to calibrate his or her spectrometer so that the absorbance readings can be converted into useful data such as molarity (M) of the sample.

The absorbance of light (A) is directly proportional to the concentration of the species in solution (C). This relationship is known as "Beer's law" and may be expressed as:

$A = \varepsilon b C$ (Equation 2)

where ε is the molar absorptivity (or the molar extinction coefficient), *b* is the path length or distance the light travels through the sample, and C is the concentration of the solution in terms of molarity. In most instruments *b* is a constant (1.0 cm in our experiment) and can therefore be factored into ε.

Thus, Equation 2 can be simplified as:

$A = \varepsilon C$ $y = mx$ (Equation 3)

which is the equation of a line where the intercept is through the origin (0,0).

The molar absorptivity, ε, can therefore be determined by finding the slope of a plot of the Absorbance (A) as a function of concentration (C) for a series of standard solutions of known concentrations. This is known as a *calibration curve*.

Once ε has been determined from the calibration curve, Equation 3 can be used to determine the concentration of an unknown solution by measuring its absorbance under

the same conditions.

N.B. A new calibration curve is required if a different instrument, wavelength, type of solution, or procedure is used. It is also a good idea to check an instrument's calibration on a periodic basis.

Overall, this experimental project consists of:

- Carefully preparing five (5) iron(II) solutions of known concentrations
- Determining the wavelength at which the solution is absorbing the strongest, λ_{max}
- Measuring the absorbance of each known solution to prepare a standard calibration curve
- Carefully preparing an iron(II) solution from a given dietary supplement
- Calculating the amount of iron in one tablet (from the standard calibration curve) and comparing it with the manufacturer’s label

Further Reading

Technique I: Use of Spectrophotometer (from our laboratory manual)

Technique E: Volumetric Transfer of Reagent Using a Pipet

Expt 2501 Using Excel for Graphical Analysis of Data

2.0 SAFETY PRECAUTIONS AND WASTE DISPOSAL

3.0 CHEMICALS AND SolutionS

Chemical	**Concentration**	**Approximate Amount**	**Notes**
Laboratory water		500 mL	
Hydroxylamine solution	2% w/v	25 mL	
$Fe(NH_4)_2(SO_4)_2.6H_2O$		0.2 g	Light green powder
1,10-phenanthroline		50 mL	
Sodium acetate	10% w/v	100 mL	
Hydrochloric acid	6 M	25 mL	
Sulfuric acid	6 M	6 mL	
Iron supplement tablet	As-is		

4.0 GLASSWARE AND APPARATUS

250 mL volumetric flask	100 mL volumetric flasks (7 needed)
500 mL volumetric flask	250 mL beaker (2 needed)
Volumetric pipets	Graduated cylinder
600 mL beaker (waste container)	Filter paper
Funnel	Watch glass
Spec 20/200	Balance

5.0 PROCEDURE

1. Using an Iron Supplement Tablet to Prepare a Sample Solution

NOTE: If unfamiliar, ask your instructor to describe and demonstrate the use of volumetric pipets and volumetric flasks.

1. Read the label of the bottle of iron supplement to find the amount of Fe in each tablet. Record these values in section 6.4 on your data collection sheet.
2. Transfer the tablet into a 250-mL beaker. Add 25 mL of 6 M HCl into the beaker. Check the pH of the solution using a pH paper. This pH should be around 2.
3. Boil the solution gently for about 15 mins on a hot plate to COMPLETELY dissolve the solids. Do not let the volume of the mixture to fall below 15 mL, in which case, add some laboratory water. Turn off heat and let it cool.
4. Gravity filtration: Filter the solution into a clean volumetric flask. (***Note****: Use 100-mL capacity if the iron supplement tablet contains 28 mg or less; 250-mL capacity if the iron supplement tablet contains 65 mg.*) Make sure no particles of dust, paper, or other impurities get into the flask. Pour the iron-containing liquid from the beaker onto the filter paper in the funnel. The clear solution that comes through the filter contains your sample. The solid that is left on the filter paper will eventually be discarded. Rinse the beaker three times with laboratory water (about 30 mL total) and pour these rinsings into the filter funnel also.
5. When the sample has been filtered, remove the funnel, and dispose in the trash the filter paper and undissolved residue from the iron supplement tablet.
6. Carefully add enough laboratory water to bring the solution level exactly to the mark on the flask. Mix the sample thoroughly by carefully inverting the stoppered flask repeatedly. (Hold the glass or rubber stopper tightly in place with your thumb. Make sure no liquid leaks out before the solution is completely mixed.)
7. The solution at this point may be too concentrated to be used. Prepare a dilution of it by using a volumetric pipet to transfer exactly 1.00 mL of it into a clean 100.0 mL volumetric flask. (The flask may be wet as long as it is clean.) Carefully add 10.0 mL of sodium acetate, 2.0 mL hydroxylamine, and 4.0 mL phenanthroline in this order. Add laboratory water to bring the solution level up to the 100.0-mL mark on the flask. Mix thoroughly. This will be your sample solution. You may label this as ‘Sample #1’. Set aside on your lab bench.

8. Repeat this preparation for another iron supplement tablet as an additional trial. Depending on time constraints, students may perform a total of three trials, or they may borrow data from another lab group for statistical comparisons. *Make sure the other lab group is using the same tablet brand as your group!*

5.2 Preparation of Iron(II) Stock Solution

1. Into a 250 mL beaker containing around 10 mL of laboratory water, slowly add 6 mL of 6 M sulfuric acid.
2. Weigh approximately 0.140 g of $Fe(NH_4)_2(SO_4)_2.6H_2O$ (record its exact mass) and transfer into the beaker containing the acid. Mix thoroughly to dissolve the solid completely.
3. Transfer the solution into a 500 mL volumetric flask. Rinse the beaker three times with laboratory water (about 30 mL total) and pour these rinses into the volumetric flask. Add laboratory water exactly to the mark. Mix the sample thoroughly. Label this as 'Iron(II) stock solution'.
4. Calculate the exact concentration of this iron(II) stock solution and record on your data recording sheet.

5.3 Preparation of Solutions of Known Concentration

1. Obtain the following solutions and transfer into small beakers: 75 mL iron(II) stock solution, 100 mL sodium acetate, 25 mL hydroxylamine, and 50 mL phenanthroline.
2. Label five (S1, S2, S3, S4, S5) clean 100 mL volumetric flasks for the preparation of your standard solutions.
3. Using a volumetric pipet, carefully measure 1.00 mL of the stock solution you made in section 5.2 and place it in a volumetric flask (S1). Add mL 10.0 mL sodium acetate, 2.0 mL hydroxylamine, and 4.0 mL phenanthroline (in this order). Carefully fill the flask with laboratory water exactly to the 100 mL mark. Mix the sample thoroughly. Allow at least 5 minutes for maximum color development. This is your standard solution S1.
4. Repeat step 3 to prepare the other standard solutions - using 2.00, 3.00, 5.00, and 10.00 mL of the stock solution from section 5.4 into each volumetric flask. Calculate the concentrations of these standard solutions and record on your data recording sheet.
5. Prepare a blank solution (labeled S0) by combining 10.0 mL sodium acetate, 2.0 mL hydroxylamine, and 4.0 mL phenanthroline in a clean 100.0 mL volumetric flask and diluting with laboratory water to the mark. Mix thoroughly.

5.4 Absorbance Measurements

1. Gather all the prepared solutions (standards and sample) for spectroscopic analysis. The proper use and operation of the spectrophotometer can be found in Appendix: Technique I Use of Spec 20/Spec 200.
2. Obtain solution from S5 to obtain the λ_{max} that will be used for the determinations. Use the 'Scan' mode of the spectrophotometer.

3. Once λ_{max} is determined, adjust the spectrophotometer to measure the absorbance of the solutions at this wavelength.
4. Use the ‘blank’ solution (S0) as the reference liquid - that is, set the instrument for zero absorbance when the light path is passing through this ‘blank’ solution.
5. Measure the absorbance for each of your solutions of known concentrations (S1- S5) and for your sample solutions (the 100 mL volumetric flasks labeled "Sample#").
6. Rinse the cuvette (sample holder) thoroughly with laboratory water, then with each solution in between measurements.

6.0 DATA RECORDING SHEET

Last Name	**First Name**		**Partner Name(s)**	**Date**

6.0 DATA COLLECTION

6.1 Preparation of the Iron(II) Stock Solution (procedure section 5.2)

Mass of $Fe(NH_4)_2(SO_4)_2.6H_2O$ (green powder)	
Concentration of Fe^{2+} stock solution	

Show your calculations below for the exact concentration of your iron(II) stock solution.

6.2 Absorbance Measurements (procedure section 5.4)

λ_{max} = ________________________________

6.3 Absorbances for Standard Solutions

Solution	**Concentration**	**Absorbance**
S1		
S2		
S3		
S4		
S5		

Show your calculations below for one of the standard solutions.

6.4 Absorbances for Tablets

Solution	**Mass of Fe** **(According to Manufacturer)**	**Absorbance**
Tablet #1		
Tablet #2		
Tablet #3 (or class data)		

7.0 CALCULATIONS

1. Using Excel, construct a graph of absorbance (A) vs. concentration (C) for your standard solutions. Provide a comprehensive title for the graph and label your axes completely. Using the trendline function, add a best-fit line to your plotted data and display the equation of this line (*in the form: $y = mx + b$* or A = εC + *b*) and its R^2 value on your graph. **Submit this graph with your report.**

Equation of the line____

Use this graph to determine the value of the molar absorptivity, ε. (Be certain to include the proper units!)

Molar absorptivity, ε: ______________________________

2 Use your graph to determine the concentration and mass of Fe^{2+} in your tablets.

	Tablet #1	Tablet #2	Tablet #3 or class data
Absorbance			
Concentration of Fe^{2+}in the diluted solution (from calibration curve)			
Concentration of Fe^{2+}(aq) in the original undiluted sample			
Mass of Fe in the tablet*			
Average mass of Fe in the tablet			

*use the undiluted original concentration

Show your calculations for one of the trials.

1. POST-LAB QUESTIONS

1. Discuss your experimental results. Do your results agree with the iron supplement manufacturer's claim regarding the milligrams of Fe in each tablet? Discuss any of your experimental errors that may affect your conclusions.

2. How does the structure of the Fe-phen complex become more useful in the assay of iron using spectrophotometry. Consider the color and the stability of the complex.

3. Using the analytical method we used in this lab, what other types of samples could you assay the Fe content of? Discuss any limitations or interferences with the assay process that you may encounter. How would you plan to address them?

2618 Nuclear Chemistry

Last Name	First Name	Partner Name(s)	Date

1.0 INTRODUCTION

1.1 Objectives

After this “dry” exercise, students will know how to measure radiation from nuclear decay, will recognize the presence of background radiation, will determine the half-life of a radioisotope, and will determine the degree of shielding lead provides against radiation.

1.2 Background

Nuclear reactions generate three main types of radiation: alpha, beta, and gamma. Alpha radiation consists of He^{+2} nuclei; beta radiation consists of electrons; and gamma radiation is energy. These different forms of radiation are detected with a Geiger counter.

The experiment is performed by Linda Nuss and recorded in three you-tube videos. Further background information is provided in each video.

PART A: BACKGROUND RADIATION

Watch the following.

https://www.youtube.com/watch?v=RNVRfSPJAv0&t=169s

2.0 Safety and Waste Disposal

List three (3) safety precautions for doing nuclear chemistry (you may need to do an extra search outside of the video).

3.0 Chemicals and Solutions Used

What are the 4 sources of background radiation?

4.0 Glassware and Apparatus

What does a Geiger Counter actually count? (Hint: it isn’t the radiation.)

6.0 Data Recording Sheet and 7.0 Calculations

Collect the data from the video for the background radiation and determine the average.

Minute	Average Geiger Counter Count
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Average over 10 minutes	

PART B: DETERMINING THE HALF-LIFE OF A RADIOISOTOPE

Watch the video https://www.youtube.com/watch?v=43BwBAaDt0k

3.0 Chemicals and Solutions

What is the decay process being studied?

6.0 Data Recording Sheet

Record the average count of the Geiger Counter over the time period, then calculate the average minus the average background from Part A

Time (minutes)	Average Count	Average Count minus Background from Part A	Ln of Average Count minus Background
0			
1			
2			
3			
4			
5			
6			
7			
8			

7.0 Calculations

1. Generate the graph of ln(average count – background) versus time as shown in the video and attach it or paste it here.
2. Determine the slope of your graph, which is –k.
3. Determine the half-life, which is 0.693/k.

PART C: DETERMINING THE QUANTITY OF LEAD NECESSARY TO SHIELD AGAINST RADIATION

Watch the video https://www.youtube.com/watch?v=_Y6EGJpc-VQ

6.0 Data Collection

Number of Pb Sheets	Thickness of Pb Sheets Total (mm)	Geiger Counter Readings (Enter all 5)	Average of the five readings	Average Counter Reading minus Background from Part A	Ln of Average minus Background
1					
2					
3					
4					
5					

7.0 Calculations

Plot ln(counter reading – background) versus mm of Pb and attach the graph or paste it here.

How many mm of Pb are required to block all of the radiation from the Co-60?

(Note that ln(0) is undefined. We can use ln(1) where the counter reading is just one more than the background count.)

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D

Glossary

Sample Word 1 | Sample Definition 1

Detailed Licensing

Overview

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